FHIR-STU3 — Medications Module (sections 11 – 11.8.3)

Source: HL7 FHIR STU3 (v3.0.2). Text-only edition for AI ingestion (NotebookLM); pages follow the source order (its table of contents when one exists).

11 Medications Module

URL: https://hl7.org/fhir/STU3/medications-module.html

There are other resources that are of particular interest in the medication domain.

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Work Group Pharmacy & Public Health and Emergency Response | Ballot Status: Informative

11.0 Medications Module

11.0.1 Introduction

This module is concerned with resources and functionality in 2 main domains:

* The ordering, dispensing, administration of medications and recording statements of medication use.

* Recording of Immunizations given (or not given), and recommendations for an individual patient at a point in time.

11.0.2 Index

MedicationRequest MedicationDispense MedicationAdministration | MedicationStatement Medication | Immunization ImmunizationRecommendation

11.0.2.1 Medications

Name | Description

MedicationRequest | Represents an instruction for the administration of medication to a patient - both in the inpatient (hospital) and community setting. It can also include instructions for the dispensing, the reasons why the administration should occur and other data. It is called an 'Request' to be consistent with other FHIR resources and the workflow pattern, but a common alias for this resource is a 'Prescription' or an 'Order'. The Order itself represents the content of the instruction and is not, by itself, actionable. The workflow process around 'fulfilling' the order is part of the generic FHIR workflow (see below), with the MedicationRequest representing the contents.

MedicationDispense | The provision of a supply of a medication with the intention that it is subsequently consumed by a patient (usually in response to a prescription).

MedicationAdministration | A record of a patient actually consuming a medicine, or if it has otherwise been administered to them

MedicationStatement | This is a record indicating that a patient may be taking a medication now, has taken the medication in the past, or will be taking the medication in the future. The source for this information can be the patient, significant other (such as a family member or spouse), or a clinician. A common scenario where this information is captured is during the history taking process during a patient visit or stay. A medication statement is not a part of the prescribe->dispense->administer sequence, but is a report that such a sequence (or at least a part of it) did take place, resulting in a belief that the patient has received a particular medication. It may be used to construct a patients 'Current Medications' list.

Medication | The medication resource represents an actual medication that can be given to a patient, and referenced by the other medication resources. In many cases, this resource is not needed and the drug is indicated by a reference to the appropriate terminology and so can be represented using a codeable concept. In other cases, however, it may be desired to indicate more details than the simple drug (such as the packaging, whether it is a generic medication or the active and inactive ingredients) and so the Medication resource can be used for this.

11.0.2.2 Immunizations

Name | Description

Immunization | The Immunization resource is intended to cover the recording of current and historical administration of vaccines to patients across all healthcare disciplines in all care settings and all regions. This includes immunization of both humans and animals, but does not include the administration of non-vaccine agents, even those that may have or claim to have immunological effects.

ImmunizationRecommendation | A patient's point-in-time immunization and recommendation (i.e. forecasting a patient's immunization eligibility according to a published schedule) with optional supporting justification

11.0.3 Common Use Cases

* Placing a Medication Request (aka Prescription or Order). The MedicationRequest resource represents the details of what medication the prescriber intends the patient to receive and other details such as the dose, timing and route. However, while it represents that order, the actual workflow around the supply (dispensing) and administration of that medication is managed by the common workflow functionality, resulting in the creation of the other medication resources (MedicationDispense, MedicationAdministration) as it executes.

* Listing a patient's current medications. This is represented using the List resource, which allows an author to construct a 'curated' list. This has features beyond a simple collection of resources - such as the ability to state that the patient is not currently taking any medications at the time that the list is constructed, or changes that are made at a point in time (sometimes called medication reconciliation). FHIR defines a number of specific functional lists for this purpose, and a specific operation to retrieve it. Note that FHIR does not describe how the list should be maintained by the individual system, as that will vary widely between implementations.

* Retrieving a list of Immunizations. This can be achieved using a simple query against the Immunization resource type. Note that an Immunization resource can specifically state that it was not given, as well as any reactions that occurred. Note that if the reaction is considered to be an indication of an allergy or intolerance, then a separate AllergyIntolerance resource should be created.

* Retrieving recommendations for a specific patient at a point in time. The recommendation for immunizations to give would be represented by an ImmunizationRecommendation resource, but the determination of that recommendation is a part of Decision Support, with the actual immunization schedule being represented by a generic Plan Definition, individual Care Plan or some other record.

* Dispensing a medication based on a Medication Request (aka Prescription or Order). The MedicationDispense resource represents the details of the dispensing event including the actual product provided to the patient and the dosage to be administered or consumed.

* Retrieving the planned medication therapy for specific patient at a poin in time. The intended medication therapy may be represented by an individual Care Plan.

11.0.4 Other resources

* SupplyRequest and SupplyDelivery which, like MedicationRequest, are 'detail' resources used as part of a workflow. They are concerned with the request of supplies used in the healthcare process. This includes supplies specifically used in the treatment of patients as well as supply movement within an institution (transport a set of supplies from materials management to a service unit (nurse station).

11.0.5 Security and Privacy

As with all clinical data, Medications (in particular) can be sensitive information as specific medications can indicate the presence of private information such as mental health disorders or HIV. However, withholding information about what medications a person is taking can lead to catastrophic results, and so needs to be considered very carefully. At the least, a clinician should be made aware that there is information available that they have not been given when making clinical decisions.

For more general considerations, see the Security and Privacy module.

11.0.6 Developmental Roadmap

The Pharmacy workgroup has plans to improve all existing resources e.g. adding in features that support detailing our conditional orders in a structured way; evaluating requirements for supporting drug formularies.

Drug formulary work may involve updates to the Medication Resource.

11.1 Resource MedicationRequest - Content

URL: https://hl7.org/fhir/STU3/medicationrequest.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Pharmacy Work Group | Maturity Level: 3 | Trial Use | Compartments: Encounter, Patient, Practitioner

An order or request for both supply of the medication and the instructions for administration of the medication to a patient. The resource is called "MedicationRequest" rather than "MedicationPrescription" or "MedicationOrder" to generalize the use across inpatient and outpatient settings, including care plans, etc., and to harmonize with workflow patterns.

11.1.1 Scope and Usage

This resource covers all orders for medications for a patient. This includes in-patient medication orders as well as community orders (whether filled by the prescriber or by a pharmacy). It also includes orders for over-the-counter medications (e.g. Aspirin), total parenteral nutrition and diet/ vitamin supplements. It may be used to support the order of medication-related devices. It is not intended for use in prescribing particular diets, or for ordering non-medication-related items (eye-glasses, supplies, etc.).

The MedicationRequest resource is a "request" resource from a FHIR workflow perspective - see Workflow Request.

The MedicationRequest resource allows requesting only a single medication. If a workflow requires requesting multiple items simultaneously, this is done using multiple instances of this resource. These instances can be linked in different ways, depending on the needs of the workflow. For guidance, refer to the Request pattern

11.1.2 Boundaries and Relationships

The MedicationRequest resource is used to request or order medication for a subject. When requesting supplies or devices when there is a patient focus or instructions regarding their use, SupplyRequest or DeviceRequest should be used instead

The Medication domain includes a number of related resources

MedicationRequest | An order for both supply of the medication and the instructions for administration of the medicine to a patient.

MedicationDispense | Provision of a supply of a medication with the intention that it is subsequently consumed by a patient (usually in response to a prescription).

MedicationAdministration | When a patient actually consumes a medicine, or it is otherwise administered to them

MedicationStatement | This is a record of medication being taken by a patient, or that the medication has been given to a patient where the record is the result of a report from the patient, or another clinician. A medication statement is not a part of the prescribe->dispense->administer sequence but is a report that such a sequence (or at least a part of it) did take place resulting in a belief that the patient has received a particular medication.

This resource is referenced by CarePlan, Claim, ClinicalImpression, DiagnosticReport, ExplanationOfBenefit, MedicationAdministration, MedicationDispense, MedicationStatement and Observation

11.1.3 Resource Content

* Structure
* UML
* XML
* JSON
* Turtle
* R2 Diff
* All

Structure

Name | Flags | Card. | Type | Description & Constraints
MedicationRequest | | | DomainResource | Ordering of medication for patient or group Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | External ids for this request
definition | ? | 0..* | Reference(ActivityDefinition | PlanDefinition) | Protocol or definition
basedOn | ? | 0..* | Reference(CarePlan | MedicationRequest | ProcedureRequest | ReferralRequest) | What request fulfills
groupIdentifier | ? | 0..1 | Identifier | Composite request this is part of
status | ?!? | 0..1 | code | active | on-hold | cancelled | completed | entered-in-error | stopped | draft | unknown MedicationRequestStatus (Required)
intent | ?!? | 1..1 | code | proposal | plan | order | instance-order MedicationRequestIntent (Required)
category | | 0..1 | CodeableConcept | Type of medication usage MedicationRequestCategory (Preferred)
priority | ? | 0..1 | code | routine | urgent | stat | asap MedicationRequestPriority (Required)
medication[x] | ? | 1..1 | | Medication to be taken SNOMED CT Medication Codes (Example)
medicationCodeableConcept | | | CodeableConcept
medicationReference | | | Reference(Medication)
subject | ? | 1..1 | Reference(Patient | Group) | Who or group medication request is for
context | | 0..1 | Reference(Encounter | EpisodeOfCare) | Created during encounter/admission/stay
supportingInformation | | 0..* | Reference(Any) | Information to support ordering of the medication
authoredOn | ? | 0..1 | dateTime | When request was initially authored
requester | ?I | 0..1 | BackboneElement | Who/What requested the Request + onBehalfOf can only be specified if agent is practitioner or device
agent | ? | 1..1 | Reference(Practitioner | Organization | Patient | RelatedPerson | Device) | Who ordered the initial medication(s)
onBehalfOf | ?I | 0..1 | Reference(Organization) | Organization agent is acting for
recorder | | 0..1 | Reference(Practitioner) | Person who entered the request
reasonCode | | 0..* | CodeableConcept | Reason or indication for writing the prescription Condition/Problem/Diagnosis Codes (Example)
reasonReference | | 0..* | Reference(Condition | Observation) | Condition or Observation that supports why the prescription is being written
note | | 0..* | Annotation | Information about the prescription
dosageInstruction | | 0..* | Dosage | How the medication should be taken
dispenseRequest | | 0..1 | BackboneElement | Medication supply authorization
validityPeriod | | 0..1 | Period | Time period supply is authorized for
numberOfRepeatsAllowed | | 0..1 | positiveInt | Number of refills authorized
quantity | | 0..1 | SimpleQuantity | Amount of medication to supply per dispense
expectedSupplyDuration | | 0..1 | Duration | Number of days supply per dispense
performer | | 0..1 | Reference(Organization) | Intended dispenser
substitution | | 0..1 | BackboneElement | Any restrictions on medication substitution
allowed | ?! | 1..1 | boolean | Whether substitution is allowed or not
reason | | 0..1 | CodeableConcept | Why should (not) substitution be made SubstanceAdminSubstitutionReason (Example)
priorPrescription | | 0..1 | Reference(MedicationRequest) | An order/prescription that is being replaced
detectedIssue | | 0..* | Reference(DetectedIssue) | Clinical Issue with action
eventHistory | | 0..* | Reference(Provenance) | A list of events of interest in the lifecycle
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicationRequest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External ids for this request --></identifier>
 <definition><!-- 0..* Reference(ActivityDefinition|PlanDefinition) Protocol or definition --></definition>
 <basedOn><!-- 0..* Reference(CarePlan|MedicationRequest|ProcedureRequest|
   ReferralRequest) What request fulfills --></basedOn>
 <groupIdentifier><!-- 0..1 Identifier Composite request this is part of --></groupIdentifier>
 <status value="[code]"/><!-- 0..1 active | on-hold | cancelled | completed | entered-in-error | stopped | draft |
    unknown -->
 <intent value="[code]"/><!-- 1..1 proposal | plan | order | instance-order -->
 <category><!-- 0..1 CodeableConcept Type of medication usage --></category>
 <priority value="[code]"/><!-- 0..1 routine | urgent | stat | asap -->
 <medication[x]><!-- 1..1 CodeableConcept|Reference(Medication) Medication to be taken --></medication[x]>
 <subject><!-- 1..1 Reference(Patient|Group) Who or group medication request is for --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Created during encounter/admission/stay --></context>
 <supportingInformation><!-- 0..* Reference(Any) Information to support ordering of the medication
    --></supportingInformation>
 <authoredOn value="[dateTime]"/><!-- 0..1 When request was initially authored -->
 <requester>  <!-- 0..1 Who/What requested the Request -->
  <agent><!-- 1..1 Reference(Practitioner|Organization|Patient|RelatedPerson|
    Device) Who ordered the initial medication(s) --></agent>
```

```
  <onBehalfOf><!--  0..1 Reference(Organization) Organization agent is acting for --></onBehalfOf>
 </requester>
 <recorder><!-- 0..1 Reference(Practitioner) Person who entered the request --></recorder>
 <reasonCode><!-- 0..* CodeableConcept Reason or indication for writing the prescription --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation) Condition or Observation that supports why the prescription
     is being written --></reasonReference>
 <note><!-- 0..* Annotation Information about the prescription --></note>
 <dosageInstruction><!-- 0..* Dosage How the medication should be taken --></dosageInstruction>
 <dispenseRequest>  <!-- 0..1 Medication supply authorization -->
  <validityPeriod><!-- 0..1 Period Time period supply is authorized for --></validityPeriod>
  <numberOfRepeatsAllowed value="[positiveInt]"/><!-- 0..1 Number of refills authorized -->
  <quantity><!-- 0..1 Quantity(SimpleQuantity) Amount of medication to supply per dispense --></quantity>
  <expectedSupplyDuration><!-- 0..1 Duration Number of days supply per dispense --></expectedSupplyDuration>
  <performer><!-- 0..1 Reference(Organization) Intended dispenser --></performer>
 </dispenseRequest>
 <substitution>  <!-- 0..1 Any restrictions on medication substitution -->
  <allowed value="[boolean]"/><!-- 1..1 Whether substitution is allowed or not -->
  <reason><!-- 0..1 CodeableConcept Why should (not) substitution be made --></reason>
 </substitution>
 <priorPrescription><!-- 0..1 Reference(MedicationRequest) An order/prescription that is being replaced
    --></priorPrescription>
 <detectedIssue><!-- 0..* Reference(DetectedIssue) Clinical Issue with action --></detectedIssue>
 <eventHistory><!-- 0..* Reference(Provenance) A list of events of interest in the lifecycle --></eventHistory>
</MedicationRequest>
```

JSON Template

```
{
  "resourceType" : "MedicationRequest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External ids for this request
  "definition" : [{ Reference(ActivityDefinition|PlanDefinition) }], // Protocol or definition
  "basedOn" : [{ Reference(CarePlan|MedicationRequest|ProcedureRequest|
   ReferralRequest) }], // What request fulfills
  "groupIdentifier" : { Identifier }, // Composite request this is part of
  "status" : "<code>", // active | on-hold | cancelled | completed | entered-in-error | stopped | draft | unknown
  "intent" : "<code>", // R!  proposal | plan | order | instance-order
  "category" : { CodeableConcept }, // Type of medication usage
  "priority" : "<code>", // routine | urgent | stat | asap
  // medication[x]: Medication to be taken. One of these 2:
  "medicationCodeableConcept" : { CodeableConcept },
  "medicationReference" : { Reference(Medication) },
  "subject" : { Reference(Patient|Group) }, // R!  Who or group medication request is for
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Created during encounter/admission/stay
  "supportingInformation" : [{ Reference(Any) }], // Information to support ordering of the medication
  "authoredOn" : "<dateTime>", // When request was initially authored
  "requester" : { // Who/What requested the Request
    "agent" : { Reference(Practitioner|Organization|Patient|RelatedPerson|
    Device) }, // R!  Who ordered the initial medication(s)
    "onBehalfOf" : { Reference(Organization) } // C? Organization agent is acting for
  },
  "recorder" : { Reference(Practitioner) }, // Person who entered the request
  "reasonCode" : [{ CodeableConcept }], // Reason or indication for writing the prescription
  "reasonReference" : [{ Reference(Condition|Observation) }], // Condition or Observation that supports why the
    prescription is being written
  "note" : [{ Annotation }], // Information about the prescription
  "dosageInstruction" : [{ Dosage }], // How the medication should be taken
  "dispenseRequest" : { // Medication supply authorization
    "validityPeriod" : { Period }, // Time period supply is authorized for
    "numberOfRepeatsAllowed" : "<positiveInt>", // Number of refills authorized
    "quantity" : { Quantity(SimpleQuantity) }, // Amount of medication to supply per dispense
    "expectedSupplyDuration" : { Duration }, // Number of days supply per dispense
    "performer" : { Reference(Organization) } // Intended dispenser
  },
  "substitution" : { // Any restrictions on medication substitution
    "allowed" : <boolean>, // R!  Whether substitution is allowed or not
    "reason" : { CodeableConcept } // Why should (not) substitution be made
  },
  "priorPrescription" : { Reference(MedicationRequest) }, // An order/prescription that is being replaced
  "detectedIssue" : [{ Reference(DetectedIssue) }], // Clinical Issue with action
  "eventHistory" : [{ Reference(Provenance) }] // A list of events of interest in the lifecycle
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicationRequest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicationRequest.identifier [ Identifier ], ... ; # 0..* External ids for this request
  fhir:MedicationRequest.definition [ Reference(ActivityDefinition|PlanDefinition) ], ... ; # 0..* Protocol or
    definition
  fhir:MedicationRequest.basedOn [ Reference(CarePlan|MedicationRequest|ProcedureRequest|ReferralRequest) ], ... ; #
    0..* What request fulfills
  fhir:MedicationRequest.groupIdentifier [ Identifier ]; # 0..1 Composite request this is part of
  fhir:MedicationRequest.status [ code ]; # 0..1 active | on-hold | cancelled | completed | entered-in-error | stopped |
```

```
    draft | unknown
  fhir:MedicationRequest.intent [ code ]; # 1..1 proposal | plan | order | instance-order
  fhir:MedicationRequest.category [ CodeableConcept ]; # 0..1 Type of medication usage
  fhir:MedicationRequest.priority [ code ]; # 0..1 routine | urgent | stat | asap
  # MedicationRequest.medication[x] : 1..1 Medication to be taken. One of these 2
    fhir:MedicationRequest.medicationCodeableConcept [ CodeableConcept ]
    fhir:MedicationRequest.medicationReference [ Reference(Medication) ]
  fhir:MedicationRequest.subject [ Reference(Patient|Group) ]; # 1..1 Who or group medication request is for
  fhir:MedicationRequest.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Created during encounter/admission/stay
  fhir:MedicationRequest.supportingInformation [ Reference(Any) ], ... ; # 0..* Information to support ordering of the
    medication
  fhir:MedicationRequest.authoredOn [ dateTime ]; # 0..1 When request was initially authored
  fhir:MedicationRequest.requester [ # 0..1 Who/What requested the Request
    fhir:MedicationRequest.requester.agent [ Reference(Practitioner|Organization|Patient|RelatedPerson|Device) ]; # 1..1
     Who ordered the initial medication(s)
    fhir:MedicationRequest.requester.onBehalfOf [ Reference(Organization) ]; # 0..1 Organization agent is acting for
  ];
  fhir:MedicationRequest.recorder [ Reference(Practitioner) ]; # 0..1 Person who entered the request
  fhir:MedicationRequest.reasonCode [ CodeableConcept ], ... ; # 0..* Reason or indication for writing the prescription
  fhir:MedicationRequest.reasonReference [ Reference(Condition|Observation) ], ... ; # 0..* Condition or Observation
    that supports why the prescription is being written
  fhir:MedicationRequest.note [ Annotation ], ... ; # 0..* Information about the prescription
  fhir:MedicationRequest.dosageInstruction [ Dosage ], ... ; # 0..* How the medication should be taken
  fhir:MedicationRequest.dispenseRequest [ # 0..1 Medication supply authorization
    fhir:MedicationRequest.dispenseRequest.validityPeriod [ Period ]; # 0..1 Time period supply is authorized for
    fhir:MedicationRequest.dispenseRequest.numberOfRepeatsAllowed [ positiveInt ]; # 0..1 Number of refills authorized
    fhir:MedicationRequest.dispenseRequest.quantity [ Quantity(SimpleQuantity) ]; # 0..1 Amount of medication to supply
    per dispense
    fhir:MedicationRequest.dispenseRequest.expectedSupplyDuration [ Duration ]; # 0..1 Number of days supply per
    dispense
    fhir:MedicationRequest.dispenseRequest.performer [ Reference(Organization) ]; # 0..1 Intended dispenser
  ];
  fhir:MedicationRequest.substitution [ # 0..1 Any restrictions on medication substitution
    fhir:MedicationRequest.substitution.allowed [ boolean ]; # 1..1 Whether substitution is allowed or not
    fhir:MedicationRequest.substitution.reason [ CodeableConcept ]; # 0..1 Why should (not) substitution be made
  ];
  fhir:MedicationRequest.priorPrescription [ Reference(MedicationRequest) ]; # 0..1 An order/prescription that is being
    replaced
  fhir:MedicationRequest.detectedIssue [ Reference(DetectedIssue) ], ... ; # 0..* Clinical Issue with action
  fhir:MedicationRequest.eventHistory [ Reference(Provenance) ], ... ; # 0..* A list of events of interest in the
    lifecycle
]
```

Changes since DSTU2

MedicationRequest | Name Changed from MedicationOrder to MedicationRequest

MedicationRequest.definition | Added Element

MedicationRequest.basedOn | Added Element

MedicationRequest.groupIdentifier | Added Element

MedicationRequest.status | Change value set from http://hl7.org/fhir/ValueSet/medication-order-status to http://hl7.org/fhir/ValueSet/medication-request-status

MedicationRequest.intent | Added Element

MedicationRequest.category | Added Element

MedicationRequest.priority | Added Element

MedicationRequest.subject | Added Element

MedicationRequest.context | Added Element

MedicationRequest.supportingInformation | Added Element

MedicationRequest.authoredOn | Added Element

MedicationRequest.requester | Added Element

MedicationRequest.requester.agent | Added Element

MedicationRequest.requester.onBehalfOf | Added Element

MedicationRequest.recorder | Added Element

MedicationRequest.reasonCode | Added Element

MedicationRequest.reasonReference | Added Element

MedicationRequest.note | Max Cardinality changed from 1 to * Type changed from string to Annotation

MedicationRequest.dosageInstruction | Type changed from BackboneElement to Dosage

MedicationRequest.dispenseRequest.expectedSupplyDuration | Type changed from Quantity{http://hl7.org/fhir/StructureDefinition/Duration} to Duration

MedicationRequest.dispenseRequest.performer | Added Element

MedicationRequest.substitution.allowed | Added Element

MedicationRequest.priorPrescription | Type changed from Reference(MedicationOrder) to Reference(MedicationRequest)

MedicationRequest.detectedIssue | Added Element

MedicationRequest.eventHistory | Added Element

MedicationOrder.dateWritten | deleted

MedicationOrder.dateEnded | deleted

MedicationOrder.reasonEnded | deleted

MedicationOrder.patient | deleted

MedicationOrder.prescriber | deleted

MedicationOrder.encounter | deleted

MedicationOrder.reason[x] | deleted

MedicationOrder.dosageInstruction.text | deleted

MedicationOrder.dosageInstruction.additionalInstructions | deleted

MedicationOrder.dosageInstruction.timing | deleted
MedicationOrder.dosageInstruction.asNeeded[x] | deleted
MedicationOrder.dosageInstruction.site[x] | deleted
MedicationOrder.dosageInstruction.route | deleted
MedicationOrder.dosageInstruction.method | deleted
MedicationOrder.dosageInstruction.dose[x] | deleted
MedicationOrder.dosageInstruction.rate[x] | deleted
MedicationOrder.dosageInstruction.maxDosePerPeriod | deleted
MedicationOrder.dispenseRequest.medication[x] | deleted
MedicationOrder.substitution.type | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints
MedicationRequest | | | DomainResource | Ordering of medication for patient or group Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | External ids for this request
definition | ? | 0..* | Reference(ActivityDefinition | PlanDefinition) | Protocol or definition
basedOn | ? | 0..* | Reference(CarePlan | MedicationRequest | ProcedureRequest | ReferralRequest) | What request fulfills
groupIdentifier | ? | 0..1 | Identifier | Composite request this is part of
status | ?!? | 0..1 | code | active | on-hold | cancelled | completed | entered-in-error | stopped | draft | unknown MedicationRequestStatus (Required)
intent | ?!? | 1..1 | code | proposal | plan | order | instance-order MedicationRequestIntent (Required)
category | | 0..1 | CodeableConcept | Type of medication usage MedicationRequestCategory (Preferred)
priority | ? | 0..1 | code | routine | urgent | stat | asap MedicationRequestPriority (Required)
medication[x] | ? | 1..1 | | Medication to be taken SNOMED CT Medication Codes (Example)
medicationCodeableConcept | | | CodeableConcept
medicationReference | | | Reference(Medication)
subject | ? | 1..1 | Reference(Patient | Group) | Who or group medication request is for
context | | 0..1 | Reference(Encounter | EpisodeOfCare) | Created during encounter/admission/stay
supportingInformation | | 0..* | Reference(Any) | Information to support ordering of the medication
authoredOn | ? | 0..1 | dateTime | When request was initially authored
requester | ?I | 0..1 | BackboneElement | Who/What requested the Request + onBehalfOf can only be specified if agent is practitioner or device
agent | ? | 1..1 | Reference(Practitioner | Organization | Patient | RelatedPerson | Device) | Who ordered the initial medication(s)
onBehalfOf | ?I | 0..1 | Reference(Organization) | Organization agent is acting for
recorder | | 0..1 | Reference(Practitioner) | Person who entered the request
reasonCode | | 0..* | CodeableConcept | Reason or indication for writing the prescription Condition/Problem/Diagnosis Codes (Example)
reasonReference | | 0..* | Reference(Condition | Observation) | Condition or Observation that supports why the prescription is being written
note | | 0..* | Annotation | Information about the prescription
dosageInstruction | | 0..* | Dosage | How the medication should be taken
dispenseRequest | | 0..1 | BackboneElement | Medication supply authorization
validityPeriod | | 0..1 | Period | Time period supply is authorized for
numberOfRepeatsAllowed | | 0..1 | positiveInt | Number of refills authorized
quantity | | 0..1 | SimpleQuantity | Amount of medication to supply per dispense
expectedSupplyDuration | | 0..1 | Duration | Number of days supply per dispense
performer | | 0..1 | Reference(Organization) | Intended dispenser
substitution | | 0..1 | BackboneElement | Any restrictions on medication substitution
allowed | ?! | 1..1 | boolean | Whether substitution is allowed or not
reason | | 0..1 | CodeableConcept | Why should (not) substitution be made SubstanceAdminSubstitutionReason (Example)
priorPrescription | | 0..1 | Reference(MedicationRequest) | An order/prescription that is being replaced
detectedIssue | | 0..* | Reference(DetectedIssue) | Clinical Issue with action
eventHistory | | 0..* | Reference(Provenance) | A list of events of interest in the lifecycle
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicationRequest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External ids for this request --></identifier>
 <definition><!-- 0..* Reference(ActivityDefinition|PlanDefinition) Protocol or definition --></definition>
 <basedOn><!-- 0..* Reference(CarePlan|MedicationRequest|ProcedureRequest|
   ReferralRequest) What request fulfills --></basedOn>
 <groupIdentifier><!-- 0..1 Identifier Composite request this is part of --></groupIdentifier>
 <status value="[code]"/><!-- 0..1 active | on-hold | cancelled | completed | entered-in-error | stopped | draft |
    unknown -->
 <intent value="[code]"/><!-- 1..1 proposal | plan | order | instance-order -->
 <category><!-- 0..1 CodeableConcept Type of medication usage --></category>
 <priority value="[code]"/><!-- 0..1 routine | urgent | stat | asap -->
 <medication[x]><!-- 1..1 CodeableConcept|Reference(Medication) Medication to be taken --></medication[x]>
 <subject><!-- 1..1 Reference(Patient|Group) Who or group medication request is for --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Created during encounter/admission/stay --></context>
 <supportingInformation><!-- 0..* Reference(Any) Information to support ordering of the medication
    --></supportingInformation>
 <authoredOn value="[dateTime]"/><!-- 0..1 When request was initially authored -->
```

```
 <requester>  <!-- 0..1 Who/What requested the Request -->
  <agent><!-- 1..1 Reference(Practitioner|Organization|Patient|RelatedPerson|
    Device) Who ordered the initial medication(s) --></agent>
  <onBehalfOf><!--  0..1 Reference(Organization) Organization agent is acting for --></onBehalfOf>
 </requester>
 <recorder><!-- 0..1 Reference(Practitioner) Person who entered the request --></recorder>
 <reasonCode><!-- 0..* CodeableConcept Reason or indication for writing the prescription --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation) Condition or Observation that supports why the prescription
     is being written --></reasonReference>
 <note><!-- 0..* Annotation Information about the prescription --></note>
 <dosageInstruction><!-- 0..* Dosage How the medication should be taken --></dosageInstruction>
 <dispenseRequest>  <!-- 0..1 Medication supply authorization -->
  <validityPeriod><!-- 0..1 Period Time period supply is authorized for --></validityPeriod>
  <numberOfRepeatsAllowed value="[positiveInt]"/><!-- 0..1 Number of refills authorized -->
  <quantity><!-- 0..1 Quantity(SimpleQuantity) Amount of medication to supply per dispense --></quantity>
  <expectedSupplyDuration><!-- 0..1 Duration Number of days supply per dispense --></expectedSupplyDuration>
  <performer><!-- 0..1 Reference(Organization) Intended dispenser --></performer>
 </dispenseRequest>
 <substitution>  <!-- 0..1 Any restrictions on medication substitution -->
  <allowed value="[boolean]"/><!-- 1..1 Whether substitution is allowed or not -->
  <reason><!-- 0..1 CodeableConcept Why should (not) substitution be made --></reason>
 </substitution>
 <priorPrescription><!-- 0..1 Reference(MedicationRequest) An order/prescription that is being replaced
    --></priorPrescription>
 <detectedIssue><!-- 0..* Reference(DetectedIssue) Clinical Issue with action --></detectedIssue>
 <eventHistory><!-- 0..* Reference(Provenance) A list of events of interest in the lifecycle --></eventHistory>
</MedicationRequest>
```

JSON Template

```
{
  "resourceType" : "MedicationRequest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External ids for this request
  "definition" : [{ Reference(ActivityDefinition|PlanDefinition) }], // Protocol or definition
  "basedOn" : [{ Reference(CarePlan|MedicationRequest|ProcedureRequest|
   ReferralRequest) }], // What request fulfills
  "groupIdentifier" : { Identifier }, // Composite request this is part of
  "status" : "<code>", // active | on-hold | cancelled | completed | entered-in-error | stopped | draft | unknown
  "intent" : "<code>", // R!  proposal | plan | order | instance-order
  "category" : { CodeableConcept }, // Type of medication usage
  "priority" : "<code>", // routine | urgent | stat | asap
  // medication[x]: Medication to be taken. One of these 2:
  "medicationCodeableConcept" : { CodeableConcept },
  "medicationReference" : { Reference(Medication) },
  "subject" : { Reference(Patient|Group) }, // R!  Who or group medication request is for
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Created during encounter/admission/stay
  "supportingInformation" : [{ Reference(Any) }], // Information to support ordering of the medication
  "authoredOn" : "<dateTime>", // When request was initially authored
  "requester" : { // Who/What requested the Request
    "agent" : { Reference(Practitioner|Organization|Patient|RelatedPerson|
    Device) }, // R!  Who ordered the initial medication(s)
    "onBehalfOf" : { Reference(Organization) } // C? Organization agent is acting for
  },
  "recorder" : { Reference(Practitioner) }, // Person who entered the request
  "reasonCode" : [{ CodeableConcept }], // Reason or indication for writing the prescription
  "reasonReference" : [{ Reference(Condition|Observation) }], // Condition or Observation that supports why the
    prescription is being written
  "note" : [{ Annotation }], // Information about the prescription
  "dosageInstruction" : [{ Dosage }], // How the medication should be taken
  "dispenseRequest" : { // Medication supply authorization
    "validityPeriod" : { Period }, // Time period supply is authorized for
    "numberOfRepeatsAllowed" : "<positiveInt>", // Number of refills authorized
    "quantity" : { Quantity(SimpleQuantity) }, // Amount of medication to supply per dispense
    "expectedSupplyDuration" : { Duration }, // Number of days supply per dispense
    "performer" : { Reference(Organization) } // Intended dispenser
  },
  "substitution" : { // Any restrictions on medication substitution
    "allowed" : <boolean>, // R!  Whether substitution is allowed or not
    "reason" : { CodeableConcept } // Why should (not) substitution be made
  },
  "priorPrescription" : { Reference(MedicationRequest) }, // An order/prescription that is being replaced
  "detectedIssue" : [{ Reference(DetectedIssue) }], // Clinical Issue with action
  "eventHistory" : [{ Reference(Provenance) }] // A list of events of interest in the lifecycle
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicationRequest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicationRequest.identifier [ Identifier ], ... ; # 0..* External ids for this request
  fhir:MedicationRequest.definition [ Reference(ActivityDefinition|PlanDefinition) ], ... ; # 0..* Protocol or
    definition
  fhir:MedicationRequest.basedOn [ Reference(CarePlan|MedicationRequest|ProcedureRequest|ReferralRequest) ], ... ; #
```

```
    0..* What request fulfills
  fhir:MedicationRequest.groupIdentifier [ Identifier ]; # 0..1 Composite request this is part of
  fhir:MedicationRequest.status [ code ]; # 0..1 active | on-hold | cancelled | completed | entered-in-error | stopped |
     draft | unknown
  fhir:MedicationRequest.intent [ code ]; # 1..1 proposal | plan | order | instance-order
  fhir:MedicationRequest.category [ CodeableConcept ]; # 0..1 Type of medication usage
  fhir:MedicationRequest.priority [ code ]; # 0..1 routine | urgent | stat | asap
  # MedicationRequest.medication[x] : 1..1 Medication to be taken. One of these 2
    fhir:MedicationRequest.medicationCodeableConcept [ CodeableConcept ]
    fhir:MedicationRequest.medicationReference [ Reference(Medication) ]
  fhir:MedicationRequest.subject [ Reference(Patient|Group) ]; # 1..1 Who or group medication request is for
  fhir:MedicationRequest.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Created during encounter/admission/stay
  fhir:MedicationRequest.supportingInformation [ Reference(Any) ], ... ; # 0..* Information to support ordering of the
    medication
  fhir:MedicationRequest.authoredOn [ dateTime ]; # 0..1 When request was initially authored
  fhir:MedicationRequest.requester [ # 0..1 Who/What requested the Request
    fhir:MedicationRequest.requester.agent [ Reference(Practitioner|Organization|Patient|RelatedPerson|Device) ]; # 1..1
     Who ordered the initial medication(s)
    fhir:MedicationRequest.requester.onBehalfOf [ Reference(Organization) ]; # 0..1 Organization agent is acting for
  ];
  fhir:MedicationRequest.recorder [ Reference(Practitioner) ]; # 0..1 Person who entered the request
  fhir:MedicationRequest.reasonCode [ CodeableConcept ], ... ; # 0..* Reason or indication for writing the prescription
  fhir:MedicationRequest.reasonReference [ Reference(Condition|Observation) ], ... ; # 0..* Condition or Observation
    that supports why the prescription is being written
  fhir:MedicationRequest.note [ Annotation ], ... ; # 0..* Information about the prescription
  fhir:MedicationRequest.dosageInstruction [ Dosage ], ... ; # 0..* How the medication should be taken
  fhir:MedicationRequest.dispenseRequest [ # 0..1 Medication supply authorization
    fhir:MedicationRequest.dispenseRequest.validityPeriod [ Period ]; # 0..1 Time period supply is authorized for
    fhir:MedicationRequest.dispenseRequest.numberOfRepeatsAllowed [ positiveInt ]; # 0..1 Number of refills authorized
    fhir:MedicationRequest.dispenseRequest.quantity [ Quantity(SimpleQuantity) ]; # 0..1 Amount of medication to supply
    per dispense
    fhir:MedicationRequest.dispenseRequest.expectedSupplyDuration [ Duration ]; # 0..1 Number of days supply per
    dispense
    fhir:MedicationRequest.dispenseRequest.performer [ Reference(Organization) ]; # 0..1 Intended dispenser
  ];
  fhir:MedicationRequest.substitution [ # 0..1 Any restrictions on medication substitution
    fhir:MedicationRequest.substitution.allowed [ boolean ]; # 1..1 Whether substitution is allowed or not
    fhir:MedicationRequest.substitution.reason [ CodeableConcept ]; # 0..1 Why should (not) substitution be made
  ];
  fhir:MedicationRequest.priorPrescription [ Reference(MedicationRequest) ]; # 0..1 An order/prescription that is being
    replaced
  fhir:MedicationRequest.detectedIssue [ Reference(DetectedIssue) ], ... ; # 0..* Clinical Issue with action
  fhir:MedicationRequest.eventHistory [ Reference(Provenance) ], ... ; # 0..* A list of events of interest in the
    lifecycle
]
```

Changes since DSTU2

MedicationRequest | Name Changed from MedicationOrder to MedicationRequest

MedicationRequest.definition | Added Element

MedicationRequest.basedOn | Added Element

MedicationRequest.groupIdentifier | Added Element

MedicationRequest.status | Change value set from http://hl7.org/fhir/ValueSet/medication-order-status to http://hl7.org/fhir/ValueSet/medication-request-status

MedicationRequest.intent | Added Element

MedicationRequest.category | Added Element

MedicationRequest.priority | Added Element

MedicationRequest.subject | Added Element

MedicationRequest.context | Added Element

MedicationRequest.supportingInformation | Added Element

MedicationRequest.authoredOn | Added Element

MedicationRequest.requester | Added Element

MedicationRequest.requester.agent | Added Element

MedicationRequest.requester.onBehalfOf | Added Element

MedicationRequest.recorder | Added Element

MedicationRequest.reasonCode | Added Element

MedicationRequest.reasonReference | Added Element

MedicationRequest.note | Max Cardinality changed from 1 to * Type changed from string to Annotation

MedicationRequest.dosageInstruction | Type changed from BackboneElement to Dosage

MedicationRequest.dispenseRequest.expectedSupplyDuration | Type changed from Quantity{http://hl7.org/fhir/StructureDefinition/Duration} to Duration

MedicationRequest.dispenseRequest.performer | Added Element

MedicationRequest.substitution.allowed | Added Element

MedicationRequest.priorPrescription | Type changed from Reference(MedicationOrder) to Reference(MedicationRequest)

MedicationRequest.detectedIssue | Added Element

MedicationRequest.eventHistory | Added Element

MedicationOrder.dateWritten | deleted

MedicationOrder.dateEnded | deleted

MedicationOrder.reasonEnded | deleted

MedicationOrder.patient | deleted

MedicationOrder.prescriber | deleted

MedicationOrder.encounter | deleted

MedicationOrder.reason[x] | deleted
MedicationOrder.dosageInstruction.text | deleted
MedicationOrder.dosageInstruction.additionalInstructions | deleted
MedicationOrder.dosageInstruction.timing | deleted
MedicationOrder.dosageInstruction.asNeeded[x] | deleted
MedicationOrder.dosageInstruction.site[x] | deleted
MedicationOrder.dosageInstruction.route | deleted
MedicationOrder.dosageInstruction.method | deleted
MedicationOrder.dosageInstruction.dose[x] | deleted
MedicationOrder.dosageInstruction.rate[x] | deleted
MedicationOrder.dosageInstruction.maxDosePerPeriod | deleted
MedicationOrder.dispenseRequest.medication[x] | deleted
MedicationOrder.substitution.type | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

11.1.3.1 Terminology Bindings

Path | Definition | Type | Reference
MedicationRequest.status | A coded concept specifying the state of the prescribing event. Describes the lifecycle of the prescription | Required | MedicationRequestStatus
MedicationRequest.intent | The kind of medication order | Required | MedicationRequestIntent
MedicationRequest.category | A coded concept identifying where the medication ordered is expected to be consumed or administered | Preferred | MedicationRequestCategory
MedicationRequest.priority | Identifies the level of importance to be assigned to actioning the request | Required | MedicationRequestPriority
MedicationRequest.medication[x] | A coded concept identifying substance or product that can be ordered. | Example | SNOMED CT Medication Codes
MedicationRequest.reasonCode | A coded concept indicating why the medication was ordered. | Example | Condition/Problem/Diagnosis Codes
MedicationRequest.substitution.reason | A coded concept describing the reason that a different medication should (or should not) be substituted from what was prescribed. | Example | SubstanceAdminSubstitutionReason

11.1.3.2 Constraints

* mps-1: On MedicationRequest.requester: onBehalfOf can only be specified if agent is practitioner or device (expression on MedicationRequest.requester: (agent.resolve().empty()) or (agent.resolve() is Device) or (agent.resolve() is Practitioner) or onBehalfOf.exists().not())

11.1.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
authoredon | date | Return prescriptions written on this date | MedicationRequest.authoredOn
category | token | Returns prescriptions with different categories | MedicationRequest.category
code | token | Return prescriptions of this medication code | MedicationRequest.medication.as(CodeableConcept) | 4 Resources
context | reference | Return prescriptions with this encounter or episode of care identifier | MedicationRequest.context (EpisodeOfCare, Encounter)
date | date | Returns medication request to be administered on a specific date | MedicationRequest.dosageInstruction.timing.event | 3 Resources
identifier | token | Return prescriptions with this external identifier | MedicationRequest.identifier | 3 Resources
intended-dispenser | reference | Returns prescriptions intended to be dispensed by this Organization | MedicationRequest.dispenseRequest.performer (Organization)
intent | token | Returns prescriptions with different intents | MedicationRequest.intent
medication | reference | Return prescriptions of this medication reference | MedicationRequest.medication.as(Reference) (Medication) | 3 Resources
patient | reference | Returns prescriptions for a specific patient | MedicationRequest.subject (Patient) | 3 Resources
priority | token | Returns prescriptions with different priorities | MedicationRequest.priority
requester | reference | Returns prescriptions prescribed by this prescriber | MedicationRequest.requester.agent (Practitioner, Organization, Device, Patient, RelatedPerson)
status | token | Status of the prescription | MedicationRequest.status | 3 Resources
subject | reference | The identity of a patient to list orders for | MedicationRequest.subject (Group, Patient)

11.1.5 Resource MedicationRequest - Examples

URL: https://hl7.org/fhir/STU3/medicationrequest-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.1.5 Resource MedicationRequest - Examples

Pharmacy Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Encounter, Patient, Practitioner

ORAL DOSAGE FORMS

The following examples are MedicationRequest resources for Oral Dosage Forms that reference Medication Resources for generic formulations:

* Active Order with link to encounter, reasonCode, note multiple dosage lines, dispenseRequest and substitution - Azithromycin

* Order with as needed (PRN) dosage with pre-condition - Oxycodone - active with reasonCode, note, dispenseRequest and substitution

* Order for Tapering Dose - Prednisone - active with note and multiple DosageInstructions with boundsPeriod

* Order for Oral Drops - Nystatin Suspension - completed - with Dispense Request

* Order with Dosage with a Pre-Condition - Alprazolam - dosageInstructions includes boundsPeriod with periodMax and asNeededCodeableConcept

* Order for a Dosage that needs to be dispensed as different strength tablets - Warfarin - active

The following examples are MedicationRequest resources for Oral Dosage Forms that reference Medication Resources for manufactured products:

* Oral Chemotherapy Order - Myleran - completed - with reasonCode

* Order with as needed (PRN) dosage - Percocet - completed with link to encounter, dispenseRequest and substitution

* Order for as needed (PRN) dosage - Vicodin - completed with dosageInstruction with additionalInstruction (coded) and as needed with reason, dispenseRequest and substitution

* Oral Chemotherapy Order - Capecitabine - active medication with dosing by Body Surface Area (BSA) (mg/m2)

* Order for Over the Counter Medication - Tylenol PM - with dosage range as needed with pre-condition for use

* Order for a medication that includes the dosage of a prescription in text

* Fully populated example of a MedicationRequest - Chlorthalidone - active - with link to encounter, reasonReference, note, dispenseRequest, substitution and eventHistory

The following examples are MedicationRequest resources for Oral Dosage Forms that include a Medication Code

* Liquid Medication - Phenytoin with link to prior prescription

* Order with a single dosage line - Azithromycin - completed - with link to prior prescription and reason code

* Order - Levothyroxine - completed with reasonCode, dosageInstructions and dispenseRequest

INJECTABLE DOSAGE FORMS

The following examples of MedicationRequest resources for Injectable Dosage Forms that reference a Medication Resource for a generic formulation

* Order for an injectable medication - Lorazepam - active - with link to encounter, reasonCode and dosageInstruction

The following examples of MedicationRequest resources for Injectable Dosage Forms that reference a Medication Resource for a manufactured product:

* Order for Intravenous Chemotherapy (inpatient order) - Brentuximab Vedotin - completed

* Order for Intravenous Chemotherapy (inpatient order) - Alemtuzumab - completed with multiple dosageInstruction lines

* Order for Intravenous Antibiotic - Vancomycin - active - with link to encounter, reasonCode, and note

* Order for Intravenous Antibiotic with a Dosage Rate - Piperacillin - completed with reason code

* Order for Subcutaneous injection with multiple dosage lines - Insulin - Novolog - active - includes reasonCode and note. Dosage includes additionalInstructions as text

The following examples of MedicationRequest resources for Injectable Dosage Forms that include a Medication Code:

* Order for Subcutaneous injection - Insulin - Lantus - completed, with reason code and dispense request

The following examples of MedicationRequest resources for Injectable Dosage Forms for Injectable Dosage Forms where the medication is a compounded (aka extemporaneous or magistral) product:

* Order for Intravenous Admixture - Potassium Chloride in D5W - completed with reason code

* Order for TPN Solution - completed

MISCELLAENOUS DOSAGE FORMS

The following examples of MedicationRequest resources including miscellaneous dosage forms:

* Order using Medication Code for Rectal Suppository - Acetaminophen - completed with reasonCode, dispenseRequest and substitution

* Order using Medication Code for Topical Ointment - Nystatin - on hold and with dispense request

* Order using Medication Code for Inhaler - Proventil - on hold with dispenseRequest and substitution

* Order using Medication Code for Patch - Fentanyl - active, with dispense request

* Medication Resource Reference - Nasal Spray - Nasonex

* Order referencing Medication Resource for Compounded Product - HC 1%, Salicylic Acid 5% in Glaxal Base - on hold with dispenseRequest

* Order referencing Medication Resource for Eye Drops - Timoptic - active - with link to encounter, dispenseRequest and substitution flag

MISCELLAENOUS DOSAGE INSTRUCTIONS

The following examples of MedicationRequest resources including miscellaneous dosage instructions:

* Order that includes a dosage instruction for a medication that is intended to be given once only as single dose

* Order for a medication that is intended to be given in the morning

* Order that includes a dosage instruction for a medication that is intended to be given once only as single dose

* Order that includes a dosage instruction for a medication that is intended to be given once only as single dose

Example List:

Order for a medication that includes the dosage of a prescription in text | medrx002 | XML | JSON | Turtle

Active Order with link to encounter, reasonCode, note multiple dosage lines, dispenseRequest and substitution - Azithromycin | medrx0302 | XML | JSON | Turtle

Order with as needed (PRN) dosage with pre-condition - Oxycodone - active with reasonCode, note, dispenseRequest and substitution | medrx0301 | XML | JSON | Turtle

Order for Tapering Dose - Prednisone - active with note and multiple DosageInstructions with boundsPeriod | medrx0303 | XML | JSON | Turtle

Order for Oral Drops - Nystatin Suspension - completed - with Dispense Request | medrx0304 | XML | JSON | Turtle

Order with Dosage with a Pre-Condition - Alprazolam - dosageInstructions includes boundsPeriod with periodMax and asNeededCodeableConcept | medrx0305 | XML | JSON | Turtle

Oral Chemotherapy Order - Myleran - completed - with reasonCode | medrx0306 | XML | JSON | Turtle

Order with as needed (PRN) dosage - Percocet - completed with link to encounter, dispenseRequest and substitution | medrx0307 | XML | JSON | Turtle

Order for as needed (PRN) dosage - Vicodin - completed with dosageInstruction with additionalInstruction (coded) and as needed with reason, dispenseRequest and substitution | medrx0308 | XML | JSON | Turtle

Oral Chemotherapy Order - Capecitabine - active medication with dosing by Body Surface Area (BSA) (mg/m2) | medrx0309 | XML | JSON | Turtle

Order for Over the Counter Medication - Tylenol PM - with dosage range as needed with pre-condition for use | medrx0310 | XML | JSON | Turtle

Fully populated example of a MedicationRequest - Chlorthalidone - active - with link to encounter, reasonReference, note, dispenseRequest, substitution and eventHistory | medrx0311 | XML | JSON | Turtle

Liquid Medication - Phenytoin with link to prior prescription | medrx0312 | XML | JSON | Turtle

Order with a single dosage line - Azithromycin - completed - with link to prior prescription and reason code | medrx0313 | XML | JSON | Turtle

Order - Levothyroxine - completed with reasonCode, dosageInstructions and dispenseRequest | medrx0314 | XML | JSON | Turtle

Order for an injectable medication - Lorazepam - active - with link to encounter, reasonCode and dosageInstruction | medrx0315 | XML | JSON | Turtle

Order for Intravenous Chemotherapy (inpatient order) - Brentuximab Vedotin - completed | medrx0316 | XML | JSON | Turtle

Order for Intravenous Chemotherapy (inpatient order) - Alemtuzumab - completed with multiple dosageInstruction lines | medrx0317 | XML | JSON | Turtle

Order for Intravenous Antibiotic - Vancomycin - active - with link to encounter, reasonCode, and note | medrx0318 | XML | JSON | Turtle

Order for Intravenous Antibiotic with a Dosage Rate - Piperacillin - completed with reason code | medrx0319 | XML | JSON | Turtle

Order for Subcutaneous injection - Insulin - Lantus - completed, with reason code and dispense request | medrx0320 | XML | JSON | Turtle

Order for Subcutaneous injection with multiple dosage lines - Insulin - Novolog - active - includes reasonCode and note. Dosage includes additionalInstructions as text | medrx0321 | XML | JSON | Turtle

Order for Intravenous Admixture - Potassium Chloride in D5W - completed with reason code | medrx0322 | XML | JSON | Turtle

Order for TPN Solution - completed | medrx0323 | XML | JSON | Turtle

Order using Medication Code for Rectal Suppository - Acetaminophen - completed with reasonCode, dispenseRequest and substitution | medrx0324 | XML | JSON | Turtle

Order using Medication Code for Topical Ointment - Nystatin - on hold and with dispense request | medrx0325 | XML | JSON | Turtle

Order using Medication Code for Inhaler - Proventil - on hold with dispenseRequest and substitution | medrx0326 | XML | JSON | Turtle

Order using Medication Code for Patch - Fentanyl - active, with dispense request | medrx0327 | XML | JSON | Turtle

Order referencing Medication Resource for Nasal Spray - Nasonex | medrx0328 | XML | JSON | Turtle

Order referencing Medication Resource for Compounded Product - HC 1%, Salicylic Acid 5% in Glaxal Base - on hold with dispenseRequest | medrx0329 | XML | JSON | Turtle

Order referencing Medication Resource for Eye Drops - Timoptic - active - with link to encounter, dispenseRequest and substitution flag | medrx0330 | XML | JSON | Turtle

Order for a Dosage that needs to be dispensed as different strength tablets - Warfarin - active | medrx0331 | XML | JSON | Turtle

Order for a medication that is intended to be given as a single, once only dose | medrx0332 | XML | JSON | Turtle

Order for a medication that is intended to be given in the morning | medrx0333 | XML | JSON | Turtle

Order for a medication that is intended to be given on a specific day | medrx0334 | XML | JSON | Turtle

Order for a medication that is intended to be given at a specific time | medrx0335 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.1.6 Resource MedicationRequest - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/medicationrequest-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.1.6 Resource MedicationRequest - Detailed Descriptions

Pharmacy Work Group | Maturity Level: 3 | Trial Use | Compartments: Encounter, Patient, Practitioner

Detailed Descriptions for the elements in the MedicationRequest resource.

MedicationRequest

Definition | An order or request for both supply of the medication and the instructions for administration of the medication to a patient. The resource is called "MedicationRequest" rather than "MedicationPrescription" or "MedicationOrder" to generalize the use across inpatient and outpatient settings,

including care plans, etc., and to harmonize with workflow patterns.
Control | 1..1
Alternate Names | Prescription; Order
MedicationRequest.identifier
Definition | This records identifiers associated with this medication request that are defined by business processes and/or used to refer to it when a direct URL reference to the resource itself is not appropriate. For example a re-imbursement system might issue its own id for each prescription that is created. This is particularly important where FHIR only provides part of an entire workflow process where records must be tracked through an entire system.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
MedicationRequest.definition
Definition | Protocol or definition followed by this request.
Control | 0..*
Type | Reference(ActivityDefinition | PlanDefinition)
Summary | true
MedicationRequest.basedOn
Definition | A plan or request that is fulfilled in whole or in part by this medication request.
Control | 0..*
Type | Reference(CarePlan | MedicationRequest | ProcedureRequest | ReferralRequest)
Summary | true
MedicationRequest.groupIdentifier
Definition | A shared identifier common to all requests that were authorized more or less simultaneously by a single author, representing the identifier of the requisition or prescription.
Control | 0..1
Type | Identifier
Requirements | Requests are linked either by a "basedOn" relationship (i.e. one request is fulfilling another) or by having a common requisition. Requests that are part of the same requisition are generally treated independently from the perspective of changing their state or maintaining them after initial creation.
Summary | true
MedicationRequest.status
Definition | A code specifying the current state of the order. Generally this will be active or completed state.
Control | 0..1
Terminology Binding | MedicationRequestStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
MedicationRequest.intent
Definition | Whether the request is a proposal, plan, or an original order.
Control | 1..1
Terminology Binding | MedicationRequestIntent (Required)
Type | code
Is Modifier | true
Summary | true
Comments | It is expected that the type of requester will be restricted for different stages of a MedicationRequest. For example, Proposals can be created by a patient, relatedPerson, Practitioner or Device. Plans can be created by Practitioners, Patients, RelatedPersons and Devices. Original orders can be created by a Practitioner only. An instance-order is an instantiation of a request or order and may be used to populate Medication Administration Record. This element is labeled as a modifier because the intent alters when and how the resource is actually applicable.
MedicationRequest.category
Definition | Indicates the type of medication order and where the medication is expected to be consumed or administered.
Control | 0..1
Terminology Binding | MedicationRequestCategory (Preferred)
Type | CodeableConcept
MedicationRequest.priority
Definition | Indicates how quickly the Medication Request should be addressed with respect to other requests.
Control | 0..1
Terminology Binding | MedicationRequestPriority (Required)
Type | code
Summary | true
MedicationRequest.medication[x]
Definition | Identifies the medication being requested. This is a link to a resource that represents the medication which may be the details of the medication or simply an attribute carrying a code that identifies the medication from a known list of medications.
Control | 1..1
Terminology Binding | SNOMED CT Medication Codes (Example)
Type | CodeableConcept|Reference(Medication)
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | If only a code is specified, then it needs to be a code for a specific product. If more information is required, then the use of the medication resource is recommended. For example, if you require form or lot number or if the medication is compounded or extemporaneously prepared, then you must reference the Medication resource. .
MedicationRequest.subject
Definition | A link to a resource representing the person or set of individuals to whom the medication will be given.

Control | 1..1
Type | Reference(Patient | Group)
Summary | true
Comments | The subject on a medication request is mandatory. For the secondary use case where the actual subject is not provided, there still must be an anonymized subject specified.
MedicationRequest.context
Definition | A link to an encounter, or episode of care, that identifies the particular occurrence or set occurrences of contact between patient and health care provider.
Control | 0..1
Type | Reference(Encounter | EpisodeOfCare)
Comments | SubstanceAdministration->component->EncounterEvent.
MedicationRequest.supportingInformation
Definition | Include additional information (for example, patient height and weight) that supports the ordering of the medication.
Control | 0..*
Type | Reference(Any)
MedicationRequest.authoredOn
Definition | The date (and perhaps time) when the prescription was initially written or authored on.
Control | 0..1
Type | dateTime
Summary | true
MedicationRequest.requester
Definition | The individual, organization or device that initiated the request and has responsibility for its activation.
Control | 0..1
Summary | true
Invariants | Defined on this element mps-1: onBehalfOf can only be specified if agent is practitioner or device (expression : (agent.resolve().empty()) or (agent.resolve() is Device) or (agent.resolve() is Practitioner) or onBehalfOf.exists().not(), xpath: contains(f:agent/f:reference/@value, '/Practitioner/') or contains(f:agent/f:reference/@value, '/Device/') or not(exists(f:onBehalfOf)))
MedicationRequest.requester.agent
Definition | The healthcare professional responsible for authorizing the initial prescription.
Control | 1..1
Type | Reference(Practitioner | Organization | Patient | RelatedPerson | Device)
Summary | true
Comments | It is expected that the type of requester will be restricted for different stages of a MedicationRequest. For example, Proposals can be created by a patient, relatedPerson, Practitioner or Device. Plans can be created by Practitioners, Patients, RelatedPersons and Devices. Original orders can be created by a Practitioner only.
MedicationRequest.requester.onBehalfOf
Definition | The organization the device or practitioner was acting on behalf of.
Control | 0..1
Type | Reference(Organization)
Requirements | Practitioners and Devices can be associated with multiple organizations. This element indicates which organization they were acting on behalf of when authoring the request.
Summary | true
Invariants | Affect this element mps-1: onBehalfOf can only be specified if agent is practitioner or device (expression : (agent.resolve().empty()) or (agent.resolve() is Device) or (agent.resolve() is Practitioner) or onBehalfOf.exists().not(), xpath: contains(f:agent/f:reference/@value, '/Practitioner/') or contains(f:agent/f:reference/@value, '/Device/') or not(exists(f:onBehalfOf)))
MedicationRequest.recorder
Definition | The person who entered the order on behalf of another individual for example in the case of a verbal or a telephone order.
Control | 0..1
Type | Reference(Practitioner)
MedicationRequest.reasonCode
Definition | The reason or the indication for ordering the medication.
Control | 0..*
Terminology Binding | Condition/Problem/Diagnosis Codes (Example)
Type | CodeableConcept
Comments | This could be a diagnosis code. If a full condition record exists or additional detail is needed, use reasonReference.
MedicationRequest.reasonReference
Definition | Condition or observation that supports why the medication was ordered.
Control | 0..*
Type | Reference(Condition | Observation)
Comments | This is a reference to a condition or observation that is the reason for the medication order. If only a code exists, use reasonCode.
MedicationRequest.note
Definition | Extra information about the prescription that could not be conveyed by the other attributes.
Control | 0..*
Type | Annotation
MedicationRequest.dosageInstruction
Definition | Indicates how the medication is to be used by the patient.
Control | 0..*
Type | Dosage
Comments | There are examples where a medication request may include the option of an oral dose or an Intravenous or Intramuscular dose. For example, "Ondansetron 8mg orally or IV twice a day as needed for nausea" or "Compazine® (prochlorperazine) 5-10mg PO or 25mg PR bid prn nausea or vomiting". In these cases, two medication requests would be created that could be grouped together. The decision on which dose and route of administration to use is based on the patient's condition at the time the dose is needed.
MedicationRequest.dispenseRequest

Definition | Indicates the specific details for the dispense or medication supply part of a medication request (also known as a Medication Prescription or Medication Order). Note that this information is not always sent with the order. There may be in some settings (e.g. hospitals) institutional or system support for completing the dispense details in the pharmacy department.
Control | 0..1
MedicationRequest.dispenseRequest.validityPeriod
Definition | This indicates the validity period of a prescription (stale dating the Prescription).
Control | 0..1
Type | Period
Requirements | Indicates when the Prescription becomes valid, and when it ceases to be a dispensable Prescription.
Comments | It reflects the prescribers' perspective for the validity of the prescription. Dispenses must not be made against the prescription outside of this period. The lower-bound of the Dispensing Window signifies the earliest date that the prescription can be filled for the first time. If an upper-bound is not specified then the Prescription is open-ended or will default to a stale-date based on regulations.
MedicationRequest.dispenseRequest.numberOfRepeatsAllowed
Definition | An integer indicating the number of times, in addition to the original dispense, (aka refills or repeats) that the patient can receive the prescribed medication. Usage Notes: This integer does not include the original order dispense. This means that if an order indicates dispense 30 tablets plus "3 repeats", then the order can be dispensed a total of 4 times and the patient can receive a total of 120 tablets.
Control | 0..1
Type | positiveInt
Comments | If displaying "number of authorized fills", add 1 to this number.
MedicationRequest.dispenseRequest.quantity
Definition | The amount that is to be dispensed for one fill.
Control | 0..1
Type | SimpleQuantity
MedicationRequest.dispenseRequest.expectedSupplyDuration
Definition | Identifies the period time over which the supplied product is expected to be used, or the length of time the dispense is expected to last.
Control | 0..1
Type | Duration
Comments | In some situations, this attribute may be used instead of quantity to identify the amount supplied by how long it is expected to last, rather than the physical quantity issued, e.g. 90 days supply of medication (based on an ordered dosage) When possible, it is always better to specify quantity, as this tends to be more precise. expectedSupplyDuration will always be an estimate that can be influenced by external factors.
MedicationRequest.dispenseRequest.performer
Definition | Indicates the intended dispensing Organization specified by the prescriber.
Control | 0..1
Type | Reference(Organization)
MedicationRequest.substitution
Definition | Indicates whether or not substitution can or should be part of the dispense. In some cases substitution must happen, in other cases substitution must not happen. This block explains the prescriber's intent. If nothing is specified substitution may be done.
Control | 0..1
MedicationRequest.substitution.allowed
Definition | True if the prescriber allows a different drug to be dispensed from what was prescribed.
Control | 1..1
Type | boolean
Is Modifier | true
Comments | This element is labeled as a modifer because whether substitution is allow or not cannot be ignored.
MedicationRequest.substitution.reason
Definition | Indicates the reason for the substitution, or why substitution must or must not be performed.
Control | 0..1
Terminology Binding | SubstanceAdminSubstitutionReason (Example)
Type | CodeableConcept
MedicationRequest.priorPrescription
Definition | A link to a resource representing an earlier order related order or prescription.
Control | 0..1
Type | Reference(MedicationRequest)
MedicationRequest.detectedIssue
Definition | Indicates an actual or potential clinical issue with or between one or more active or proposed clinical actions for a patient; e.g. Drug-drug interaction, duplicate therapy, dosage alert etc.
Control | 0..*
Type | Reference(DetectedIssue)
Alternate Names | Contraindication; Drug Utilization Review (DUR); Alert
MedicationRequest.eventHistory
Definition | Links to Provenance records for past versions of this resource or fulfilling request or event resources that identify key state transitions or updates that are likely to be relevant to a user looking at the current version of the resource.
Control | 0..*
Type | Reference(Provenance)
Comments | This may not include provenances for all versions of the request – only those deemed “relevant” or important. This SHALL NOT include the Provenance associated with this current version of the resource. (If that provenance is deemed to be a “relevant” change, it will need to be added as part of a later update. Until then, it can be queried directly as the Provenance that points to this version using _revinclude All Provenances should have some historical version of this Request as their subject.).

11.1.7 Resource MedicationRequest - Mappings

URL: https://hl7.org/fhir/STU3/medicationrequest-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.1.7 Resource MedicationRequest - Mappings

Pharmacy Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Encounter, Patient, Practitioner

Mappings:

RIM Mapping

HL7 v2 Mapping

Mapping to NCPDP SCRIPT 10.6

Workflow Mapping

Mappings for the medicationrequest resource.

11.1.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

MedicationRequest | CombinedMedicationRequest
identifier | .id
definition | .outboundRelationship[typeCode=DEFN].target[classCode=unspecified]
basedOn | .outboundRelationship[typeCode=FLFS].target[classCode=SBADM or PROC or PCPR or OBS, moodCode=RQO orPLAN or PRP]
groupIdentifier | .outboundRelationship(typeCode=COMP].target[classCode=SBADM, moodCode=INT].id
status | .statusCode
intent | .moodCode (nuances beyond PRP/PLAN/RQO would need to be elsewhere)
category | .inboundRelationship[typeCode=COMP].source[classCode=OBS, moodCode=EVN, code="type of medication usage"].value
priority | .priorityCode
medication[x] | consumable.administrableMedication
subject | .participation[typeCode=AUT].role
context | .inboundRelationship[typeCode=COMP].source[classCode=ENC, moodCode=EVN, code="type of encounter or episode"]
supportingInformation | .outboundRelationship[typeCode=PERT].target[A_SupportingClinicalStatement CMET minimal with many different choices of classCodes(ORG, ENC, PROC, SPLY, SBADM, OBS) and each of the act class codes draws from one or more of the following moodCodes (EVN, DEF, INT PRMS, RQO, PRP, APT, ARQ, GOL)]
authoredOn | author.time
requester | .participation[typeCode=AUT].role
agent | .player
onBehalfOf | .scoper
recorder | .participation[typeCode=TRANS].role[classCode=ASSIGNED].code (HealthcareProviderType)
reasonCode | reason.observation.reasonCode
reasonReference | reason.observation[code=ASSERTION].value
note | .inboundRelationship[typeCode=SUBJ]/source[classCode=OBS,moodCode=EVN,code="annotation"].value
dosageInstruction | see dosageInstruction mapping
dispenseRequest | component.supplyEvent
validityPeriod | effectiveTime
numberOfRepeatsAllowed | repeatNumber
quantity | quantity
expectedSupplyDuration | expectedUseTime
performer | .outboundRelationship[typeCode=COMP].target[classCode=SPLY, moodCode=RQO] .participation[typeCode=PRF].role[scoper.determinerCode=INSTANCE]
substitution | subjectOf.substitutionPersmission
allowed | code
reason | reasonCode
priorPrescription | .outboundRelationship[typeCode=?RPLC or ?SUCC]/target[classCode=SBADM,moodCode=RQO]
detectedIssue | .inboundRelationship[typeCode=SUBJ]/source[classCode=ALRT,moodCode=EVN].value
eventHistory | .inboundRelationship(typeCode=SUBJ].source[classCode=CACT, moodCode=EVN]

11.1.7.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

MedicationRequest
identifier | ORC-2-Placer Order Number / ORC-3-Filler Order Number
definition
basedOn
groupIdentifier
status
intent
category
priority
medication[x] | RXE-2-Give Code / RXO-1-Requested Give Code / RXC-2-Component Code
subject | PID-3-Patient ID List
context | PV1-19-Visit Number
supportingInformation
authoredOn | RXE-32-Original Order Date/Time / ORC-9-Date/Time of Transaction
requester
agent | RXE-13-Ordering Provider's DEA Number / RXO-14-Ordering Provider's DEA Number / RXE-14-Pharmacist/Treatment Supplier's Verifier ID / RXO-15-Pharmacist/Treatment Supplier's Verifier ID / ORC-12-Ordering Provider / PRT-5-Participation Person: PRT-4-Participation='OP' (all but last

deprecated)
onBehalfOf
recorder
reasonCode | ORC-16-Order Control Code Reason /RXE-27-Give Indication/RXO-20-Indication / RXD-21-Indication / RXG-22-Indication / RXA-19-Indication
reasonReference
note
dosageInstruction
dispenseRequest
validityPeriod
numberOfRepeatsAllowed | RXE-12-Number of Refills
quantity | RXD-4-Actual Dispense Amount / RXD-5.1-Actual Dispense Units.code / RXD-5.3-Actual Dispense Units.name of coding system
expectedSupplyDuration
performer
substitution
allowed | RXO-9-Allow Substitutions / RXE-9-Substitution Status
reason | RXE-9 Substition status
priorPrescription
detectedIssue
eventHistory

11.1.7.3 Mappings for Mapping to NCPDP SCRIPT 10.6 (http://ncpdp.org/SCRIPT10_6)

MedicationRequest | Message/Body/NewRx
identifier | Message/Header/PrescriberOrderNumber
definition
basedOn
groupIdentifier
status | no mapping
intent
category | Message/Body/NewRx/MedicationPrescribed/Directions or Message/Body/NewRx/MedicationPrescribed/StructuredSIG
priority
medication[x] | Message/Body/NewRx/MedicationPrescribed Medication.code.coding.code = Message/Body/NewRx/MedicationPrescribed/DrugCoded/ProductCode Medication.code.coding.system = Message/Body/NewRx/MedicationPrescribed/DrugCoded/ProductCodeQualifier Medication.code.coding.display = Message/Body/NewRx/MedicationPrescribed/DrugDescription
subject | Message/Body/NewRx/Patient (need detail to link to specific patient … Patient.Identification in SCRIPT)
context | no mapping
supportingInformation
authoredOn | Message/Body/NewRx/MedicationPrescribed/WrittenDate
requester
agent | Message/Body/NewRx/Prescriber (need detail to link to specific prescriber … Prescriber.Identification in SCRIPT)
onBehalfOf
recorder
reasonCode | Message/Body/NewRx/MedicationPrescribed/Diagnosis/Primary/Value
reasonReference | no mapping
note | Message/Body/NewRx/MedicationPrescribed/Note
dosageInstruction
dispenseRequest | Message/Body/NewRx/MedicationPrescribed/ExpirationDate
validityPeriod | Message/Body/NewRx/MedicationPrescribed/Refills
numberOfRepeatsAllowed | Message/Body/NewRx/MedicationPrescribed/Quantity
quantity | Message/Body/NewRx/MedicationPrescribed/DaysSupply
expectedSupplyDuration | Message/Body/NewRx/MedicationPrescribed/Substitutions
performer
substitution | specific values within Message/Body/NewRx/MedicationPrescribed/Substitutions
allowed | specific values within Message/Body/NewRx/MedicationPrescribed/Substitutions
reason | not mapped
priorPrescription | not mapped
detectedIssue
eventHistory

11.1.7.4 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

MedicationRequest | ..Request
identifier | …identifer
definition | …definition
basedOn | …basedOn
groupIdentifier | …groupIdentifier
status | …status
intent | …intent
category
priority | …priority
medication[x] | …code
subject | …subject

context | …context
supportingInformation
authoredOn | …authoredOn
requester | …requester
agent | ….agent
onBehalfOf | ….onBehalfOf
recorder
reasonCode | …reasonCode
reasonReference | …reasonReference
note | …note
dosageInstruction | …occurrence[x]
dispenseRequest
validityPeriod
numberOfRepeatsAllowed
quantity
expectedSupplyDuration
performer
substitution
allowed
reason
priorPrescription | …replaces
detectedIssue
eventHistory | …relevantHistory

11.1.8 Resource MedicationRequest - Design Notes

URL: https://hl7.org/fhir/STU3/medicationrequest-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.1.8 Resource MedicationRequest - Design Notes

to do

11.1.9 Resource MedicationRequest - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/medicationrequest-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.1.9 Resource MedicationRequest - Extensions & Profiles

Pharmacy Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the MedicationRequest resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

11.1.9.1 Profiles

No Profiles defined for this resource

11.1.9.2 Extensions

pharmacy-core-doseType | doseType | for International Realm Pharmacy Extensions Profile
pharmacy-core-infuseOver | infuseOver | for International Realm Pharmacy Extensions Profile
pharmacy-core-maxDeliveryRate | maxDeliveryRate | for International Realm Pharmacy Extensions Profile
pharmacy-core-maxDeliveryVolume | maxDeliveryVolume | for International Realm Pharmacy Extensions Profile
pharmacy-core-minDosePerPeriod | minDosePerPeriod | for International Realm Pharmacy Extensions Profile
pharmacy-core-rateGoal | rateGoal | for International Realm Pharmacy Extensions Profile
pharmacy-core-rateIncrement | rateIncrement | for International Realm Pharmacy Extensions Profile
pharmacy-core-rateIncrementInterval | rateIncrementInterval | for International Realm Pharmacy Extensions Profile
pharmacy-core-refillsRemaining | refillsRemaining | for International Realm Pharmacy Extensions Profile

Extensions that reference this resource:

procedure-causedBy | causedBy | for Procedure HL7 Extensions

11.1.9.3 Search Extensions

No Search Extensions defined for this resource

11.1.10 MedicationRequest HL7 Extensions

URL: https://hl7.org/fhir/STU3/medicationrequest-extensions.html

Defines common extensions used with or related to the MedicationRequest resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.1.10 MedicationRequest HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

11.1.10.1 Content

11.1.11 Resource MedicationRequest - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/medicationrequest-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.1.11 Resource MedicationRequest - Detailed Descriptions

Pharmacy Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Encounter, Patient, Practitioner

R2 : R3 Converson maps for MedicationRequest.

Functional status for this map: 18 tests of which 1 fail to execute. All tests pass round-trip testing and 5 r3 resources are invalid (6 errors). (see documentation)

11.1.11.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/MedicationOrder2to3" = "R2 to R3 Conversions for MedicationOrder"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/MedicationOrder" alias MedicationOrderR2 as source
uses "http://hl7.org/fhir/StructureDefinition/MedicationRequest" alias MedicationRequest as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types MedicationOrder extends DomainResource
  input src : MedicationOrderR2 as source
  input tgt : MedicationRequest as target

  "MedicationOrder-intent" : for src make tgt.intent = "order"
  "MedicationOrder-identifier" : for src.identifier make tgt.identifier
  "MedicationOrder-dateWritten" : for src.dateWritten make tgt.authoredOn
  // dateEnded, reasonEnded (see Provenance)
  "MedicationOrder-status" : for src.status make tgt.status
  "MedicationOrder-patient" : for src.patient make tgt.subject
  "MedicationOrder-prescriber" : for src where prescriber.exists() make tgt.requester as vt then medRequestor(src, vt)
  "MedicationOrder-encounter" : for src.encounter make tgt.context
  "MedicationOrder-reasonCode" : for src.reason where $this is CodeableConcept make tgt.reasonCode
  "MedicationOrder-reasonCode" : for src.reason where $this is Reference make tgt.reasonReference
  "MedicationOrder-note" : for src.note make tgt.note // only_one
  "MedicationOrder-medication" : for src.medication make tgt.medication

  "MedicationOrder-dosageInstruction" : for src.dosageInstruction as vs make tgt.dosageInstruction as vt then
    medDosageInstruction(vs, vt)
  "MedicationOrder-dispenseRequest" : for src.dispenseRequest as vs check medication.empty() make tgt.dispenseRequest as
     vt then medDispenseRequest(vs, vt)
  "MedicationOrder-substitution" : for src.substitution as vs make tgt.substitution as vt then medSubstitution(vs, vt)
  "MedicationOrder-priorPrescription" : for src.priorPrescription make tgt.priorPrescription
endgroup

group medDosageInstruction extends Element //? BackboneElement
  input src as source
  input tgt as target

  "MedicationOrder.dosage-text" : for src.text make tgt.text
  "MedicationOrder.dosage-additionalInstructions" : for src.additionalInstructions make tgt.additionalInstruction
  "MedicationOrder.dosage-timing" : for src.timing make tgt.timing
  "MedicationOrder.dosage-asNeeded" : for src.asNeeded make tgt.asNeeded
  "MedicationOrder.dosage-site" : for src.site where $this is CodeableConcept make tgt.site
  "MedicationOrder.dosage-route" : for src.route make tgt.route
  "MedicationOrder.dosage-method" : for src.method make tgt.method
  "MedicationOrder.dosage-dose" : for src.dose make tgt.dose
  "MedicationOrder.dosage-rate" : for src.rate make tgt.rate
  "MedicationOrder.dosage-maxDosePerPeriod" : for src.maxDosePerPeriod make tgt.maxDosePerPeriod
endgroup


group medDispenseRequest extends BackboneElement
  input src as source
  input tgt as target

  // the medication at this level is a duplicate, and should not be different to the main medication
  // (in spite of R2 containing an example that does just that). it is ignored here
  "MedicationOrder.dispenseRequest-validityPeriod" : for src.validityPeriod make tgt.validityPeriod
  "MedicationOrder.dispenseRequest-numberOfRepeatsAllowed" : for src.numberOfRepeatsAllowed make
    tgt.numberOfRepeatsAllowed
  "MedicationOrder.dispenseRequest-quantity" : for src.quantity make tgt.quantity
  "MedicationOrder.dispenseRequest-expectedSupplyDuration" : for src.expectedSupplyDuration as v make
```

```
    tgt.expectedSupplyDuration as d then QuantityToDuration(v, d)
endgroup

group medSubstitution extends BackboneElement
  input src as source
  input tgt as target

  "MedicationOrder.substitution-reason" : for src.reason make tgt.reason
endgroup

group medRequestor
  input src as source
  input tgt as target

  "MedicationOrder-prescriber2" : for src.prescriber make tgt.agent
endgroup
```

11.1.11.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/MedicationOrder3to2" = "R3 to R2 Conversions for MedicationOrder"

uses "http://hl7.org/fhir/StructureDefinition/MedicationRequest" alias MedicationRequest as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/MedicationOrder" alias MedicationOrderR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types MedicationRequest extends DomainResource
  input src : MedicationRequest as source
  input tgt : MedicationOrderR2 as target

  "MedicationOrder-identifier" : for src.identifier make tgt.identifier
  "MedicationOrder-dateWritten" : for src.authoredOn make tgt.dateWritten
  // dateEnded
  // reasonEnded
  "MedicationOrder-status" : for src.status check $this != 'cancelled' and $this != 'unknown' make tgt.status
  "MedicationOrder-patient" : for src.subject make tgt.patient
  "MedicationOrder-prescriber" : for src.requester as req make tgt.prescriber as vt then medRequestor(req, vt)
  "MedicationOrder-encounter" : for src.context make tgt.encounter
  "MedicationOrder-reasonCode" : for src.reasonCode make tgt.reason
  "MedicationOrder-reasonCode" : for src.reasonReference make tgt.reason
  "MedicationOrder-note" : for src.note only_one make tgt.note
  "MedicationOrder-medication" : for src.medication make tgt.medication

  "MedicationOrder-dosageInstruction" : for src.dosageInstruction as vs make tgt.dosageInstruction as vt then
    medDosageInstruction(vs, vt)
  "MedicationOrder-dispenseRequest" : for src.dispenseRequest as vs make tgt.dispenseRequest as vt then
    medDispenseRequest(vs, vt)
  "MedicationOrder-substitution" : for src.substitution as vs make tgt.substitution as vt then medSubstitution(vs, vt)
  "MedicationOrder-priorPrescription" : for src.priorPrescription make tgt.priorPrescription
endgroup

group medDosageInstruction extends Element //? BackboneElement
  input src as source
  input tgt as target

  "MedicationOrder.dosage-text" : for src.text make tgt.text
  "MedicationOrder.dosage-additionalInstructions" : for src.additionalInstruction only_one make
    tgt.additionalInstructions
  "MedicationOrder.dosage-timing" : for src.timing make tgt.timing
  "MedicationOrder.dosage-asNeeded" : for src.asNeeded make tgt.asNeeded
  "MedicationOrder.dosage-site" : for src.site make tgt.site
  "MedicationOrder.dosage-route" : for src.route make tgt.route
  "MedicationOrder.dosage-method" : for src.method make tgt.method
  "MedicationOrder.dosage-dose" : for src.dose make tgt.dose
  "MedicationOrder.dosage-rate" : for src.rate make tgt.rate
  "MedicationOrder.dosage-maxDosePerPeriod" : for src.maxDosePerPeriod make tgt.maxDosePerPeriod
endgroup


group medDispenseRequest extends BackboneElement
  input src as source
  input tgt as target

  "MedicationOrder.dispenseRequest-validityPeriod" : for src.validityPeriod make tgt.validityPeriod
  "MedicationOrder.dispenseRequest-numberOfRepeatsAllowed" : for src.numberOfRepeatsAllowed make
    tgt.numberOfRepeatsAllowed
  "MedicationOrder.dispenseRequest-quantity" : for src.quantity make tgt.quantity
  "MedicationOrder.dispenseRequest-expectedSupplyDuration" : for src.expectedSupplyDuration as v make
    tgt.expectedSupplyDuration as d then DurationToQuantity(v, d)
endgroup

group medSubstitution extends BackboneElement
  input src as source
  input tgt as target

  "MedicationOrder.substitution-reason" : for src.reason make tgt.reason
endgroup

group medRequestor
  input src as source
```

```
  input tgt as target

  "MedicationOrder-prescriber2" : for src.agent as sa then Reference(sa, tgt)
endgroup
```

11.1.11.3 R3 Validation Errors

medicationorder-example | ERROR @ MedicationRequest.dosageInstruction.extension[2][url='http://hl7.org/fhir/StructureDefinition/pharmacy-core-infuseOver']: The Extension 'http://hl7.org/fhir/StructureDefinition/pharmacy-core-infuseOver' definition allows for the types [Duration] but found type Quantity (src = InstanceValidator) INFORMATION @ MedicationRequest.dosageInstruction.extension[2].valueQuantity: Could not verify slice for profile http://hl7.org/fhir/StructureDefinition/pharmacy-core-infuseOver (src = InstanceValidator)

f001 | ERROR @ MedicationRequest.dispenseRequest.expectedSupplyDuration: There SHALL be a code if there is a value and it SHALL be an expression of time. If system is present, it SHALL be UCUM. [code.exists() implies ((system = %ucum) and value.exists())] (src = InstanceValidator)

medrx007 | ERROR @ MedicationRequest: Profile http://hl7.org/fhir/StructureDefinition/MedicationRequest, Element 'MedicationRequest.subject': minimum required = 1, but only found 0 (src = InstanceValidator)

medrx008 | ERROR @ MedicationRequest: Profile http://hl7.org/fhir/StructureDefinition/MedicationRequest, Element 'MedicationRequest.subject': minimum required = 1, but only found 0 (src = InstanceValidator)

medrx009 | ERROR @ MedicationRequest: Profile http://hl7.org/fhir/StructureDefinition/MedicationRequest, Element 'MedicationRequest.subject': minimum required = 1, but only found 0 (src = InstanceValidator)

11.2 Resource MedicationAdministration - Content

URL: https://hl7.org/fhir/STU3/medicationadministration.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Pharmacy Work Group | Maturity Level: 2 | Trial Use | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Describes the event of a patient consuming or otherwise being administered a medication. This may be as simple as swallowing a tablet or it may be a long running infusion. Related resources tie this event to the authorizing prescription, and the specific encounter between patient and health care practitioner.

11.2.1 Scope and Usage

This resource covers the administration of all medications and vaccines. Please refer to the Immunization Resource/Profile for the treatment of vaccines. It will principally be used within care settings (including inpatient) to record the capture of medication administrations, including self-administrations of oral medications, injections, intra-venous adjustments, etc. It can also be used in out-patient settings to record allergy shots and other non-immunization administrations. In some cases it might be used for home-health reporting, such as recording self-administered or even device-administered insulin.

MedicationAdministration is an event resource from a FHIR workflow perspective - see Workflow Event

11.2.2 Boundaries and Relationships

The Medication domain includes a number of related resources

MedicationRequest | An order for both supply of the medication and the instructions for administration of the medicine to a patient.

MedicationDispense | Provision of a supply of a medication with the intention that it is subsequently consumed by a patient (usually in response to a prescription).

MedicationAdministration | When a patient actually consumes a medicine, or it is otherwise administered to them

MedicationStatement | This is a record of a medication being taken by a patient or that a medication has been given to a patient, where the record is the result of a report from the patient or another clinician. A medication statement is not a part of the prescribe->dispense->administer sequence, but is a report that such a sequence (or at least a part of it) did take place, resulting in a belief that the patient has received a particular medication.

MedicationAdministration is intended for tracking the administration of non-vaccine medications. Administration of vaccines is intended to be handled using the Immunization resource. Some systems treat immunizations in the same way as any other medication administration. Such systems SHOULD use an immunization resource to represent these. If systems need to use a MedicationAdministration resource to capture vaccinations for workflow or other reasons, they SHOULD also create and expose an equivalent Immunization instance.

This resource is referenced by AdverseEvent, ChargeItem, MedicationStatement and Procedure

11.2.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints

MedicationAdministration | I | | DomainResource | Administration of medication to a patient + Reason not given is only permitted if NotGiven is true + Reason given is only permitted if NotGiven is false Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension,

modifierExtension
identifier | | 0..* | Identifier | External identifier
definition | ? | 0..* | Reference(PlanDefinition | ActivityDefinition) | Instantiates protocol or definition
partOf | ? | 0..* | Reference(MedicationAdministration | Procedure) | Part of referenced event
status | ?!? | 1..1 | code | in-progress | on-hold | completed | entered-in-error | stopped | unknown MedicationAdministrationStatus (Required)
category | | 0..1 | CodeableConcept | Type of medication usage MedicationAdministrationCategory (Preferred)
medication[x] | ? | 1..1 | | What was administered SNOMED CT Medication Codes (Example)
medicationCodeableConcept | | | CodeableConcept
medicationReference | | | Reference(Medication)
subject | ? | 1..1 | Reference(Patient | Group) | Who received medication
context | | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter or Episode of Care administered as part of
supportingInformation | | 0..* | Reference(Any) | Additional information to support administration
effective[x] | ? | 1..1 | | Start and end time of administration
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
performer | ? | 0..* | BackboneElement | Who administered substance
actor | ? | 1..1 | Reference(Practitioner | Patient | RelatedPerson | Device) | Individual who was performing
onBehalfOf | | 0..1 | Reference(Organization) | Organization organization was acting for
notGiven | ?!? | 0..1 | boolean | True if medication not administered
reasonNotGiven | I | 0..* | CodeableConcept | Reason administration not performed SNOMED CT Reason Medication Not Given Codes (Example)
reasonCode | I | 0..* | CodeableConcept | Reason administration performed Reason Medication Given Codes (Example)
reasonReference | | 0..* | Reference(Condition | Observation) | Condition or Observation that supports why the medication was administered
prescription | | 0..1 | Reference(MedicationRequest) | Request administration performed against
device | | 0..* | Reference(Device) | Device used to administer
note | | 0..* | Annotation | Information about the administration
dosage | I | 0..1 | BackboneElement | Details of how medication was taken + SHALL have at least one of dosage.dose or dosage.rate[x]
text | | 0..1 | string | Free text dosage instructions e.g. SIG
site | | 0..1 | CodeableConcept | Body site administered to SNOMED CT Anatomical Structure for Administration Site Codes (Example)
route | | 0..1 | CodeableConcept | Path of substance into body SNOMED CT Route Codes (Example)
method | | 0..1 | CodeableConcept | How drug was administered SNOMED CT Administration Method Codes (Example)
dose | | 0..1 | SimpleQuantity | Amount of medication per dose
rate[x] | | 0..1 | | Dose quantity per unit of time
rateRatio | | | Ratio
rateQuantity | | | SimpleQuantity
eventHistory | | 0..* | Reference(Provenance) | A list of events of interest in the lifecycle
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicationAdministration xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External identifier --></identifier>
 <definition><!-- 0..* Reference(PlanDefinition|ActivityDefinition) Instantiates protocol or definition --></definition>
 <partOf><!-- 0..* Reference(MedicationAdministration|Procedure) Part of referenced event --></partOf>
 <status value="[code]"/><!-- 1..1 in-progress | on-hold | completed | entered-in-error | stopped | unknown -->
 <category><!-- 0..1 CodeableConcept Type of medication usage --></category>
 <medication[x]><!-- 1..1 CodeableConcept|Reference(Medication) What was administered --></medication[x]>
 <subject><!-- 1..1 Reference(Patient|Group) Who received medication --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter or Episode of Care administered as part of
    --></context>
 <supportingInformation><!-- 0..* Reference(Any) Additional information to support administration
    --></supportingInformation>
 <effective[x]><!-- 1..1 dateTime|Period Start and end time of administration --></effective[x]>
 <performer>  <!-- 0..* Who administered substance -->
  <actor><!-- 1..1 Reference(Practitioner|Patient|RelatedPerson|Device) Individual who was performing --></actor>
  <onBehalfOf><!-- 0..1 Reference(Organization) Organization organization was acting for --></onBehalfOf>
 </performer>
 <notGiven value="[boolean]"/><!-- 0..1 True if medication not administered -->
 <reasonNotGiven><!--  0..* CodeableConcept Reason administration not performed --></reasonNotGiven>
 <reasonCode><!--  0..* CodeableConcept Reason administration performed --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation) Condition or Observation that supports why the medication
    was administered --></reasonReference>
 <prescription><!-- 0..1 Reference(MedicationRequest) Request administration performed against --></prescription>
 <device><!-- 0..* Reference(Device) Device used to administer --></device>
 <note><!-- 0..* Annotation Information about the administration --></note>
 <dosage>  <!-- 0..1 Details of how medication was taken -->
  <text value="[string]"/><!-- 0..1 Free text dosage instructions e.g. SIG -->
  <site><!-- 0..1 CodeableConcept Body site administered to --></site>
  <route><!-- 0..1 CodeableConcept Path of substance into body --></route>
  <method><!-- 0..1 CodeableConcept How drug was administered --></method>
  <dose><!-- 0..1 Quantity(SimpleQuantity) Amount of medication per dose --></dose>
  <rate[x]><!-- 0..1 Ratio|Quantity(SimpleQuantity) Dose quantity per unit of time --></rate[x]>
 </dosage>
 <eventHistory><!-- 0..* Reference(Provenance) A list of events of interest in the lifecycle --></eventHistory>
</MedicationAdministration>
```

JSON Template

```
{
  "resourceType" : "MedicationAdministration",
```

```
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External identifier
  "definition" : [{ Reference(PlanDefinition|ActivityDefinition) }], // Instantiates protocol or definition
  "partOf" : [{ Reference(MedicationAdministration|Procedure) }], // Part of referenced event
  "status" : "<code>", // R!  in-progress | on-hold | completed | entered-in-error | stopped | unknown
  "category" : { CodeableConcept }, // Type of medication usage
  // medication[x]: What was administered. One of these 2:
  "medicationCodeableConcept" : { CodeableConcept },
  "medicationReference" : { Reference(Medication) },
  "subject" : { Reference(Patient|Group) }, // R!  Who received medication
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter or Episode of Care administered as part of
  "supportingInformation" : [{ Reference(Any) }], // Additional information to support administration
  // effective[x]: Start and end time of administration. One of these 2:
  "effectiveDateTime" : "<dateTime>",
  "effectivePeriod" : { Period },
  "performer" : [{ // Who administered substance
    "actor" : { Reference(Practitioner|Patient|RelatedPerson|Device) }, // R!  Individual who was performing
    "onBehalfOf" : { Reference(Organization) } // Organization organization was acting for
  }],
  "notGiven" : <boolean>, // True if medication not administered
  "reasonNotGiven" : [{ CodeableConcept }], // C? Reason administration not performed
  "reasonCode" : [{ CodeableConcept }], // C? Reason administration performed
  "reasonReference" : [{ Reference(Condition|Observation) }], // Condition or Observation that supports why the
    medication was administered
  "prescription" : { Reference(MedicationRequest) }, // Request administration performed against
  "device" : [{ Reference(Device) }], // Device used to administer
  "note" : [{ Annotation }], // Information about the administration
  "dosage" : { // Details of how medication was taken
    "text" : "<string>", // Free text dosage instructions e.g. SIG
    "site" : { CodeableConcept }, // Body site administered to
    "route" : { CodeableConcept }, // Path of substance into body
    "method" : { CodeableConcept }, // How drug was administered
    "dose" : { Quantity(SimpleQuantity) }, // Amount of medication per dose
    // rate[x]: Dose quantity per unit of time. One of these 2:
    "rateRatio" : { Ratio }
    "rateQuantity" : { Quantity(SimpleQuantity) }
  },
  "eventHistory" : [{ Reference(Provenance) }] // A list of events of interest in the lifecycle
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicationAdministration;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicationAdministration.identifier [ Identifier ], ... ; # 0..* External identifier
  fhir:MedicationAdministration.definition [ Reference(PlanDefinition|ActivityDefinition) ], ... ; # 0..* Instantiates
    protocol or definition
  fhir:MedicationAdministration.partOf [ Reference(MedicationAdministration|Procedure) ], ... ; # 0..* Part of
    referenced event
  fhir:MedicationAdministration.status [ code ]; # 1..1 in-progress | on-hold | completed | entered-in-error | stopped |
     unknown
  fhir:MedicationAdministration.category [ CodeableConcept ]; # 0..1 Type of medication usage
  # MedicationAdministration.medication[x] : 1..1 What was administered. One of these 2
    fhir:MedicationAdministration.medicationCodeableConcept [ CodeableConcept ]
    fhir:MedicationAdministration.medicationReference [ Reference(Medication) ]
  fhir:MedicationAdministration.subject [ Reference(Patient|Group) ]; # 1..1 Who received medication
  fhir:MedicationAdministration.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter or Episode of Care
    administered as part of
  fhir:MedicationAdministration.supportingInformation [ Reference(Any) ], ... ; # 0..* Additional information to support
     administration
  # MedicationAdministration.effective[x] : 1..1 Start and end time of administration. One of these 2
    fhir:MedicationAdministration.effectiveDateTime [ dateTime ]
    fhir:MedicationAdministration.effectivePeriod [ Period ]
  fhir:MedicationAdministration.performer [ # 0..* Who administered substance
    fhir:MedicationAdministration.performer.actor [ Reference(Practitioner|Patient|RelatedPerson|Device) ]; # 1..1
    Individual who was performing
    fhir:MedicationAdministration.performer.onBehalfOf [ Reference(Organization) ]; # 0..1 Organization organization was
     acting for
  ], ...;
  fhir:MedicationAdministration.notGiven [ boolean ]; # 0..1 True if medication not administered
  fhir:MedicationAdministration.reasonNotGiven [ CodeableConcept ], ... ; # 0..* Reason administration not performed
  fhir:MedicationAdministration.reasonCode [ CodeableConcept ], ... ; # 0..* Reason administration performed
  fhir:MedicationAdministration.reasonReference [ Reference(Condition|Observation) ], ... ; # 0..* Condition or
    Observation that supports why the medication was administered
  fhir:MedicationAdministration.prescription [ Reference(MedicationRequest) ]; # 0..1 Request administration performed
    against
  fhir:MedicationAdministration.device [ Reference(Device) ], ... ; # 0..* Device used to administer
  fhir:MedicationAdministration.note [ Annotation ], ... ; # 0..* Information about the administration
  fhir:MedicationAdministration.dosage [ # 0..1 Details of how medication was taken
    fhir:MedicationAdministration.dosage.text [ string ]; # 0..1 Free text dosage instructions e.g. SIG
    fhir:MedicationAdministration.dosage.site [ CodeableConcept ]; # 0..1 Body site administered to
    fhir:MedicationAdministration.dosage.route [ CodeableConcept ]; # 0..1 Path of substance into body
    fhir:MedicationAdministration.dosage.method [ CodeableConcept ]; # 0..1 How drug was administered
```

```
    fhir:MedicationAdministration.dosage.dose [ Quantity(SimpleQuantity) ]; # 0..1 Amount of medication per dose
    # MedicationAdministration.dosage.rate[x] : 0..1 Dose quantity per unit of time. One of these 2
      fhir:MedicationAdministration.dosage.rateRatio [ Ratio ]
      fhir:MedicationAdministration.dosage.rateSimpleQuantity [ Quantity(SimpleQuantity) ]
  ];
  fhir:MedicationAdministration.eventHistory [ Reference(Provenance) ], ... ; # 0..* A list of events of interest in the
     lifecycle
]
```

Changes since DSTU2

MedicationAdministration
MedicationAdministration.definition | Added Element
MedicationAdministration.partOf | Added Element
MedicationAdministration.category | Added Element
MedicationAdministration.subject | Renamed from patient to subject Add Reference(Group)
MedicationAdministration.context | Renamed from encounter to context Add Reference(EpisodeOfCare)
MedicationAdministration.supportingInformation | Added Element
MedicationAdministration.effective[x] | Added Element
MedicationAdministration.performer | Renamed from practitioner to performer Max Cardinality changed from 1 to * Remove Reference(Practitioner), Remove Reference(Patient), Remove Reference(RelatedPerson)
MedicationAdministration.performer.actor | Added Element
MedicationAdministration.performer.onBehalfOf | Added Element
MedicationAdministration.notGiven | Renamed from wasNotGiven to notGiven
MedicationAdministration.reasonCode | Added Element
MedicationAdministration.reasonReference | Added Element
MedicationAdministration.prescription | Type changed from Reference(MedicationOrder) to Reference(MedicationRequest)
MedicationAdministration.note | Max Cardinality changed from 1 to * Type changed from string to Annotation
MedicationAdministration.dosage.site | Renamed from site[x] to site Remove Reference(BodySite)
MedicationAdministration.dosage.dose | Renamed from quantity to dose
MedicationAdministration.dosage.rate[x] | Remove Range, Add Quantity{http://hl7.org/fhir/StructureDefinition/SimpleQuantity}
MedicationAdministration.eventHistory | Added Element
MedicationAdministration.reasonGiven | deleted
MedicationAdministration.effectiveTime[x] | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints
MedicationAdministration | I | | DomainResource | Administration of medication to a patient + Reason not given is only permitted if NotGiven is true + Reason given is only permitted if NotGiven is false Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | External identifier
definition | ? | 0..* | Reference(PlanDefinition | ActivityDefinition) | Instantiates protocol or definition
partOf | ? | 0..* | Reference(MedicationAdministration | Procedure) | Part of referenced event
status | ?!? | 1..1 | code | in-progress | on-hold | completed | entered-in-error | stopped | unknown MedicationAdministrationStatus (Required)
category | | 0..1 | CodeableConcept | Type of medication usage MedicationAdministrationCategory (Preferred)
medication[x] | ? | 1..1 | | What was administered SNOMED CT Medication Codes (Example)
medicationCodeableConcept | | | CodeableConcept
medicationReference | | | Reference(Medication)
subject | ? | 1..1 | Reference(Patient | Group) | Who received medication
context | | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter or Episode of Care administered as part of
supportingInformation | | 0..* | Reference(Any) | Additional information to support administration
effective[x] | ? | 1..1 | | Start and end time of administration
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
performer | ? | 0..* | BackboneElement | Who administered substance
actor | ? | 1..1 | Reference(Practitioner | Patient | RelatedPerson | Device) | Individual who was performing
onBehalfOf | | 0..1 | Reference(Organization) | Organization organization was acting for
notGiven | ?!? | 0..1 | boolean | True if medication not administered
reasonNotGiven | I | 0..* | CodeableConcept | Reason administration not performed SNOMED CT Reason Medication Not Given Codes (Example)
reasonCode | I | 0..* | CodeableConcept | Reason administration performed Reason Medication Given Codes (Example)
reasonReference | | 0..* | Reference(Condition | Observation) | Condition or Observation that supports why the medication was administered
prescription | | 0..1 | Reference(MedicationRequest) | Request administration performed against
device | | 0..* | Reference(Device) | Device used to administer
note | | 0..* | Annotation | Information about the administration
dosage | I | 0..1 | BackboneElement | Details of how medication was taken + SHALL have at least one of dosage.dose or dosage.rate[x]
text | | 0..1 | string | Free text dosage instructions e.g. SIG
site | | 0..1 | CodeableConcept | Body site administered to SNOMED CT Anatomical Structure for Administration Site Codes (Example)
route | | 0..1 | CodeableConcept | Path of substance into body SNOMED CT Route Codes (Example)
method | | 0..1 | CodeableConcept | How drug was administered SNOMED CT Administration Method Codes (Example)
dose | | 0..1 | SimpleQuantity | Amount of medication per dose

rate[x] | | 0..1 | | Dose quantity per unit of time
rateRatio | | | Ratio
rateQuantity | | | SimpleQuantity
eventHistory | | 0..* | Reference(Provenance) | A list of events of interest in the lifecycle
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicationAdministration xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External identifier --></identifier>
 <definition><!-- 0..* Reference(PlanDefinition|ActivityDefinition) Instantiates protocol or definition --></definition>
 <partOf><!-- 0..* Reference(MedicationAdministration|Procedure) Part of referenced event --></partOf>
 <status value="[code]"/><!-- 1..1 in-progress | on-hold | completed | entered-in-error | stopped | unknown -->
 <category><!-- 0..1 CodeableConcept Type of medication usage --></category>
 <medication[x]><!-- 1..1 CodeableConcept|Reference(Medication) What was administered --></medication[x]>
 <subject><!-- 1..1 Reference(Patient|Group) Who received medication --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter or Episode of Care administered as part of
    --></context>
 <supportingInformation><!-- 0..* Reference(Any) Additional information to support administration
    --></supportingInformation>
 <effective[x]><!-- 1..1 dateTime|Period Start and end time of administration --></effective[x]>
 <performer>  <!-- 0..* Who administered substance -->
  <actor><!-- 1..1 Reference(Practitioner|Patient|RelatedPerson|Device) Individual who was performing --></actor>
  <onBehalfOf><!-- 0..1 Reference(Organization) Organization organization was acting for --></onBehalfOf>
 </performer>
 <notGiven value="[boolean]"/><!-- 0..1 True if medication not administered -->
 <reasonNotGiven><!--  0..* CodeableConcept Reason administration not performed --></reasonNotGiven>
 <reasonCode><!--  0..* CodeableConcept Reason administration performed --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation) Condition or Observation that supports why the medication
    was administered --></reasonReference>
 <prescription><!-- 0..1 Reference(MedicationRequest) Request administration performed against --></prescription>
 <device><!-- 0..* Reference(Device) Device used to administer --></device>
 <note><!-- 0..* Annotation Information about the administration --></note>
 <dosage>  <!-- 0..1 Details of how medication was taken -->
  <text value="[string]"/><!-- 0..1 Free text dosage instructions e.g. SIG -->
  <site><!-- 0..1 CodeableConcept Body site administered to --></site>
  <route><!-- 0..1 CodeableConcept Path of substance into body --></route>
  <method><!-- 0..1 CodeableConcept How drug was administered --></method>
  <dose><!-- 0..1 Quantity(SimpleQuantity) Amount of medication per dose --></dose>
  <rate[x]><!-- 0..1 Ratio|Quantity(SimpleQuantity) Dose quantity per unit of time --></rate[x]>
 </dosage>
 <eventHistory><!-- 0..* Reference(Provenance) A list of events of interest in the lifecycle --></eventHistory>
</MedicationAdministration>
```

JSON Template

```
{
  "resourceType" : "MedicationAdministration",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External identifier
  "definition" : [{ Reference(PlanDefinition|ActivityDefinition) }], // Instantiates protocol or definition
  "partOf" : [{ Reference(MedicationAdministration|Procedure) }], // Part of referenced event
  "status" : "<code>", // R!  in-progress | on-hold | completed | entered-in-error | stopped | unknown
  "category" : { CodeableConcept }, // Type of medication usage
  // medication[x]: What was administered. One of these 2:
  "medicationCodeableConcept" : { CodeableConcept },
  "medicationReference" : { Reference(Medication) },
  "subject" : { Reference(Patient|Group) }, // R!  Who received medication
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter or Episode of Care administered as part of
  "supportingInformation" : [{ Reference(Any) }], // Additional information to support administration
  // effective[x]: Start and end time of administration. One of these 2:
  "effectiveDateTime" : "<dateTime>",
  "effectivePeriod" : { Period },
  "performer" : [{ // Who administered substance
    "actor" : { Reference(Practitioner|Patient|RelatedPerson|Device) }, // R!  Individual who was performing
    "onBehalfOf" : { Reference(Organization) } // Organization organization was acting for
  }],
  "notGiven" : <boolean>, // True if medication not administered
  "reasonNotGiven" : [{ CodeableConcept }], // C? Reason administration not performed
  "reasonCode" : [{ CodeableConcept }], // C? Reason administration performed
  "reasonReference" : [{ Reference(Condition|Observation) }], // Condition or Observation that supports why the
    medication was administered
  "prescription" : { Reference(MedicationRequest) }, // Request administration performed against
  "device" : [{ Reference(Device) }], // Device used to administer
  "note" : [{ Annotation }], // Information about the administration
  "dosage" : { // Details of how medication was taken
    "text" : "<string>", // Free text dosage instructions e.g. SIG
    "site" : { CodeableConcept }, // Body site administered to
    "route" : { CodeableConcept }, // Path of substance into body
    "method" : { CodeableConcept }, // How drug was administered
    "dose" : { Quantity(SimpleQuantity) }, // Amount of medication per dose
    // rate[x]: Dose quantity per unit of time. One of these 2:
    "rateRatio" : { Ratio }
    "rateQuantity" : { Quantity(SimpleQuantity) }
  },
```

```
  "eventHistory" : [{ Reference(Provenance) }] // A list of events of interest in the lifecycle
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicationAdministration;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicationAdministration.identifier [ Identifier ], ... ; # 0..* External identifier
  fhir:MedicationAdministration.definition [ Reference(PlanDefinition|ActivityDefinition) ], ... ; # 0..* Instantiates
    protocol or definition
  fhir:MedicationAdministration.partOf [ Reference(MedicationAdministration|Procedure) ], ... ; # 0..* Part of
    referenced event
  fhir:MedicationAdministration.status [ code ]; # 1..1 in-progress | on-hold | completed | entered-in-error | stopped |
     unknown
  fhir:MedicationAdministration.category [ CodeableConcept ]; # 0..1 Type of medication usage
  # MedicationAdministration.medication[x] : 1..1 What was administered. One of these 2
    fhir:MedicationAdministration.medicationCodeableConcept [ CodeableConcept ]
    fhir:MedicationAdministration.medicationReference [ Reference(Medication) ]
  fhir:MedicationAdministration.subject [ Reference(Patient|Group) ]; # 1..1 Who received medication
  fhir:MedicationAdministration.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter or Episode of Care
    administered as part of
  fhir:MedicationAdministration.supportingInformation [ Reference(Any) ], ... ; # 0..* Additional information to support
     administration
  # MedicationAdministration.effective[x] : 1..1 Start and end time of administration. One of these 2
    fhir:MedicationAdministration.effectiveDateTime [ dateTime ]
    fhir:MedicationAdministration.effectivePeriod [ Period ]
  fhir:MedicationAdministration.performer [ # 0..* Who administered substance
    fhir:MedicationAdministration.performer.actor [ Reference(Practitioner|Patient|RelatedPerson|Device) ]; # 1..1
    Individual who was performing
    fhir:MedicationAdministration.performer.onBehalfOf [ Reference(Organization) ]; # 0..1 Organization organization was
     acting for
  ], ...;
  fhir:MedicationAdministration.notGiven [ boolean ]; # 0..1 True if medication not administered
  fhir:MedicationAdministration.reasonNotGiven [ CodeableConcept ], ... ; # 0..* Reason administration not performed
  fhir:MedicationAdministration.reasonCode [ CodeableConcept ], ... ; # 0..* Reason administration performed
  fhir:MedicationAdministration.reasonReference [ Reference(Condition|Observation) ], ... ; # 0..* Condition or
    Observation that supports why the medication was administered
  fhir:MedicationAdministration.prescription [ Reference(MedicationRequest) ]; # 0..1 Request administration performed
    against
  fhir:MedicationAdministration.device [ Reference(Device) ], ... ; # 0..* Device used to administer
  fhir:MedicationAdministration.note [ Annotation ], ... ; # 0..* Information about the administration
  fhir:MedicationAdministration.dosage [ # 0..1 Details of how medication was taken
    fhir:MedicationAdministration.dosage.text [ string ]; # 0..1 Free text dosage instructions e.g. SIG
    fhir:MedicationAdministration.dosage.site [ CodeableConcept ]; # 0..1 Body site administered to
    fhir:MedicationAdministration.dosage.route [ CodeableConcept ]; # 0..1 Path of substance into body
    fhir:MedicationAdministration.dosage.method [ CodeableConcept ]; # 0..1 How drug was administered
    fhir:MedicationAdministration.dosage.dose [ Quantity(SimpleQuantity) ]; # 0..1 Amount of medication per dose
    # MedicationAdministration.dosage.rate[x] : 0..1 Dose quantity per unit of time. One of these 2
      fhir:MedicationAdministration.dosage.rateRatio [ Ratio ]
      fhir:MedicationAdministration.dosage.rateSimpleQuantity [ Quantity(SimpleQuantity) ]
  ];
  fhir:MedicationAdministration.eventHistory [ Reference(Provenance) ], ... ; # 0..* A list of events of interest in the
     lifecycle
]
```

Changes since DSTU2

MedicationAdministration

MedicationAdministration.definition | Added Element

MedicationAdministration.partOf | Added Element

MedicationAdministration.category | Added Element

MedicationAdministration.subject | Renamed from patient to subject Add Reference(Group)

MedicationAdministration.context | Renamed from encounter to context Add Reference(EpisodeOfCare)

MedicationAdministration.supportingInformation | Added Element

MedicationAdministration.effective[x] | Added Element

MedicationAdministration.performer | Renamed from practitioner to performer Max Cardinality changed from 1 to * Remove Reference(Practitioner), Remove Reference(Patient), Remove Reference(RelatedPerson)

MedicationAdministration.performer.actor | Added Element

MedicationAdministration.performer.onBehalfOf | Added Element

MedicationAdministration.notGiven | Renamed from wasNotGiven to notGiven

MedicationAdministration.reasonCode | Added Element

MedicationAdministration.reasonReference | Added Element

MedicationAdministration.prescription | Type changed from Reference(MedicationOrder) to Reference(MedicationRequest)

MedicationAdministration.note | Max Cardinality changed from 1 to * Type changed from string to Annotation

MedicationAdministration.dosage.site | Renamed from site[x] to site Remove Reference(BodySite)

MedicationAdministration.dosage.dose | Renamed from quantity to dose

MedicationAdministration.dosage.rate[x] | Remove Range, Add Quantity{http://hl7.org/fhir/StructureDefinition/SimpleQuantity}

MedicationAdministration.eventHistory | Added Element

MedicationAdministration.reasonGiven | deleted

MedicationAdministration.effectiveTime[x] | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

11.2.3.1 Terminology Bindings

Path | Definition | Type | Reference

MedicationAdministration.status | A set of codes indicating the current status of a MedicationAdministration. | Required | MedicationAdministrationStatus

MedicationAdministration.category | A coded concept describing where the medication administered is expected to occur | Preferred | MedicationAdministrationCategory

MedicationAdministration.medication[x] | Codes identifying substance or product that can be administered. | Example | SNOMED CT Medication Codes

MedicationAdministration.reasonNotGiven | A set of codes indicating the reason why the MedicationAdministration is negated. | Example | SNOMED CT Reason Medication Not Given Codes

MedicationAdministration.reasonCode | A set of codes indicating the reason why the MedicationAdministration was made. | Example | Reason Medication Given Codes

MedicationAdministration.dosage.site | A coded concept describing the site location the medicine enters into or onto the body. | Example | SNOMED CT Anatomical Structure for Administration Site Codes

MedicationAdministration.dosage.route | A coded concept describing the route or physiological path of administration of a therapeutic agent into or onto the body of a subject. | Example | SNOMED CT Route Codes

MedicationAdministration.dosage.method | A coded concept describing the technique by which the medicine is administered. | Example | SNOMED CT Administration Method Codes

11.2.3.2 Constraints

* mad-1: On MedicationAdministration.dosage: SHALL have at least one of dosage.dose or dosage.rate[x] (expression on MedicationAdministration.dosage: dose.exists() or rate.exists())

* mad-2: Reason not given is only permitted if NotGiven is true (expression : reasonNotGiven.empty() or notGiven = true)

* mad-3: Reason given is only permitted if NotGiven is false (expression : reasonCode.empty() or notGiven.empty() or notGiven = 'false')

11.2.4 Known Issues

Issue | Comments

Medication Resource | A medication will typically be referred to by means of a code drawn from a suitable medication terminology. However, on occasion a product will be required for which the "recipe" must be specified. This implies a requirement to deal with a choice of either a code or a much more complete resource. Currently that resource has not been created.

Contrast Media | Is this resource adequate for administering contrast media to a patient?

Author (accountability) | Authorship (and any other accountability) is assumed to be dealt with by the standard FHIR methods.

11.2.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common

code | token | Return administrations of this medication code | MedicationAdministration.medication.as(CodeableConcept) | 4 Resources

context | reference | Return administrations that share this encounter or episode of care | MedicationAdministration.context (EpisodeOfCare, Encounter)

device | reference | Return administrations with this administration device identity | MedicationAdministration.device (Device)

effective-time | date | Date administration happened (or did not happen) | MedicationAdministration.effective

identifier | token | Return administrations with this external identifier | MedicationAdministration.identifier | 3 Resources

medication | reference | Return administrations of this medication resource | MedicationAdministration.medication.as(Reference) (Medication) | 3 Resources

not-given | token | Administrations that were not made | MedicationAdministration.notGiven

patient | reference | The identity of a patient to list administrations for | MedicationAdministration.subject (Patient) | 3 Resources

performer | reference | The identify of the individual who administered the medication | MedicationAdministration.performer.actor (Practitioner, Device, Patient, RelatedPerson)

prescription | reference | The identity of a prescription to list administrations from | MedicationAdministration.prescription (MedicationRequest) | 1 Resources

reason-given | token | Reasons for administering the medication | MedicationAdministration.reasonCode

reason-not-given | token | Reasons for not administering the medication | MedicationAdministration.reasonNotGiven

status | token | MedicationAdministration event status (for example one of active/paused/completed/nullified) | MedicationAdministration.status | 3 Resources

subject | reference | The identify of the individual or group to list administrations for | MedicationAdministration.subject (Group, Patient)

11.2.6 Resource MedicationAdministration - Examples

URL: https://hl7.org/fhir/STU3/medicationadministration-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.2.6 Resource MedicationAdministration - Examples

Pharmacy Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

The following are examples of MedicationAdministration resources:

INJECTABLE DOSAGE FORMS

The following examples of MedicationAdministration resources for Injectable Dosage Forms that reference a Medication Resource for a manufactured product:

* Intravenous Antibiotic - Vancomycin

* Intravenous Antibiotic with a Dosage Rate - Piperacillin

* Intravenous Chemotherapy (Inpatient order) - Alemtuzumab - completed

* Intravenous Chemotherapy (Inpatient order) - Brentuximab Vedotin

* Example of an IV administration not given with a note - Alemtuzumab

The following examples of MedicationAdministration resources for Injectable Dosage Forms that reference a Medication Resource for a generic formulation:

* Oral Antibiotic - Azithromycin

* Lorazepam with a Device

The following examples of MedicationDispense resources for Injectable Dosage Forms that include a Medication Code:

* Subcutaneous Injection - Insulin - Lantus

The following examples of MedicationDispense resources for Injectable Dosage Forms for Injectable Dosage Forms where the medication is a compounded (aka extemporaneous or magistral) product:

* Intravenous Admixture - Potassium Chloride in D5W

* TPN Solution with a Device

ORAL DOSAGE FORMS

The following examples are MedicationDispense resources for Oral Dosage Forms that reference Medication Resources for manufactured products:

* Oral Chemotherapy - Myleran

* Dosage with a Pre-Condition - Alprazolam

MISCELLANEOUS DOSAGE FORMS

The following examples of MedicationDispense resources including miscellaneous dosage forms:

* Medication Code - Patch - Fentanyl

* Medication Code - Rectal Suppository - Acetaminophen

Example List:

Example of an IV administration not given with a note - Alemtuzumab | medadminexample03 | XML | JSON | Turtle
Intravenous Antibiotic - Vancomycin | medadmin0301 | XML | JSON | Turtle
Intravenous Admixture - Potassium Chloride in D5W | medadmin0302 | XML | JSON | Turtle
Intravenous Antibiotic with a Dosage Rate - Piperacillin | medadmin0303 | XML | JSON | Turtle
Intravenous Chemotherapy (Inpatient order) - Alemtuzumab - completed | medadmin0304 | XML | JSON | Turtle
Intravenous Chemotherapy (Inpatient order) - Brentuximab Vedotin | medadmin0305 | XML | JSON | Turtle
Oral Antibiotic - Azithromycin | medadmin0306 | XML | JSON | Turtle
Lorazepam with a Device | medadmin0307 | XML | JSON | Turtle
Subcutaneous Injection - Insulin - Lantus | medadmin0308 | XML | JSON | Turtle
TPN Solution with a Device | medadmin0309 | XML | JSON | Turtle
Oral Chemotherapy - Myleran | medadmin0310 | XML | JSON | Turtle
Dosage with a Pre-Condition - Alprazolam | medadmin0311 | XML | JSON | Turtle
Medication Code - Patch - Fentanyl | medadmin0312 | XML | JSON | Turtle
Medication Code - Rectal Suppository - Acetaminophen | medadmin0313 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.2.7 Resource MedicationAdministration - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/medicationadministration-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.2.7 Resource MedicationAdministration - Detailed Descriptions

Pharmacy Work Group | Maturity Level: 2 | Trial Use | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the MedicationAdministration resource.

MedicationAdministration
Definition | Describes the event of a patient consuming or otherwise being administered a medication. This may be as simple as swallowing a tablet or it may be a long running infusion. Related resources tie this event to the authorizing prescription, and the specific encounter between patient and health care practitioner.
Control | 1..1
Comments | The WG will be updating the MedicationAdministration resource to adjust each affected resource to align with the workflow pattern (see workflow.html).
Invariants | Defined on this element mad-2: Reason not given is only permitted if NotGiven is true (expression : reasonNotGiven.empty() or notGiven = true, xpath: not(exists(f:reasonNotGiven) and f:notGiven/@value=false())) mad-3: Reason given is only permitted if NotGiven is false (expression : reasonCode.empty() or notGiven.empty() or notGiven = 'false', xpath: not(exists(f:reasonCode) and f:notGiven/@value=true()))
MedicationAdministration.identifier
Definition | External identifier - FHIR will generate its own internal identifiers (probably URLs) which do not need to be explicitly managed by the resource. The identifier here is one that would be used by another non-FHIR system - for example an automated medication pump would provide a record each time it operated; an administration while the patient was off the ward might be made with a different system and entered after the event. Particularly important if these records have to be updated.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
MedicationAdministration.definition
Definition | A protocol, guideline, orderset or other definition that was adhered to in whole or in part by this event.
Control | 0..*
Type | Reference(PlanDefinition | ActivityDefinition)
Summary | true
MedicationAdministration.partOf
Definition | A larger event of which this particular event is a component or step.
Control | 0..*
Type | Reference(MedicationAdministration | Procedure)
Summary | true
MedicationAdministration.status
Definition | Will generally be set to show that the administration has been completed. For some long running administrations such as infusions it is possible for an administration to be started but not completed or it may be paused while some other process is under way.
Control | 1..1
Terminology Binding | MedicationAdministrationStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
MedicationAdministration.category
Definition | Indicates the type of medication administration and where the medication is expected to be consumed or administered.
Control | 0..1
Terminology Binding | MedicationAdministrationCategory (Preferred)
Type | CodeableConcept
MedicationAdministration.medication[x]
Definition | Identifies the medication that was administered. This is either a link to a resource representing the details of the medication or a simple attribute carrying a code that identifies the medication from a known list of medications.
Control | 1..1
Terminology Binding | SNOMED CT Medication Codes (Example)
Type | CodeableConcept|Reference(Medication)
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | If only a code is specified, then it needs to be a code for a specific product. If more information is required, then the use of the medication resource is recommended. For example if you require form or lot number, then you must reference the Medication resource. .
MedicationAdministration.subject
Definition | The person or animal or group receiving the medication.
Control | 1..1
Type | Reference(Patient | Group)
Summary | true
MedicationAdministration.context
Definition | The visit, admission or other contact between patient and health care provider the medication administration was performed as part of.
Control | 0..1
Type | Reference(Encounter | EpisodeOfCare)
MedicationAdministration.supportingInformation
Definition | Additional information (for example, patient height and weight) that supports the administration of the medication.
Control | 0..*
Type | Reference(Any)
MedicationAdministration.effective[x]
Definition | A specific date/time or interval of time during which the administration took place (or did not take place, when the 'notGiven' attribute is true). For many administrations, such as swallowing a tablet the use of dateTime is more appropriate.
Control | 1..1
Type | dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
MedicationAdministration.performer

Definition | The individual who was responsible for giving the medication to the patient.
Control | 0..*
Summary | true
MedicationAdministration.performer.actor
Definition | The device, practitioner, etc. who performed the action.
Control | 1..1
Type | Reference(Practitioner | Patient | RelatedPerson | Device)
Summary | true
MedicationAdministration.performer.onBehalfOf
Definition | The organization the device or practitioner was acting on behalf of.
Control | 0..1
Type | Reference(Organization)
MedicationAdministration.notGiven
Definition | Set this to true if the record is saying that the medication was NOT administered.
Control | 0..1
Type | boolean
Is Modifier | true
Meaning if Missing | If this is missing, then the medication was administered
Summary | true
Comments | This element is labeled as a modifier because it indicates that a administration didn't happen.
MedicationAdministration.reasonNotGiven
Definition | A code indicating why the administration was not performed.
Control | 0..*
Terminology Binding | SNOMED CT Reason Medication Not Given Codes (Example)
Type | CodeableConcept
Invariants | Affect this element mad-2: Reason not given is only permitted if NotGiven is true (expression : reasonNotGiven.empty() or notGiven = true, xpath: not(exists(f:reasonNotGiven) and f:notGiven/@value=false()))
MedicationAdministration.reasonCode
Definition | A code indicating why the medication was given.
Control | 0..*
Terminology Binding | Reason Medication Given Codes (Example)
Type | CodeableConcept
Invariants | Affect this element mad-3: Reason given is only permitted if NotGiven is false (expression : reasonCode.empty() or notGiven.empty() or notGiven = 'false', xpath: not(exists(f:reasonCode) and f:notGiven/@value=true()))
MedicationAdministration.reasonReference
Definition | Condition or observation that supports why the medication was administered.
Control | 0..*
Type | Reference(Condition | Observation)
Comments | This is a reference to a condition that is the reason for the medication request. If only a code exists, use reasonCode.
MedicationAdministration.prescription
Definition | The original request, instruction or authority to perform the administration.
Control | 0..1
Type | Reference(MedicationRequest)
MedicationAdministration.device
Definition | The device used in administering the medication to the patient. For example, a particular infusion pump.
Control | 0..*
Type | Reference(Device)
MedicationAdministration.note
Definition | Extra information about the medication administration that is not conveyed by the other attributes.
Control | 0..*
Type | Annotation
MedicationAdministration.dosage
Definition | Describes the medication dosage information details e.g. dose, rate, site, route, etc.
Control | 0..1
Invariants | Defined on this element mad-1: SHALL have at least one of dosage.dose or dosage.rate[x] (expression : dose.exists() or rate.exists(), xpath: exists(f:dose) or exists(f:*[starts-with(local-name(.), 'rate')]))
MedicationAdministration.dosage.text
Definition | Free text dosage can be used for cases where the dosage administered is too complex to code. When coded dosage is present, the free text dosage may still be present for display to humans. The dosage instructions should reflect the dosage of the medication that was administered.
Control | 0..1
Type | string
MedicationAdministration.dosage.site
Definition | A coded specification of the anatomic site where the medication first entered the body. For example, "left arm".
Control | 0..1
Terminology Binding | SNOMED CT Anatomical Structure for Administration Site Codes (Example)
Type | CodeableConcept
Comments | If the use case requires attributes from the BodySite resource (e.g. to identify and track separately) then use the standard extension body-site-instance. May be a summary code, or a reference to a very precise definition of the location, or both.
MedicationAdministration.dosage.route
Definition | A code specifying the route or physiological path of administration of a therapeutic agent into or onto the patient. For example, topical, intravenous, etc.
Control | 0..1
Terminology Binding | SNOMED CT Route Codes (Example)

Type | CodeableConcept
MedicationAdministration.dosage.method
Definition | A coded value indicating the method by which the medication is intended to be or was introduced into or on the body. This attribute will most often NOT be populated. It is most commonly used for injections. For example, Slow Push, Deep IV.
Control | 0..1
Terminology Binding | SNOMED CT Administration Method Codes (Example)
Type | CodeableConcept
Comments | One of the reasons this attribute is not used often, is that the method is often pre-coordinated with the route and/or form of administration. This means the codes used in route or form may pre-coordinate the method in the route code or the form code. The implementation decision about what coding system to use for route or form code will determine how frequently the method code will be populated e.g. if route or form code pre-coordinate method code, then this attribute will not be populated often; if there is no pre-coordination then method code may be used frequently.
MedicationAdministration.dosage.dose
Definition | The amount of the medication given at one administration event. Use this value when the administration is essentially an instantaneous event such as a swallowing a tablet or giving an injection.
Control | 0..1
Type | SimpleQuantity
Comments | If the administration is not instantaneous (rate is present), this can be specified to convey the total amount administered over period of time of a single administration.
MedicationAdministration.dosage.rate[x]
Definition | Identifies the speed with which the medication was or will be introduced into the patient. Typically the rate for an infusion e.g. 100 ml per 1 hour or 100 ml/hr. May also be expressed as a rate per unit of time e.g. 500 ml per 2 hours. Other examples: 200 mcg/min or 200 mcg/1 minute; 1 liter/8 hours.
Control | 0..1
Type | Ratio|SimpleQuantity
[x] Note | See Choice of Data Types for further information about how to use [x]
Comments | If the rate changes over time, and you want to capture this in MedicationAdministration, then each change should be captured as a distinct MedicationAdministration, with a specific MedicationAdministration.dosage.rate, and the date time when the rate change occurred. Typically, the MedicationAdministration.dosage.rate element is not used to convey an average rate.
MedicationAdministration.eventHistory
Definition | A summary of the events of interest that have occurred, such as when the administration was verified.
Control | 0..*
Type | Reference(Provenance)
Comments | This may not include provenances for all versions of the request – only those deemed “relevant” or important. This SHALL NOT include the Provenance associated with this current version of the resource. (If that provenance is deemed to be a “relevant” change, it will need to be added as part of a later update. Until then, it can be queried directly as the Provenance that points to this version using _revinclude All Provenances should have some historical version of this Request as their subject.

11.2.8 Resource MedicationAdministration - Mappings

URL: https://hl7.org/fhir/STU3/medicationadministration-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.2.8 Resource MedicationAdministration - Mappings

Pharmacy Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Mappings:

RIM Mapping

HL7 v2 Mapping

W3C PROV

Workflow Mapping

Mappings for the medicationadministration resource.

11.2.8.1 Mappings for RIM Mapping (http://hl7.org/v3)

MedicationAdministration | SubstanceAdministration
identifier | .id
definition | .outboundRelationship[typeCode=DEFN].target[classCode=unspecified]
partOf | .outboundRelationship[typeCode=COMP]/target[classCode=SBADM or PROC,moodCode=EVN]
status | .statusCode
category | .inboundRelationship[typeCode=COMP].source[classCode=OBS, moodCode=EVN, code="type of medication usage"].value
medication[x] | .participation[typeCode=CSM].role[classCode=ADMM]
subject | .participation[typeCode=SBJ].role[classCode=PAT]
context | .inboundRelationship[typeCode=COMP].source[classCode=ENC, moodCode=EVN, code="type of encounter or episode"]
supportingInformation | .outboundRelationship[typeCode=PERT].target[A_SupportingClinicalStatement CMET minimal with many different choices of classCodes(ORG, ENC, PROC, SPLY, SBADM, OBS) and each of the act class codes draws from one or more of the following moodCodes (EVN, DEF, INT PRMS, RQO, PRP, APT, ARQ, GOL)]
effective[x] | .effectiveTime
performer | .participation[typeCode=PRF]
actor | .role
onBehalfOf | .role.scopingOrganization

notGiven | .actionNegationInd
reasonNotGiven | .outboundRelationship[typeCode=RSON]/target[classCode=OBS,moodCode=EVN, code="reason not given"].value
reasonCode | .reasonCode
reasonReference | .outboundRelationship[typeCode=RSON]/target[classCode=OBS,moodCode=EVN, code="reason for use ASSERTION"].value
prescription | InFullfillmentOf->SubstanceAdministration
device | device->Access OR device->AssignedDevice
note | .inboundRelationship[typeCode=SUBJ]/source[classCode=OBS,moodCode=EVN,code="annotation"].value
dosage | component->SubstanceAdministrationEvent
text | .text
site | .approachSiteCode
route | .routeCode
method | .methodCode
dose | .doseQuantity
rate[x] | .rateQuantity
eventHistory | .inboundRelationship(typeCode=SUBJ].source[classCode=CACT, moodCode=EVN]

11.2.8.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

MedicationAdministration
identifier
definition
partOf
status | RXA-20-Completion Status
category
medication[x] | RXA-5-Administered Code
subject | PID-3-Patient ID List
context | PV1-19-Visit Number
supportingInformation
effective[x] | RXA-3-Date/Time Start of Administration / RXA-4-Date/Time End of Administration
performer | RXA-10-Administering Provider / PRT-5-Participation Person: PRT-4-Participation='AP' (RXA-10 is deprecated)
actor
onBehalfOf
notGiven | RXA-20-Completion Status='NA'
reasonNotGiven | RXA-9-Administration Notes / RXA-18 Substance/Treatment Refusal Reason
reasonCode | RXE-27 Give Indication
reasonReference
prescription | ORC-2-Placer Order Number / ORC-3-Filler Order Number
device | PRT-10 Participation Device / RXR-3 Administrative Device
note
dosage
text
site | RXR-2 Administration Site / RXR-6 Administration Site Modifier
route | RXR-1-Route
method | RXR-4-Administration Method
dose | RXA-6 Administered Amount / RXA-7 Administered Units
rate[x] | RXA-12-Administered Per (Time Unit)
eventHistory

11.2.8.3 Mappings for W3C PROV (http://www.w3.org/ns/prov)

The provenance resource is based on known practices in the HL7 implementation space, particularly those found in the v2 EVN segment, the v3 ControlAct Wrapper, the CDA header, and IHE ATNA. The conceptual model underlying the design is the W3C provenance Specification . Though the content and format of the resource is designed to meet specific requirements for FHIR, all the parts of the resource are formally mapped to the PROV-O specification, and FHIR resources can be transformed to their W3C PROV equivalent.

MedicationAdministration
identifier
definition
partOf
status
category
medication[x]
subject
context
supportingInformation
effective[x]
performer
actor
onBehalfOf
notGiven
reasonNotGiven
reasonCode
reasonReference | no mapping

prescription
device
note
dosage
text
site
route
method
dose
rate[x]
eventHistory

11.2.8.4 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

MedicationAdministration | ..Event
identifier | …identifier
definition | …definition
partOf | …part of
status | …status
category
medication[x] | …code
subject | …subject
context | …context
supportingInformation
effective[x] | …occurrence[x]
performer | …performer
actor | ….actor
onBehalfOf | ….onBehalfOf
notGiven | …notDone
reasonNotGiven | …notDoneReason
reasonCode | …reasonGivenCodeableConcept
reasonReference | …reasonReference
prescription | …basedOn
device
note | …note
dosage
text
site
route
method
dose
rate[x]
eventHistory

11.2.9 Resource MedicationAdministration - Design Notes

URL: https://hl7.org/fhir/STU3/medicationadministration-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.2.9 Resource MedicationAdministration - Design Notes

to do

11.2.10 Resource MedicationAdministration - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/medicationadministration-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.2.10 Resource MedicationAdministration - Extensions & Profiles

Pharmacy Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the MedicationAdministration resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

11.2.10.1 Profiles

No Profiles defined for this resource

11.2.10.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

condition-dueTo | dueTo | for Condition HL7 Extensions
condition-occurredFollowing | occurredFollowing | for Condition HL7 Extensions
openEHR-administration | administration | for openEHR Archetype Profile
procedure-causedBy | causedBy | for Procedure HL7 Extensions

11.2.10.3 Search Extensions

No Search Extensions defined for this resource

11.2.11 MedicationAdministration HL7 Extensions

URL: https://hl7.org/fhir/STU3/medicationadministration-extensions.html

Defines common extensions used with or related to the MedicationAdministration resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.2.11 MedicationAdministration HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

11.2.11.1 Content

11.2.12 Resource MedicationAdministration - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/medicationadministration-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.2.12 Resource MedicationAdministration - Detailed Descriptions

Pharmacy Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

R2 : R3 Converson maps for MedicationAdministration.

Functional status for this map: 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

11.2.12.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/MedicationAdministration2to3" = "R2 to R3 Conversions for
    MedicationAdministration"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/MedicationAdministration" alias MedicationAdministrationR2 as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/MedicationAdministration" alias RatioR2 as source
uses "http://hl7.org/fhir/StructureDefinition/MedicationAdministration" alias MedicationAdministration as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  MedicationAdministration extends DomainResource
  input src : MedicationAdministrationR2 as source
  input tgt : MedicationAdministration as target

  "MedicationAdministration-identifier" : for src.identifier make tgt.identifier
  "MedicationAdministration-status" : for src.status make tgt.status
  "MedicationAdministration-patient" : for src.patient make tgt.subject
  "MedicationAdministration-practitioner" : for src.practitioner as v make tgt.performer as vd, vd.actor = v
  "MedicationAdministration-encounter" : for src.encounter make tgt.context
  "MedicationAdministration-prescription" : for src.prescription make tgt.prescription
  "MedicationAdministration-notGiven" : for src.wasNotGiven make tgt.notGiven
  "MedicationAdministration-reasonNotGiven" : for src.reasonNotGiven make tgt.reasonNotGiven
  "MedicationAdministration-effectiveTime" : for src.effectiveTime make tgt.effective
  "MedicationAdministration-medication" : for src.medication make tgt.medication
  "MedicationAdministration-device" : for src.device make tgt.device
  "MedicationAdministration-note" : for src.note make tgt.note
  "MedicationAdministration-dosage" : for src.dosage as vs0 make tgt.dosage as vt0 then mAdDosage(vs0, vt0)
endgroup

group mAdDosage extends BackboneElement
  input src : MedicationAdministrationR2 as source
  input tgt : MedicationAdministration as target

  "MedicationAdministration.dosage-text" : for src.text make tgt.text
  "MedicationAdministration.dosage-site" : for src.site make tgt.site
  "MedicationAdministration.dosage-route" : for src.route make tgt.route
  "MedicationAdministration.dosage-method" : for src.method make tgt.method
  "MedicationAdministration.dosage-dose" : for src.quantity make tgt.dose
  "MedicationAdministration.dosage-rate-Ratio" : for src.rate where $this is Ratio make tgt.rate
  // Range is not valid; should not use it, and cannot convert to R3
endgroup
```

11.2.12.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/MedicationAdministration3to2" = "R3 to R2 Conversion for MedicationAdministration"

uses "http://hl7.org/fhir/StructureDefinition/MedicationAdministration" alias MedicationAdministration as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/MedicationAdministration" alias MedicationAdministrationR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  MedicationAdministration extends DomainResource
  input src : MedicationAdministration as source
  input tgt : MedicationAdministrationR2 as target

  "MedicationAdministration-identifier" : for src.identifier make tgt.identifier
  "MedicationAdministration-status" : for src.status make tgt.status
  "MedicationAdministration-patient" : for src.subject make tgt.patient
  "MedicationAdministration-performer" : for src.performer as vs0 then mAdPerformer(vs0, tgt)
  "MedicationAdministration-encounter" : for src.context make tgt.encounter
  "MedicationAdministration-prescription" : for src.prescription make tgt.prescription
  "MedicationAdministration-notGiven" : for src.notGiven make tgt.wasNotGiven
  "MedicationAdministration-reasonNotGiven" : for src.reasonNotGiven make tgt.reasonNotGiven
  "MedicationAdministration-effectiveTime" : for src.effective make tgt.effectiveTime
  "MedicationAdministration-medication" : for src.medication make tgt.medication
  "MedicationAdministration-device" : for src.device make tgt.device
  "MedicationAdministration-note" : for src.note only_one make tgt.note
  "MedicationAdministration-dosage" : for src.dosage as vs0 make tgt.dosage as vt0 then mAdDosage(vs0, vt0)
endgroup

group mAdPerformer extends BackboneElement
  input src as source
  input tgt as target

  "MedicationAdministration-performer-actor" : for src.actor make tgt.practitioner
endgroup

group mAdDosage extends BackboneElement
  input src : MedicationAdministrationR2 as source
  input tgt : MedicationAdministration as target

  "MedicationAdministration.dosage-text" : for src.text make tgt.text
  "MedicationAdministration.dosage-site" : for src.site make tgt.site
  "MedicationAdministration.dosage-route" : for src.route make tgt.route
  "MedicationAdministration.dosage-method" : for src.method make tgt.method
  "MedicationAdministration.dosage-dose" : for src.dose make tgt.quantity
  "MedicationAdministration.dosage-rate-Ratio" : for src.rate where $this is Ratio make tgt.rate
endgroup
```

11.2.12.3 R3 Validation Errors

11.3 Resource MedicationDispense - Content

URL: https://hl7.org/fhir/STU3/medicationdispense.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Pharmacy Work Group | Maturity Level: 2 | Trial Use | Compartments: Patient, Practitioner

Indicates that a medication product is to be or has been dispensed for a named person/patient. This includes a description of the medication product (supply) provided and the instructions for administering the medication. The medication dispense is the result of a pharmacy system responding to a medication order.

11.3.1 Scope and Usage

This resource covers the supply of medications to a patient. Examples include dispensing and pick-up from an out-patient or community pharmacy, dispensing patient-specific medications from in-patient pharmacy to ward, as well as issuing a single dose from ward stock to a patient for consumption. The medication dispense is the result of a pharmacy system responding to a medication order.

MedicationDispense is an event resource from a FHIR workflow perspective - see Workflow Event

11.3.2 Boundaries and Relationships

The Medication domain includes a number of related resources

MedicationRequest | An order for both supply of the medication and the instructions for administration of the medicine to a patient.

MedicationDispense | Provision of a supply of a medication with the intention that it is subsequently consumed by a patient (usually in response to a prescription).

MedicationAdministration | When a patient actually consumes a medicine, or it is otherwise administered to them

MedicationStatement | This is a record of a medication being taken by a patient or that a medication has been given to a patient, where the record is the result of a report from the patient or another clinician. A medication statement is not a part of the prescribe->dispense->administer sequence, but is a report that such a sequence (or at least a part of it) did take place, resulting in a belief that the patient has received a particular medication.

This resource does not deal with the supply or transfer of non-medication related items to a patient.

11.3.3 Background and Context

The supply and the associated administration instructions may not exactly follow the original order (prescription), either because some details were left for completion at this point in the process or because the dispenser exercised their clinical judgment to make some appropriate modification.

This resource is referenced by ChargeItem and MedicationStatement

11.3.4 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
MedicationDispense | I | | DomainResource | Dispensing a medication to a named patient + whenHandedOver cannot be before whenPrepared
Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | External identifier
partOf | | 0..* | Reference(Procedure) | Event that dispense is part of
status | ?!? | 0..1 | code | preparation | in-progress | on-hold | completed | entered-in-error | stopped MedicationDispenseStatus (Required)
category | | 0..1 | CodeableConcept | Type of medication dispense MedicationDispenseCategory (Preferred)
medication[x] | ? | 1..1 | | What medication was supplied SNOMED CT Medication Codes (Example)
medicationCodeableConcept | | | CodeableConcept
medicationReference | | | Reference(Medication)
subject | ? | 0..1 | Reference(Patient | Group) | Who the dispense is for
context | | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter / Episode associated with event
supportingInformation | | 0..* | Reference(Any) | Information that supports the dispensing of the medication
performer | | 0..* | BackboneElement | Who performed event
actor | | 1..1 | Reference(Practitioner | Organization | Patient | Device | RelatedPerson) | Individual who was performing
onBehalfOf | | 0..1 | Reference(Organization) | Organization organization was acting for
authorizingPrescription | | 0..* | Reference(MedicationRequest) | Medication order that authorizes the dispense
type | | 0..1 | CodeableConcept | Trial fill, partial fill, emergency fill, etc. ActPharmacySupplyType (Example)
quantity | | 0..1 | SimpleQuantity | Amount dispensed
daysSupply | | 0..1 | SimpleQuantity | Amount of medication expressed as a timing amount
whenPrepared | ? | 0..1 | dateTime | When product was packaged and reviewed
whenHandedOver | | 0..1 | dateTime | When product was given out
destination | | 0..1 | Reference(Location) | Where the medication was sent
receiver | | 0..* | Reference(Patient | Practitioner) | Who collected the medication
note | | 0..* | Annotation | Information about the dispense
dosageInstruction | | 0..* | Dosage | How the medication is to be used by the patient or administered by the caregiver
substitution | | 0..1 | BackboneElement | Whether a substitution was performed on the dispense
wasSubstituted | | 1..1 | boolean | Whether a substitution was or was not performed on the dispense
type | | 0..1 | CodeableConcept | Code signifying whether a different drug was dispensed from what was prescribed
ActSubstanceAdminSubstitutionCode (Example)
reason | | 0..* | CodeableConcept | Why was substitution made SubstanceAdminSubstitutionReason (Example)
responsibleParty | | 0..* | Reference(Practitioner) | Who is responsible for the substitution
detectedIssue | | 0..* | Reference(DetectedIssue) | Clinical issue with action
notDone | | 0..1 | boolean | Whether the dispense was or was not performed
notDoneReason[x] | | 0..1 | | Why a dispense was not performed
notDoneReasonCodeableConcept | | | CodeableConcept
notDoneReasonReference | | | Reference(DetectedIssue)
eventHistory | | 0..* | Reference(Provenance) | A list of releveant lifecycle events
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicationDispense xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External identifier --></identifier>
 <partOf><!-- 0..* Reference(Procedure) Event that dispense is part of --></partOf>
 <status value="[code]"/><!-- 0..1 preparation | in-progress | on-hold | completed | entered-in-error | stopped -->
 <category><!-- 0..1 CodeableConcept Type of medication dispense --></category>
 <medication[x]><!-- 1..1 CodeableConcept|Reference(Medication) What medication was supplied --></medication[x]>
 <subject><!-- 0..1 Reference(Patient|Group) Who the dispense is for --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter / Episode associated with event --></context>
 <supportingInformation><!-- 0..* Reference(Any) Information that supports the dispensing of the medication
    --></supportingInformation>
 <performer>  <!-- 0..* Who performed event -->
  <actor><!-- 1..1 Reference(Practitioner|Organization|Patient|Device|
    RelatedPerson) Individual who was performing --></actor>
```

```
  <onBehalfOf><!-- 0..1 Reference(Organization) Organization organization was acting for --></onBehalfOf>
 </performer>
 <authorizingPrescription><!-- 0..* Reference(MedicationRequest) Medication order that authorizes the dispense
    --></authorizingPrescription>
 <type><!-- 0..1 CodeableConcept Trial fill, partial fill, emergency fill, etc. --></type>
 <quantity><!-- 0..1 Quantity(SimpleQuantity) Amount dispensed --></quantity>
 <daysSupply><!-- 0..1 Quantity(SimpleQuantity) Amount of medication expressed as a timing amount --></daysSupply>
 <whenPrepared value="[dateTime]"/><!-- 0..1 When product was packaged and reviewed -->
 <whenHandedOver value="[dateTime]"/><!-- 0..1 When product was given out -->
 <destination><!-- 0..1 Reference(Location) Where the medication was sent --></destination>
 <receiver><!-- 0..* Reference(Patient|Practitioner) Who collected the medication --></receiver>
 <note><!-- 0..* Annotation Information about the dispense --></note>
 <dosageInstruction><!-- 0..* Dosage How the medication is to be used by the patient or administered by the caregiver
    --></dosageInstruction>
 <substitution>  <!-- 0..1 Whether a substitution was performed on the dispense -->
  <wasSubstituted value="[boolean]"/><!-- 1..1 Whether a substitution was or was not performed on the dispense -->
  <type><!-- 0..1 CodeableConcept Code signifying whether a different drug was dispensed from what was prescribed
    --></type>
  <reason><!-- 0..* CodeableConcept Why was substitution made --></reason>
  <responsibleParty><!-- 0..* Reference(Practitioner) Who is responsible for the substitution --></responsibleParty>
 </substitution>
 <detectedIssue><!-- 0..* Reference(DetectedIssue) Clinical issue with action --></detectedIssue>
 <notDone value="[boolean]"/><!-- 0..1 Whether the dispense was or was not performed -->
 <notDoneReason[x]><!-- 0..1 CodeableConcept|Reference(DetectedIssue) Why a dispense was not performed
    --></notDoneReason[x]>
 <eventHistory><!-- 0..* Reference(Provenance) A list of releveant lifecycle events --></eventHistory>
</MedicationDispense>
```

JSON Template

```
{
  "resourceType" : "MedicationDispense",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External identifier
  "partOf" : [{ Reference(Procedure) }], // Event that dispense is part of
  "status" : "<code>", // preparation | in-progress | on-hold | completed | entered-in-error | stopped
  "category" : { CodeableConcept }, // Type of medication dispense
  // medication[x]: What medication was supplied. One of these 2:
  "medicationCodeableConcept" : { CodeableConcept },
  "medicationReference" : { Reference(Medication) },
  "subject" : { Reference(Patient|Group) }, // Who the dispense is for
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter / Episode associated with event
  "supportingInformation" : [{ Reference(Any) }], // Information that supports the dispensing of the medication
  "performer" : [{ // Who performed event
    "actor" : { Reference(Practitioner|Organization|Patient|Device|
    RelatedPerson) }, // R!  Individual who was performing
    "onBehalfOf" : { Reference(Organization) } // Organization organization was acting for
  }],
  "authorizingPrescription" : [{ Reference(MedicationRequest) }], // Medication order that authorizes the dispense
  "type" : { CodeableConcept }, // Trial fill, partial fill, emergency fill, etc.
  "quantity" : { Quantity(SimpleQuantity) }, // Amount dispensed
  "daysSupply" : { Quantity(SimpleQuantity) }, // Amount of medication expressed as a timing amount
  "whenPrepared" : "<dateTime>", // When product was packaged and reviewed
  "whenHandedOver" : "<dateTime>", // When product was given out
  "destination" : { Reference(Location) }, // Where the medication was sent
  "receiver" : [{ Reference(Patient|Practitioner) }], // Who collected the medication
  "note" : [{ Annotation }], // Information about the dispense
  "dosageInstruction" : [{ Dosage }], // How the medication is to be used by the patient or administered by the
    caregiver
  "substitution" : { // Whether a substitution was performed on the dispense
    "wasSubstituted" : <boolean>, // R!  Whether a substitution was or was not performed on the dispense
    "type" : { CodeableConcept }, // Code signifying whether a different drug was dispensed from what was prescribed
    "reason" : [{ CodeableConcept }], // Why was substitution made
    "responsibleParty" : [{ Reference(Practitioner) }] // Who is responsible for the substitution
  },
  "detectedIssue" : [{ Reference(DetectedIssue) }], // Clinical issue with action
  "notDone" : <boolean>, // Whether the dispense was or was not performed
  // notDoneReason[x]: Why a dispense was not performed. One of these 2:
  "notDoneReasonCodeableConcept" : { CodeableConcept },
  "notDoneReasonReference" : { Reference(DetectedIssue) },
  "eventHistory" : [{ Reference(Provenance) }] // A list of releveant lifecycle events
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicationDispense;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicationDispense.identifier [ Identifier ], ... ; # 0..* External identifier
  fhir:MedicationDispense.partOf [ Reference(Procedure) ], ... ; # 0..* Event that dispense is part of
  fhir:MedicationDispense.status [ code ]; # 0..1 preparation | in-progress | on-hold | completed | entered-in-error |
    stopped
  fhir:MedicationDispense.category [ CodeableConcept ]; # 0..1 Type of medication dispense
  # MedicationDispense.medication[x] : 1..1 What medication was supplied. One of these 2
    fhir:MedicationDispense.medicationCodeableConcept [ CodeableConcept ]
```

```
    fhir:MedicationDispense.medicationReference [ Reference(Medication) ]
  fhir:MedicationDispense.subject [ Reference(Patient|Group) ]; # 0..1 Who the dispense is for
  fhir:MedicationDispense.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter / Episode associated with
    event
  fhir:MedicationDispense.supportingInformation [ Reference(Any) ], ... ; # 0..* Information that supports the
    dispensing of the medication
  fhir:MedicationDispense.performer [ # 0..* Who performed event
    fhir:MedicationDispense.performer.actor [ Reference(Practitioner|Organization|Patient|Device|RelatedPerson) ]; #
    1..1 Individual who was performing
    fhir:MedicationDispense.performer.onBehalfOf [ Reference(Organization) ]; # 0..1 Organization organization was
    acting for
  ], ...;
  fhir:MedicationDispense.authorizingPrescription [ Reference(MedicationRequest) ], ... ; # 0..* Medication order that
    authorizes the dispense
  fhir:MedicationDispense.type [ CodeableConcept ]; # 0..1 Trial fill, partial fill, emergency fill, etc.
  fhir:MedicationDispense.quantity [ Quantity(SimpleQuantity) ]; # 0..1 Amount dispensed
  fhir:MedicationDispense.daysSupply [ Quantity(SimpleQuantity) ]; # 0..1 Amount of medication expressed as a timing
    amount
  fhir:MedicationDispense.whenPrepared [ dateTime ]; # 0..1 When product was packaged and reviewed
  fhir:MedicationDispense.whenHandedOver [ dateTime ]; # 0..1 When product was given out
  fhir:MedicationDispense.destination [ Reference(Location) ]; # 0..1 Where the medication was sent
  fhir:MedicationDispense.receiver [ Reference(Patient|Practitioner) ], ... ; # 0..* Who collected the medication
  fhir:MedicationDispense.note [ Annotation ], ... ; # 0..* Information about the dispense
  fhir:MedicationDispense.dosageInstruction [ Dosage ], ... ; # 0..* How the medication is to be used by the patient or
    administered by the caregiver
  fhir:MedicationDispense.substitution [ # 0..1 Whether a substitution was performed on the dispense
    fhir:MedicationDispense.substitution.wasSubstituted [ boolean ]; # 1..1 Whether a substitution was or was not
    performed on the dispense
    fhir:MedicationDispense.substitution.type [ CodeableConcept ]; # 0..1 Code signifying whether a different drug was
    dispensed from what was prescribed
    fhir:MedicationDispense.substitution.reason [ CodeableConcept ], ... ; # 0..* Why was substitution made
    fhir:MedicationDispense.substitution.responsibleParty [ Reference(Practitioner) ], ... ; # 0..* Who is responsible
    for the substitution
  ];
  fhir:MedicationDispense.detectedIssue [ Reference(DetectedIssue) ], ... ; # 0..* Clinical issue with action
  fhir:MedicationDispense.notDone [ boolean ]; # 0..1 Whether the dispense was or was not performed
  # MedicationDispense.notDoneReason[x] : 0..1 Why a dispense was not performed. One of these 2
    fhir:MedicationDispense.notDoneReasonCodeableConcept [ CodeableConcept ]
    fhir:MedicationDispense.notDoneReasonReference [ Reference(DetectedIssue) ]
  fhir:MedicationDispense.eventHistory [ Reference(Provenance) ], ... ; # 0..* A list of releveant lifecycle events
]
```

Changes since DSTU2

MedicationDispense
MedicationDispense.identifier | Max Cardinality changed from 1 to *
MedicationDispense.partOf | Added Element
MedicationDispense.category | Added Element
MedicationDispense.subject | Renamed from patient to subject Add Reference(Group)
MedicationDispense.context | Added Element
MedicationDispense.supportingInformation | Added Element
MedicationDispense.performer | Added Element
MedicationDispense.performer.actor | Added Element
MedicationDispense.performer.onBehalfOf | Added Element
MedicationDispense.authorizingPrescription | Type changed from Reference(MedicationOrder) to Reference(MedicationRequest)
MedicationDispense.note | Max Cardinality changed from 1 to * Type changed from string to Annotation
MedicationDispense.dosageInstruction | Type changed from BackboneElement to Dosage
MedicationDispense.substitution.wasSubstituted | Added Element
MedicationDispense.substitution.type | Min Cardinality changed from 1 to 0
MedicationDispense.detectedIssue | Added Element
MedicationDispense.notDone | Added Element
MedicationDispense.notDoneReason[x] | Added Element
MedicationDispense.eventHistory | Added Element
MedicationDispense.dispenser | deleted
MedicationDispense.dosageInstruction.text | deleted
MedicationDispense.dosageInstruction.additionalInstructions | deleted
MedicationDispense.dosageInstruction.timing | deleted
MedicationDispense.dosageInstruction.asNeeded[x] | deleted
MedicationDispense.dosageInstruction.site[x] | deleted
MedicationDispense.dosageInstruction.route | deleted
MedicationDispense.dosageInstruction.method | deleted
MedicationDispense.dosageInstruction.dose[x] | deleted
MedicationDispense.dosageInstruction.rate[x] | deleted
MedicationDispense.dosageInstruction.maxDosePerPeriod | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 7 tests that all execute ok. 7 fail round-trip testing and 1 r3 resources are invalid (2 errors).).

Structure

Name | Flags | Card. | Type | Description & Constraints
MedicationDispense | I | | DomainResource | Dispensing a medication to a named patient + whenHandedOver cannot be before whenPrepared
Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | External identifier
partOf | | 0..* | Reference(Procedure) | Event that dispense is part of
status | ?!? | 0..1 | code | preparation | in-progress | on-hold | completed | entered-in-error | stopped MedicationDispenseStatus (Required)
category | | 0..1 | CodeableConcept | Type of medication dispense MedicationDispenseCategory (Preferred)
medication[x] | ? | 1..1 | | What medication was supplied SNOMED CT Medication Codes (Example)
medicationCodeableConcept | | | CodeableConcept
medicationReference | | | Reference(Medication)
subject | ? | 0..1 | Reference(Patient | Group) | Who the dispense is for
context | | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter / Episode associated with event
supportingInformation | | 0..* | Reference(Any) | Information that supports the dispensing of the medication
performer | | 0..* | BackboneElement | Who performed event
actor | | 1..1 | Reference(Practitioner | Organization | Patient | Device | RelatedPerson) | Individual who was performing
onBehalfOf | | 0..1 | Reference(Organization) | Organization organization was acting for
authorizingPrescription | | 0..* | Reference(MedicationRequest) | Medication order that authorizes the dispense
type | | 0..1 | CodeableConcept | Trial fill, partial fill, emergency fill, etc. ActPharmacySupplyType (Example)
quantity | | 0..1 | SimpleQuantity | Amount dispensed
daysSupply | | 0..1 | SimpleQuantity | Amount of medication expressed as a timing amount
whenPrepared | ? | 0..1 | dateTime | When product was packaged and reviewed
whenHandedOver | | 0..1 | dateTime | When product was given out
destination | | 0..1 | Reference(Location) | Where the medication was sent
receiver | | 0..* | Reference(Patient | Practitioner) | Who collected the medication
note | | 0..* | Annotation | Information about the dispense
dosageInstruction | | 0..* | Dosage | How the medication is to be used by the patient or administered by the caregiver
substitution | | 0..1 | BackboneElement | Whether a substitution was performed on the dispense
wasSubstituted | | 1..1 | boolean | Whether a substitution was or was not performed on the dispense
type | | 0..1 | CodeableConcept | Code signifying whether a different drug was dispensed from what was prescribed
ActSubstanceAdminSubstitutionCode (Example)
reason | | 0..* | CodeableConcept | Why was substitution made SubstanceAdminSubstitutionReason (Example)
responsibleParty | | 0..* | Reference(Practitioner) | Who is responsible for the substitution
detectedIssue | | 0..* | Reference(DetectedIssue) | Clinical issue with action
notDone | | 0..1 | boolean | Whether the dispense was or was not performed
notDoneReason[x] | | 0..1 | | Why a dispense was not performed
notDoneReasonCodeableConcept | | | CodeableConcept
notDoneReasonReference | | | Reference(DetectedIssue)
eventHistory | | 0..* | Reference(Provenance) | A list of releveant lifecycle events
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicationDispense xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External identifier --></identifier>
 <partOf><!-- 0..* Reference(Procedure) Event that dispense is part of --></partOf>
 <status value="[code]"/><!-- 0..1 preparation | in-progress | on-hold | completed | entered-in-error | stopped -->
 <category><!-- 0..1 CodeableConcept Type of medication dispense --></category>
 <medication[x]><!-- 1..1 CodeableConcept|Reference(Medication) What medication was supplied --></medication[x]>
 <subject><!-- 0..1 Reference(Patient|Group) Who the dispense is for --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter / Episode associated with event --></context>
 <supportingInformation><!-- 0..* Reference(Any) Information that supports the dispensing of the medication
    --></supportingInformation>
 <performer>  <!-- 0..* Who performed event -->
  <actor><!-- 1..1 Reference(Practitioner|Organization|Patient|Device|
    RelatedPerson) Individual who was performing --></actor>
  <onBehalfOf><!-- 0..1 Reference(Organization) Organization organization was acting for --></onBehalfOf>
 </performer>
 <authorizingPrescription><!-- 0..* Reference(MedicationRequest) Medication order that authorizes the dispense
    --></authorizingPrescription>
 <type><!-- 0..1 CodeableConcept Trial fill, partial fill, emergency fill, etc. --></type>
 <quantity><!-- 0..1 Quantity(SimpleQuantity) Amount dispensed --></quantity>
 <daysSupply><!-- 0..1 Quantity(SimpleQuantity) Amount of medication expressed as a timing amount --></daysSupply>
 <whenPrepared value="[dateTime]"/><!-- 0..1 When product was packaged and reviewed -->
 <whenHandedOver value="[dateTime]"/><!-- 0..1 When product was given out -->
 <destination><!-- 0..1 Reference(Location) Where the medication was sent --></destination>
 <receiver><!-- 0..* Reference(Patient|Practitioner) Who collected the medication --></receiver>
 <note><!-- 0..* Annotation Information about the dispense --></note>
 <dosageInstruction><!-- 0..* Dosage How the medication is to be used by the patient or administered by the caregiver
    --></dosageInstruction>
 <substitution>  <!-- 0..1 Whether a substitution was performed on the dispense -->
  <wasSubstituted value="[boolean]"/><!-- 1..1 Whether a substitution was or was not performed on the dispense -->
  <type><!-- 0..1 CodeableConcept Code signifying whether a different drug was dispensed from what was prescribed
    --></type>
  <reason><!-- 0..* CodeableConcept Why was substitution made --></reason>
  <responsibleParty><!-- 0..* Reference(Practitioner) Who is responsible for the substitution --></responsibleParty>
 </substitution>
 <detectedIssue><!-- 0..* Reference(DetectedIssue) Clinical issue with action --></detectedIssue>
```

```
 <notDone value="[boolean]"/><!-- 0..1 Whether the dispense was or was not performed -->
 <notDoneReason[x]><!-- 0..1 CodeableConcept|Reference(DetectedIssue) Why a dispense was not performed
    --></notDoneReason[x]>
 <eventHistory><!-- 0..* Reference(Provenance) A list of releveant lifecycle events --></eventHistory>
</MedicationDispense>
```

JSON Template

```
{
  "resourceType" : "MedicationDispense",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External identifier
  "partOf" : [{ Reference(Procedure) }], // Event that dispense is part of
  "status" : "<code>", // preparation | in-progress | on-hold | completed | entered-in-error | stopped
  "category" : { CodeableConcept }, // Type of medication dispense
  // medication[x]: What medication was supplied. One of these 2:
  "medicationCodeableConcept" : { CodeableConcept },
  "medicationReference" : { Reference(Medication) },
  "subject" : { Reference(Patient|Group) }, // Who the dispense is for
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter / Episode associated with event
  "supportingInformation" : [{ Reference(Any) }], // Information that supports the dispensing of the medication
  "performer" : [{ // Who performed event
    "actor" : { Reference(Practitioner|Organization|Patient|Device|
    RelatedPerson) }, // R!  Individual who was performing
    "onBehalfOf" : { Reference(Organization) } // Organization organization was acting for
  }],
  "authorizingPrescription" : [{ Reference(MedicationRequest) }], // Medication order that authorizes the dispense
  "type" : { CodeableConcept }, // Trial fill, partial fill, emergency fill, etc.
  "quantity" : { Quantity(SimpleQuantity) }, // Amount dispensed
  "daysSupply" : { Quantity(SimpleQuantity) }, // Amount of medication expressed as a timing amount
  "whenPrepared" : "<dateTime>", // When product was packaged and reviewed
  "whenHandedOver" : "<dateTime>", // When product was given out
  "destination" : { Reference(Location) }, // Where the medication was sent
  "receiver" : [{ Reference(Patient|Practitioner) }], // Who collected the medication
  "note" : [{ Annotation }], // Information about the dispense
  "dosageInstruction" : [{ Dosage }], // How the medication is to be used by the patient or administered by the
    caregiver
  "substitution" : { // Whether a substitution was performed on the dispense
    "wasSubstituted" : <boolean>, // R!  Whether a substitution was or was not performed on the dispense
    "type" : { CodeableConcept }, // Code signifying whether a different drug was dispensed from what was prescribed
    "reason" : [{ CodeableConcept }], // Why was substitution made
    "responsibleParty" : [{ Reference(Practitioner) }] // Who is responsible for the substitution
  },
  "detectedIssue" : [{ Reference(DetectedIssue) }], // Clinical issue with action
  "notDone" : <boolean>, // Whether the dispense was or was not performed
  // notDoneReason[x]: Why a dispense was not performed. One of these 2:
  "notDoneReasonCodeableConcept" : { CodeableConcept },
  "notDoneReasonReference" : { Reference(DetectedIssue) },
  "eventHistory" : [{ Reference(Provenance) }] // A list of releveant lifecycle events
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicationDispense;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicationDispense.identifier [ Identifier ], ... ; # 0..* External identifier
  fhir:MedicationDispense.partOf [ Reference(Procedure) ], ... ; # 0..* Event that dispense is part of
  fhir:MedicationDispense.status [ code ]; # 0..1 preparation | in-progress | on-hold | completed | entered-in-error |
    stopped
  fhir:MedicationDispense.category [ CodeableConcept ]; # 0..1 Type of medication dispense
  # MedicationDispense.medication[x] : 1..1 What medication was supplied. One of these 2
    fhir:MedicationDispense.medicationCodeableConcept [ CodeableConcept ]
    fhir:MedicationDispense.medicationReference [ Reference(Medication) ]
  fhir:MedicationDispense.subject [ Reference(Patient|Group) ]; # 0..1 Who the dispense is for
  fhir:MedicationDispense.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter / Episode associated with
    event
  fhir:MedicationDispense.supportingInformation [ Reference(Any) ], ... ; # 0..* Information that supports the
    dispensing of the medication
  fhir:MedicationDispense.performer [ # 0..* Who performed event
    fhir:MedicationDispense.performer.actor [ Reference(Practitioner|Organization|Patient|Device|RelatedPerson) ]; #
    1..1 Individual who was performing
    fhir:MedicationDispense.performer.onBehalfOf [ Reference(Organization) ]; # 0..1 Organization organization was
    acting for
  ], ...;
  fhir:MedicationDispense.authorizingPrescription [ Reference(MedicationRequest) ], ... ; # 0..* Medication order that
    authorizes the dispense
  fhir:MedicationDispense.type [ CodeableConcept ]; # 0..1 Trial fill, partial fill, emergency fill, etc.
  fhir:MedicationDispense.quantity [ Quantity(SimpleQuantity) ]; # 0..1 Amount dispensed
  fhir:MedicationDispense.daysSupply [ Quantity(SimpleQuantity) ]; # 0..1 Amount of medication expressed as a timing
    amount
  fhir:MedicationDispense.whenPrepared [ dateTime ]; # 0..1 When product was packaged and reviewed
  fhir:MedicationDispense.whenHandedOver [ dateTime ]; # 0..1 When product was given out
  fhir:MedicationDispense.destination [ Reference(Location) ]; # 0..1 Where the medication was sent
  fhir:MedicationDispense.receiver [ Reference(Patient|Practitioner) ], ... ; # 0..* Who collected the medication
```

```
  fhir:MedicationDispense.note [ Annotation ], ... ; # 0..* Information about the dispense
  fhir:MedicationDispense.dosageInstruction [ Dosage ], ... ; # 0..* How the medication is to be used by the patient or
    administered by the caregiver
  fhir:MedicationDispense.substitution [ # 0..1 Whether a substitution was performed on the dispense
    fhir:MedicationDispense.substitution.wasSubstituted [ boolean ]; # 1..1 Whether a substitution was or was not
    performed on the dispense
    fhir:MedicationDispense.substitution.type [ CodeableConcept ]; # 0..1 Code signifying whether a different drug was
    dispensed from what was prescribed
    fhir:MedicationDispense.substitution.reason [ CodeableConcept ], ... ; # 0..* Why was substitution made
    fhir:MedicationDispense.substitution.responsibleParty [ Reference(Practitioner) ], ... ; # 0..* Who is responsible
    for the substitution
  ];
  fhir:MedicationDispense.detectedIssue [ Reference(DetectedIssue) ], ... ; # 0..* Clinical issue with action
  fhir:MedicationDispense.notDone [ boolean ]; # 0..1 Whether the dispense was or was not performed
  # MedicationDispense.notDoneReason[x] : 0..1 Why a dispense was not performed. One of these 2
    fhir:MedicationDispense.notDoneReasonCodeableConcept [ CodeableConcept ]
    fhir:MedicationDispense.notDoneReasonReference [ Reference(DetectedIssue) ]
  fhir:MedicationDispense.eventHistory [ Reference(Provenance) ], ... ; # 0..* A list of releveant lifecycle events
]
```

Changes since DSTU2

MedicationDispense
MedicationDispense.identifier | Max Cardinality changed from 1 to *
MedicationDispense.partOf | Added Element
MedicationDispense.category | Added Element
MedicationDispense.subject | Renamed from patient to subject Add Reference(Group)
MedicationDispense.context | Added Element
MedicationDispense.supportingInformation | Added Element
MedicationDispense.performer | Added Element
MedicationDispense.performer.actor | Added Element
MedicationDispense.performer.onBehalfOf | Added Element
MedicationDispense.authorizingPrescription | Type changed from Reference(MedicationOrder) to Reference(MedicationRequest)
MedicationDispense.note | Max Cardinality changed from 1 to * Type changed from string to Annotation
MedicationDispense.dosageInstruction | Type changed from BackboneElement to Dosage
MedicationDispense.substitution.wasSubstituted | Added Element
MedicationDispense.substitution.type | Min Cardinality changed from 1 to 0
MedicationDispense.detectedIssue | Added Element
MedicationDispense.notDone | Added Element
MedicationDispense.notDoneReason[x] | Added Element
MedicationDispense.eventHistory | Added Element
MedicationDispense.dispenser | deleted
MedicationDispense.dosageInstruction.text | deleted
MedicationDispense.dosageInstruction.additionalInstructions | deleted
MedicationDispense.dosageInstruction.timing | deleted
MedicationDispense.dosageInstruction.asNeeded[x] | deleted
MedicationDispense.dosageInstruction.site[x] | deleted
MedicationDispense.dosageInstruction.route | deleted
MedicationDispense.dosageInstruction.method | deleted
MedicationDispense.dosageInstruction.dose[x] | deleted
MedicationDispense.dosageInstruction.rate[x] | deleted
MedicationDispense.dosageInstruction.maxDosePerPeriod | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 7 tests that all execute ok. 7 fail round-trip testing and 1 r3 resources are invalid (2 errors).).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

11.3.4.1 Terminology Bindings

Path | Definition | Type | Reference
MedicationDispense.status | A coded concept specifying the state of the dispense event. | Required | MedicationDispenseStatus
MedicationDispense.category | A code describing where the dispensed medication is expected to be consumed or administered | Preferred | MedicationDispenseCategory
MedicationDispense.medication[x] | A coded concept identifying which substance or product can be dispensed | Example | SNOMED CT Medication Codes
MedicationDispense.type | Indicates the type of dispensing event that is performed. For example, Trial Fill, Completion of Trial, Partial Fill, Emergency Fill, Samples, etc. | Example | ActPharmacySupplyType
MedicationDispense.substitution.type | A coded concept describing whether a different medicinal product may be dispensed other than the product as specified exactly in the prescription | Example | ActSubstanceAdminSubstitutionCode
MedicationDispense.substitution.reason | A coded concept describing the reason that a different medication should (or should not) be substituted from what was prescribed | Example | SubstanceAdminSubstitutionReason

11.3.4.2 Constraints

* mdd-1: whenHandedOver cannot be before whenPrepared (expression : whenHandedOver.empty() or whenPrepared.empty() or whenHandedOver >= whenPrepared)

11.3.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
code | token | Return dispenses of this medicine code | MedicationDispense.medication.as(CodeableConcept) | 4 Resources
context | reference | Returns dispenses with a specific context (episode or episode of care) | MedicationDispense.context (EpisodeOfCare, Encounter)
destination | reference | Return dispenses that should be sent to a specific destination | MedicationDispense.destination (Location)
identifier | token | Return dispenses with this external identifier | MedicationDispense.identifier | 3 Resources
medication | reference | Return dispenses of this medicine resource | MedicationDispense.medication.as(Reference) (Medication) | 3 Resources
patient | reference | The identity of a patient to list dispenses for | MedicationDispense.subject (Patient) | 3 Resources
performer | reference | Return dispenses performed by a specific individual | MedicationDispense.performer.actor (Practitioner, Organization, Device, Patient, RelatedPerson)
prescription | reference | The identity of a prescription to list dispenses from | MedicationDispense.authorizingPrescription (MedicationRequest) | 1 Resources
receiver | reference | The identity of a receiver to list dispenses for | MedicationDispense.receiver (Practitioner, Patient)
responsibleparty | reference | Return dispenses with the specified responsible party | MedicationDispense.substitution.responsibleParty (Practitioner)
status | token | Return dispenses with a specified dispense status | MedicationDispense.status | 3 Resources
subject | reference | The identity of a patient to list dispenses for | MedicationDispense.subject (Group, Patient)
type | token | Return dispenses of a specific type | MedicationDispense.type
whenhandedover | date | Returns dispenses handed over on this date | MedicationDispense.whenHandedOver
whenprepared | date | Returns dispenses prepared on this date | MedicationDispense.whenPrepared

11.3.6 Resource MedicationDispense - Examples

URL: https://hl7.org/fhir/STU3/medicationdispense-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.3.6 Resource MedicationDispense - Examples

Pharmacy Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner

ORAL DOSAGE FORMS

The following examples are MedicationDispense resources for Oral Dosage Forms that reference Medication Resources for a manufactured product:

* Dispense for Oral Chemotherapy - Capecitabine (RxNorm code)

* Dispense for Oxycodone (RxNorm code) - PRN dose with dose range, coded additional instructions and PRN coded pre-condition

* Dispense for Prednisone (RxNorm code) Tapering Dose - entered-in-error with multiple Dosage lines that include coded additionalInstructions and multiple EventHistory elements

* Dispense for Percocet (NDC Code) - in-progress with additionalInstructions, and as needed dosage with condition

* Dispense for Over the Counter medication - Tylenol PM (NDC code) - on hold with as needed dosage with a condition, a dose Range and an Event History

* Dispense for Alprazolam (Rx Norm code) - in progress

* Dispense for Azithromycin (NDC code) - completed with multiple dosage lines

* Dispense for Warfarin 5mg (NDC Code) where the product is dispensed as multiple strengths based on prescribe dosage

* Dispense for Warfarin 2mg (NDC Code) where the product is dispensed as multiple strengths based on prescribe dosage

The following examples are MedicationDispenses resources for Oral Dosage Forms that include a Medication Code

* Dispense for Levothyroxine (NDC code) in-progress with authorizing prescription

* Dispense for Azithromycin (RxNorm Code) - in-progress with coded additionalInstructions

* Dispense for Dilantin Oral Suspension (NDC code) - completed

* Dispense for Busulfan (Myleran) - completed

* Dispense for Vicodin - in progress - PRN with pre-condition

INJECTABLE DOSAGE FORMS

The following examples of MedicationDispense resources for Injectable Dosage Forms that reference a Medication Resource for a manufactured product:

* Dispense for IV Antibiotic - in progress - Vancomycin IV (NDC Code) - in-progress

* Dispense for Subcutaneous injection with multiple dosage lines - Insulin - Novolog (NDC Code) - active - includes Destination, Receiver, note. Dosage includes additionalInstructions as text

* Dispense for Intravenous Antibiotic with a Dosage Rate - Piperacillin (NDC code) - completed with a coded site and route of administration and a rate expressed as a ratio

* Dispense for Intravenous Chemotherapy (inpatient order) - Alemtuzumab (RxNorm Code) - completed with multiple dosageInstruction lines

* Dispense for Lorazepam (Rx norm code) - completed with rate range

* ispense for Brentuximab Vedotin (NDC code) - IV Chemo - stopped

The following examples of MedicationDispense resources for Injectable Dosage Forms that include a Medication Code:

* Dispense for Insulin (Lantus) (NDC code) - in-progress with SC dosage

The following examples of MedicationDispense resources for Injectable Dosage Forms for Injectable Dosage Forms where the medication is a compounded (aka extemporaneous or magistral) product:

* Dispense for Potassium Chloride Admixture - stopped - with eventHistory and rate range for dosage

* Dispense for a TPN solution - completed

MISCELLAENOUS DOSAGE FORMS

The following examples of MedicationDispense resources including miscellaneous dosage forms:

* Dispense for Timoptic Opthalmic Drops - (NDC code) - completed

* Dispense for Nystatin drops (Snomed Code) - entered-in-error

* Dispense for Proventil (RxNorm Code) - completed - with additionalInstructions as text

* Dispense for Nystatin Topical Ointment (RxNorm code) - completed

* Dispense for Nasonex Inhaler (RxNorm Code) - in progress with additionalInstructions as text

* Dispense for Compounded ointment (HC 1%, SA 5% in White Petrolatum) - completed with additionalInstructions as text and a coded site in dosageInstructions

* Dispense for Fentanyl Patch (NDC code) - in progress

* Dispense for Paracetamol Suppositories - completed

Example List:

Dispense for Oral Chemotherapy - Capecitabine (RxNorm code) | meddisp008 | XML | JSON | Turtle
Dispense for IV Antibiotic - in progress - Vancomycin IV (NDC Code) - in-progress | meddisp0301 | XML | JSON | Turtle
Dispense for Subcutaneous injection with multiple dosage lines - Insulin - Novolog (NDC Code) - active - includes Destination, Receiver, note. Dosage includes additionalInstructions as text | meddisp0302 | XML | JSON | Turtle
Dispense for Oxycodone (RxNorm code) - PRN dose with dose range, coded additional instructions and PRN coded pre-condition | meddisp0303 | XML | JSON | Turtle
Dispense for Intravenous Antibiotic with a Dosage Rate - Piperacillin (NDC code) - completed with a coded site and route of administration and a rate expressed as a ratio | meddisp0304 | XML | JSON | Turtle
Dispense for Prednisone (RxNorm code) Tapering Dose - entered-in-error with multiple Dosage lines that include coded additionalInstructions and multiple EventHistory elements | meddisp0305 | XML | JSON | Turtle
Dispense for Intravenous Chemotherapy (inpatient order) - Alemtuzumab (RxNorm Code) - completed with multiple dosageInstruction lines | meddisp0306 | XML | JSON | Turtle
Dispense for Busulfan (Myleran) - completed | meddisp0307 | XML | JSON | Turtle
Dispense for Timoptic Opthalmic Drops - (NDC code) - completed | meddisp0308 | XML | JSON | Turtle
Dispense for Nystatin drops (Snomed Code) - entered-in-error | meddisp0309 | XML | JSON | Turtle
Dispense for Percocet (NDC Code) - in-progress with additionalInstructions, and as needed dosage with condition | meddisp0310 | XML | JSON | Turtle
Dispense for Paracetamol Suppositories - completed | meddisp0311 | XML | JSON | Turtle
Dispense for Over the Counter medication - Tylenol PM (NDC code) - on hold with as needed dosage with a condition, a dose Range and an Event History | meddisp0312 | XML | JSON | Turtle
Dispense for Potassium Chloride Admixture - stopped - with eventHistory and rate range for dosage | meddisp0313 | XML | JSON | Turtle
Dispense for Lorazepam (Rx norm code) - completed with rate range | meddisp0314 | XML | JSON | Turtle
Dispense for Alprazolam (Rx Norm code) - in progress | meddisp0315 | XML | JSON | Turtle
Dispense for Insulin (Lantus) (NDC code) - in-progress with SC dosage | meddisp0316 | XML | JSON | Turtle
Dispense for Brentuximab Vedotin (NDC code) - IV Chemo - stopped | meddisp0317 | XML | JSON | Turtle
Dispense for Levothyroxine (NDC code) in-progress with authorizing prescription | meddisp0318 | XML | JSON | Turtle
Dispense for Azithromycin (NDC code) - completed with multiple dosage lines | meddisp0319 | XML | JSON | Turtle
Dispense for a TPN solution - completed | meddisp0320 | XML | JSON | Turtle
Dispense for Vicodin - in progress - PRN with pre-condition | meddisp0321 | XML | JSON | Turtle
Dispense for Dilantin Oral Suspension (NDC code) - completed | meddisp0322 | XML | JSON | Turtle
Dispense for Nystatin Topical Ointment (RxNorm code) - completed | meddisp0324 | XML | JSON | Turtle
Dispense for Fentanyl Patch (NDC code) - in progress | meddisp0325 | XML | JSON | Turtle
Dispense for Azithromycin (RxNorm Code) - in-progress with coded additionalInstructions | meddisp0326 | XML | JSON | Turtle
Dispense for Proventil (RxNorm Code) - completed - with additionalInstructions as text | meddisp0327 | XML | JSON | Turtle
Dispense for Nasonex Inhaler (RxNorm Code) - in progress with additionalInstructions as text | meddisp0328 | XML | JSON | Turtle
Dispense for compounded ointment - completed with additionalInstructions as text and a coded site in dosageInstructions | meddisp0329 | XML | JSON | Turtle
Dispense for Warfarin 5mg (NDC Code) where the product is dispensed as multiple strengths based on prescribe dosage | meddisp0330 | XML | JSON | Turtle
Dispense for Warfarin 2mg (NDC Code) where the product is dispensed as multiple strengths based on prescribe dosage | meddisp0331 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.3.7 Resource MedicationDispense - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/medicationdispense-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.3.7 Resource MedicationDispense - Detailed Descriptions

Pharmacy Work Group | Maturity Level: 2 | Trial Use | Compartments: Patient, Practitioner

Detailed Descriptions for the elements in the MedicationDispense resource.

MedicationDispense
Definition | Indicates that a medication product is to be or has been dispensed for a named person/patient. This includes a description of the medication product (supply) provided and the instructions for administering the medication. The medication dispense is the result of a pharmacy system responding to a medication order.
Control | 1..1
Comments | The WG will be updating the MedicationDispense resource to adjust each affected resource to align with the workflow pattern (see workflow.html).
Invariants | Defined on this element mdd-1: whenHandedOver cannot be before whenPrepared (expression : whenHandedOver.empty() or whenPrepared.empty() or whenHandedOver >= whenPrepared, xpath: not(exists(f:whenHandedOver/@value)) or not(exists(f:whenPrepared/@value)) or (f:whenHandedOver/@value >= f:whenPrepared/@value))
MedicationDispense.identifier
Definition | Identifier assigned by the dispensing facility - this is an identifier assigned outside FHIR.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
MedicationDispense.partOf
Definition | The procedure that the dispense is done because of.
Control | 0..*
Type | Reference(Procedure)
MedicationDispense.status
Definition | A code specifying the state of the set of dispense events.
Control | 0..1
Terminology Binding | MedicationDispenseStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
MedicationDispense.category
Definition | Indicates type of medication dispense and where the medication is expected to be consumed or administered.
Control | 0..1
Terminology Binding | MedicationDispenseCategory (Preferred)
Type | CodeableConcept
MedicationDispense.medication[x]
Definition | Identifies the medication being administered. This is either a link to a resource representing the details of the medication or a simple attribute carrying a code that identifies the medication from a known list of medications.
Control | 1..1
Terminology Binding | SNOMED CT Medication Codes (Example)
Type | CodeableConcept|Reference(Medication)
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | If only a code is specified, then it needs to be a code for a specific product. If more information is required, then the use of the medication resource is recommended. For example if you require form or lot number, then you must reference the Medication resource. .
MedicationDispense.subject
Definition | A link to a resource representing the person or the group to whom the medication will be given.
Control | 0..1
Type | Reference(Patient | Group)
Summary | true
Comments | SubstanceAdministration->subject->Patient.
MedicationDispense.context
Definition | The encounter or episode of care that establishes the context for this event.
Control | 0..1
Type | Reference(Encounter | EpisodeOfCare)
MedicationDispense.supportingInformation
Definition | Additional information that supports the medication being dispensed.
Control | 0..*
Type | Reference(Any)
MedicationDispense.performer
Definition | Indicates who or what performed the event. It should be assumed that the performer is the dispenser of the medication.
Control | 0..*
Comments | The performer will be reviewed as part of STU4.
MedicationDispense.performer.actor
Definition | The device, practitioner, etc. who performed the action. It should be assumed that the actor is the dispenser of the medication.

Control | 1..1
Type | Reference(Practitioner | Organization | Patient | Device | RelatedPerson)
MedicationDispense.performer.onBehalfOf
Definition | The organization the device or practitioner was acting on behalf of.
Control | 0..1
Type | Reference(Organization)
MedicationDispense.authorizingPrescription
Definition | Indicates the medication order that is being dispensed against.
Control | 0..*
Type | Reference(MedicationRequest)
Comments | Maps to basedOn in Event logical model.
MedicationDispense.type
Definition | Indicates the type of dispensing event that is performed. For example, Trial Fill, Completion of Trial, Partial Fill, Emergency Fill, Samples, etc.
Control | 0..1
Terminology Binding | ActPharmacySupplyType (Example)
Type | CodeableConcept
MedicationDispense.quantity
Definition | The amount of medication that has been dispensed. Includes unit of measure.
Control | 0..1
Type | SimpleQuantity
MedicationDispense.daysSupply
Definition | The amount of medication expressed as a timing amount.
Control | 0..1
Type | SimpleQuantity
MedicationDispense.whenPrepared
Definition | The time when the dispensed product was packaged and reviewed.
Control | 0..1
Type | dateTime
Summary | true
MedicationDispense.whenHandedOver
Definition | The time the dispensed product was provided to the patient or their representative.
Control | 0..1
Type | dateTime
MedicationDispense.destination
Definition | Identification of the facility/location where the medication was shipped to, as part of the dispense event.
Control | 0..1
Type | Reference(Location)
MedicationDispense.receiver
Definition | Identifies the person who picked up the medication. This will usually be a patient or their caregiver, but some cases exist where it can be a healthcare professional.
Control | 0..*
Type | Reference(Patient | Practitioner)
MedicationDispense.note
Definition | Extra information about the dispense that could not be conveyed in the other attributes.
Control | 0..*
Type | Annotation
MedicationDispense.dosageInstruction
Definition | Indicates how the medication is to be used by the patient.
Control | 0..*
Type | Dosage
Comments | When the dose or rate is intended to change over the entire administration period (e.g. Tapering dose prescriptions), multiple instances of dosage instructions will need to be supplied to convey the different doses/rates. The pharmacist reviews the medication order prior to dispense and updates the dosageInstruction based on the actual product being dispensed.
MedicationDispense.substitution
Definition | Indicates whether or not substitution was made as part of the dispense. In some cases substitution will be expected but does not happen, in other cases substitution is not expected but does happen. This block explains what substitution did or did not happen and why. If nothing is specified, substitution was not done.
Control | 0..1
MedicationDispense.substitution.wasSubstituted
Definition | True if the dispenser dispensed a different drug or product from what was prescribed.
Control | 1..1
Type | boolean
MedicationDispense.substitution.type
Definition | A code signifying whether a different drug was dispensed from what was prescribed.
Control | 0..1
Terminology Binding | ActSubstanceAdminSubstitutionCode (Example)
Type | CodeableConcept
MedicationDispense.substitution.reason
Definition | Indicates the reason for the substitution of (or lack of substitution) from what was prescribed.
Control | 0..*
Terminology Binding | SubstanceAdminSubstitutionReason (Example)
Type | CodeableConcept

MedicationDispense.substitution.responsibleParty
Definition | The person or organization that has primary responsibility for the substitution.
Control | 0..*
Type | Reference(Practitioner)
MedicationDispense.detectedIssue
Definition | Indicates an actual or potential clinical issue with or between one or more active or proposed clinical actions for a patient; e.g. Drug-drug interaction, duplicate therapy, dosage alert etc.
Control | 0..*
Type | Reference(DetectedIssue)
Alternate Names | Contraindication; Drug Utilization Review (DUR); Alert
MedicationDispense.notDone
Definition | True if the dispense was not performed for some reason.
Control | 0..1
Type | boolean
MedicationDispense.notDoneReason[x]
Definition | Indicates the reason why a dispense was not performed.
Control | 0..1
Type | CodeableConcept|Reference(DetectedIssue)
[x] Note | See Choice of Data Types for further information about how to use [x]
MedicationDispense.eventHistory
Definition | A summary of the events of interest that have occurred, such as when the dispense was verified.
Control | 0..*
Type | Reference(Provenance)
Comments | This may not include provenances for all versions of the request – only those deemed “relevant” or important. This SHALL NOT include the Provenance associated with this current version of the resource. (If that provenance is deemed to be a “relevant” change, it will need to be added as part of a later update. Until then, it can be queried directly as the Provenance that points to this version using _revinclude All Provenances should have some historical version of this Request as their subject.).

11.3.8 Resource MedicationDispense - Mappings

URL: https://hl7.org/fhir/STU3/medicationdispense-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.3.8 Resource MedicationDispense - Mappings

Pharmacy Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner

Mappings:

RIM Mapping

HL7 v2 Mapping

Workflow Mapping

Mappings for the medicationdispense resource.

11.3.8.1 Mappings for RIM Mapping (http://hl7.org/v3)

MedicationDispense | Supply[moodCode=EVN]
identifier | .id
partOf | .outboundRelationship[typeCode=COMP]/target[classCode=PROC,moodCode=EVN]
status | .statusCode
category | .inboundRelationship[typeCode=COMP].source[classCode=OBS, moodCode=EVN, code="type of medication dispense"].value
medication[x] | .participation[typeCode=PRD].role
subject | .participation[typeCode=SBJ].role[classCode=PAT]
context | .inboundRelationship[typeCode=COMP].source[classCode=ENC, moodCode=EVN, code="type of encounter or episode"]
supportingInformation | .outboundRelationship[typeCode=PERT].target[A_SupportingClinicalStatement CMET minimal with many different choices of classCodes(ORG, ENC, PROC, SPLY, SBADM, OBS) and each of the act class codes draws from one or more of the following moodCodes (EVN, DEF, INT PRMS, RQO, PRP, APT, ARQ, GOL)]
performer | .participation[typeCode=PRF]
actor | .role
onBehalfOf | .role.scopingOrganization
authorizingPrescription | .outboundRelationship[typeCode=FLFS].target[moodCode=RQO]
type | .code
quantity | .quantity
daysSupply | effectiveUseTime
whenPrepared | .effectiveTime[xmi:type=IVL_TS].low
whenHandedOver | .effectiveTime[xmi:type=IVL_TS].high
destination | .participation[typeCode=DST].role
receiver | .participation[typeCode=RCV].role
note | .inboundRelationship[typeCode=SUBJ]/source[classCode=ACT,moodCode=EVN]
dosageInstruction | .outboundRelationship[typeCode=COMP].target[classCode=SBADM, moodCode=INT]
substitution | .outboundRelationship[typeCode=COMP].target[classCode=SUBST, moodCode=EVN]
wasSubstituted | .actionNegationInd
type | .code

reason | .reasonCode
responsibleParty | .participation[typeCode=PFM].role[classCode=ASSIGN]
detectedIssue | .inboundRelationship[typeCode=SUBJ]/source[classCode=ALRT,moodCode=EVN].value
notDone | .actionNegationInd
notDoneReason[x] | .outboundRelationship[typeCode=RSON]/target[classCode=OBS,moodCode=EVN, code="not done reason"].value
eventHistory | .inboundRelationship(typeCode=SUBJ].source[classCode=CACT, moodCode=EVN]

11.3.8.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

MedicationDispense
identifier
partOf
status
category
medication[x] | RXD-2-Dispense/Give Code
subject | PID-3-Patient ID List
context
supportingInformation
performer
actor
onBehalfOf
authorizingPrescription | ORC-2 Placer Order Number
type | RXD-33-Dispense Type
quantity | RXD-4-Actual Dispense Amount / RXD-5.1-Actual Dispense Units.code / RXD-5.3-Actual Dispense Units.name of coding system
daysSupply | TQ1.6 Timing/Quantity Segment Service Duration. Prior to v2.5, ORC.7.3 Common Order Segment / Quantity/Timing / Duration component. This is a formatted string, first character for the time unit (e.g., D=days), followed by the value. For example, “D14” represents “14 days supply” From v2.5 on, TQ1.6 Timing/Quantity Segment / Service Duration. This is a CQ data type (<Quantity (NM)> ^ <Units (CWE)>), thus for days supply, assuming the unit of measure is “days”, the numeric value is TQ1.6.1 (…|14^ For backwards compatibility, ORC.7 was permitted through v2.6. Both forms (field and segment) may be present in v2.5, v2.5.1, and v2.6
whenPrepared | RXD-3-Date/Time Dispensed
whenHandedOver | RXD-3-Date/Time Dispensed
destination | RXD-13 Dispense-to Location; RXD-30 Dispense to Pharmacy; RXD-31 Dispense to Pharmacy Address
receiver
note | RXD-9 Dispense Notes
dosageInstruction
substitution
wasSubstituted
type | RXO-9-Allow Substitutions / RXE-9-Substitution Status
reason
responsibleParty
detectedIssue
notDone
notDoneReason[x]
eventHistory

11.3.8.3 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

MedicationDispense | ..Event
identifier | …identifier
partOf | …part of
status | …status
category
medication[x] | …code
subject | …subject
context | …context
supportingInformation
performer | …performer
actor | ….actor
onBehalfOf | ….onBehalfOf
authorizingPrescription | …basedOn
type
quantity
daysSupply
whenPrepared
whenHandedOver | …occurrenceDateTime
destination
receiver
note | …note
dosageInstruction
substitution
wasSubstituted
type
reason
responsibleParty

detectedIssue
notDone | …notDone
notDoneReason[x] | …notDoneReason
eventHistory

11.3.9 Resource MedicationDispense - Design Notes

URL: https://hl7.org/fhir/STU3/medicationdispense-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.3.9 Resource MedicationDispense - Design Notes

to do

11.3.10 Resource MedicationDispense - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/medicationdispense-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.3.10 Resource MedicationDispense - Extensions & Profiles

Pharmacy Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the MedicationDispense resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

11.3.10.1 Profiles

No Profiles defined for this resource

11.3.10.2 Extensions

medicationdispense-validityPeriod | validityPeriod | for MedicationDispense HL7 Extensions
pharmacy-core-doseType | doseType | for International Realm Pharmacy Extensions Profile
pharmacy-core-infuseOver | infuseOver | for International Realm Pharmacy Extensions Profile
pharmacy-core-maxDeliveryRate | maxDeliveryRate | for International Realm Pharmacy Extensions Profile
pharmacy-core-maxDeliveryVolume | maxDeliveryVolume | for International Realm Pharmacy Extensions Profile
pharmacy-core-minDosePerPeriod | minDosePerPeriod | for International Realm Pharmacy Extensions Profile
pharmacy-core-rateGoal | rateGoal | for International Realm Pharmacy Extensions Profile
pharmacy-core-rateIncrement | rateIncrement | for International Realm Pharmacy Extensions Profile
pharmacy-core-rateIncrementInterval | rateIncrementInterval | for International Realm Pharmacy Extensions Profile
pharmacy-core-refillsRemaining | refillsRemaining | for International Realm Pharmacy Extensions Profile

Extensions that reference this resource:

No Extensions refer to this resource

11.3.10.3 Search Extensions

No Search Extensions defined for this resource

11.3.11 MedicationDispense HL7 Extensions

URL: https://hl7.org/fhir/STU3/medicationdispense-extensions.html

Defines common extensions used with or related to the MedicationDispense resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.3.11 MedicationDispense HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

11.3.11.1 Content

Extensions:
medicationdispense-validityPeriod | validityPeriod : The time interval when the medication must be administered. The start and end times must be expressed as dateTimes, not times relative to the dispensing time (e.g. use within 24 hours).

11.3.12 Resource MedicationDispense - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/medicationdispense-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.3.12 Resource MedicationDispense - Detailed Descriptions

Pharmacy Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner

R2 : R3 Converson maps for MedicationDispense.

Functional status for this map: 7 tests that all execute ok. 7 fail round-trip testing and 1 r3 resources are invalid (2 errors). (see documentation)

11.3.12.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/MedicationDispense2to3" = "R2 to R3 Conversions for MedicationDispense"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/MedicationDispense" alias MedicationDispenseR2 as source
uses "http://hl7.org/fhir/StructureDefinition/MedicationDispense" alias MedicationDispense as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  MedicationDispense extends DomainResource
  input src : MedicationDispenseR2 as source
  input tgt : MedicationDispense as target

  "MedicationDispense-identifier" : for src.identifier make tgt.identifier
  "MedicationDispense-status" : for src.status make tgt.status
  "MedicationDispense-patient" : for src.patient make tgt.subject
  "MedicationDispense-dispenser" : for src.dispenser as v make tgt.performer as d, d.actor as v
  "MedicationDispense-authorizingPrescription" : for src.authorizingPrescription make tgt.authorizingPrescription
  "MedicationDispense-type" : for src.type make tgt.type
  "MedicationDispense-quantity" : for src.quantity make tgt.quantity
  "MedicationDispense-daysSupply" : for src.daysSupply make tgt.daysSupply
  "MedicationDispense-medication" : for src.medication make tgt.medication
  "MedicationDispense-whenPrepared" : for src.whenPrepared make tgt.whenPrepared
  "MedicationDispense-whenHandedOver" : for src.whenHandedOver make tgt.whenHandedOver
  "MedicationDispense-destination" : for src.destination make tgt.destination
  "MedicationDispense-receiver" : for src.receiver make tgt.receiver
  "MedicationDispense-note" : for src.note make tgt.note

  "MedicationDispense-dosageInstruction" : for src.dosageInstruction as vs0 make tgt.dosageInstruction as vt0 then
    mddDosage(vs0, vt0)
  "MedicationDispense-substitution" : for src.substitution as vs0 make tgt.substitution as vt0 then mddSubstitution(vs0,
     vt0)
endgroup

group mddSubstitution extends BackboneElement
  input src as source
  input tgt as target

  "MedicationDispense.substitution-was" : for src as su make tgt.wasSubstituted = true
  "MedicationDispense.substitution-type" : for src.type make tgt.type
  "MedicationDispense.substitution-reason" : for src.reason make tgt.reason
  "MedicationDispense.substitution-responsibleParty" : for src.responsibleParty make tgt.responsibleParty
endgroup

group mddDosage extends Element // ? BackboneElement
  input src as source
  input tgt as target

  "MedicationDispense.dosage-text" : for src.text make tgt.text
  "MedicationDispense.dosage-additionalInstructions" : for src.additionalInstructions make tgt.additionalInstruction
  "MedicationDispense.dosage-timing" : for src.timing make tgt.timing
  "MedicationDispense.dosage-asNeeded" : for src.asNeeded make tgt.asNeeded
  "MedicationDispense.dosage-site" : for src.site where $this is CodeableConcept make tgt.site
  "MedicationDispense.dosage-route" : for src.route make tgt.route
  "MedicationDispense.dosage-method" : for src.method make tgt.method
  "MedicationDispense.dosage-dose" : for src.dose make tgt.dose
  "MedicationDispense.dosage-rate" : for src.rate make tgt.rate
  "MedicationDispense.dosage-maxDosePerPeriod" : for src.maxDosePerPeriod make tgt.maxDosePerPeriod
endgroup
```

11.3.12.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/MedicationDispense3to2" = "R3 to R2 Conversion for MedicationDispense"

uses "http://hl7.org/fhir/StructureDefinition/MedicationDispense" alias MedicationDispense as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/MedicationDispense" alias MedicationDispenseR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  MedicationDispense extends DomainResource
  input src : MedicationDispense as source
  input tgt : MedicationDispenseR2 as target

  "MedicationDispense-identifier" : for src.identifier make tgt.identifier
  "MedicationDispense-status" : for src.status make tgt.status
  "MedicationDispense-patient" : for src.subject make tgt.patient
  "MedicationDispense-dispenser" : for src.performer as vs0 then mddPerformer(vs0, tgt)
  "MedicationDispense-authorizingPrescription" : for src.authorizingPrescription make tgt.authorizingPrescription
  "MedicationDispense-type" : for src.type make tgt.type
  "MedicationDispense-quantity" : for src.quantity make tgt.quantity
  "MedicationDispense-daysSupply" : for src.daysSupply make tgt.daysSupply
```

```
  "MedicationDispense-medication" : for src.medication make tgt.medication
  "MedicationDispense-whenPrepared" : for src.whenPrepared make tgt.whenPrepared
  "MedicationDispense-whenHandedOver" : for src.whenHandedOver make tgt.whenHandedOver
  "MedicationDispense-destination" : for src.destination make tgt.destination
  "MedicationDispense-receiver" : for src.receiver make tgt.receiver
  "MedicationDispense-note" : for src.note only_one make tgt.note

  "MedicationDispense-dosageInstruction" : for src.dosageInstruction as vs0 make tgt.dosageInstruction as vt0 then
    mddDosage(vs0, vt0)
  "MedicationDispense-substitution" : for src.substitution as vs0 make tgt.substitution as vt0 then mddSubstitution(vs0,
     vt0)
endgroup

group mddPerformer extends BackboneElement
  input src as source
  input tgt as target

  "MedicationDispense.performer-actor" : for src.actor make tgt.dispenser
endgroup

group mddSubstitution extends BackboneElement
  input src as source
  input tgt as target

  "MedicationDispense.substitution-type" : for src.type make tgt.type
  "MedicationDispense.substitution-reason" : for src.reason make tgt.reason
  "MedicationDispense.substitution-responsibleParty" : for src.responsibleParty make tgt.responsibleParty
endgroup

group mddDosage extends Element // ? BackboneElement
  input src as source
  input tgt as target

  "MedicationDispense.dosage-text" : for src.text make tgt.text
  "MedicationDispense.dosage-additionalInstructions" : for src.additionalInstruction  only_one make
    tgt.additionalInstructions
  "MedicationDispense.dosage-timing" : for src.timing make tgt.timing
  "MedicationDispense.dosage-asNeeded" : for src.asNeeded make tgt.asNeeded
  "MedicationDispense.dosage-site" : for src.site make tgt.site
  "MedicationDispense.dosage-route" : for src.route make tgt.route
  "MedicationDispense.dosage-method" : for src.method make tgt.method
  "MedicationDispense.dosage-dose" : for src.dose make tgt.dose
  "MedicationDispense.dosage-rate" : for src.rate make tgt.rate
  "MedicationDispense.dosage-maxDosePerPeriod" : for src.maxDosePerPeriod make tgt.maxDosePerPeriod
endgroup
```

11.3.12.3 R3 Validation Errors

medicationdispense-example | ERROR @ MedicationDispense.dosageInstruction.extension[2][url='http://hl7.org/fhir/StructureDefinition/pharmacy-core-infuseOver']: The Extension 'http://hl7.org/fhir/StructureDefinition/pharmacy-core-infuseOver' definition allows for the types [Duration] but found type Quantity (src = InstanceValidator) INFORMATION @ MedicationDispense.dosageInstruction.extension[2].valueQuantity: Could not verify slice for profile http://hl7.org/fhir/StructureDefinition/pharmacy-core-infuseOver (src = InstanceValidator)

11.4 Resource MedicationStatement - Content

URL: https://hl7.org/fhir/STU3/medicationstatement.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Pharmacy Work Group | Maturity Level: 3 | Trial Use | Compartments: Patient, Practitioner, RelatedPerson

A record of a medication that is being consumed by a patient. A MedicationStatement may indicate that the patient may be taking the medication now, or has taken the medication in the past or will be taking the medication in the future. The source of this information can be the patient, significant other (such as a family member or spouse), or a clinician. A common scenario where this information is captured is during the history taking process during a patient visit or stay. The medication information may come from sources such as the patient's memory, from a prescription bottle, or from a list of medications the patient, clinician or other party maintains The primary difference between a medication statement and a medication administration is that the medication administration has complete administration information and is based on actual administration information from the person who administered the medication. A medication statement is often, if not always, less specific. There is no required date/time when the medication was administered, in fact we only know that a source has reported the patient is taking this medication, where details such as time, quantity, or rate or even medication product may be incomplete or missing or less precise. As stated earlier, the medication statement information may come from the patient's memory, from a prescription bottle or from a list of medications the patient, clinician or other party maintains. Medication administration is more formal and is not missing detailed information.

11.4.1 Scope and Usage

Common usage includes:

* the recording of non-prescription and/or recreational drugs

* the recording of an intake medication list upon admission to hospital

* the summarization of a patient's "active medications" in a patient profile

A MedicationStatement may be used to record substance abuse or the use of other agents such as tobacco or alcohol. This would typically be done if these substances are intended to be inluded in clinical decision support checking (for example, interaction checking) and as part of an active medication list. If the intent is to populate social history and/or to include additional information (for example, desire to quit, amount per day, negative health effects), then it is better to record as an Observation that could then be used to populate Social History.

This resource does not produce a medication list, but it does produce individual medication statements that may be used in the List resource to construct various types of medication lists. Note that other medication lists can also be constructed from the other Pharmacy resources (e.g., MedicationRequest, MedicationAdministration).

A medication statement is not a part of the prescribe -> dispense -> administer sequence, but is a report by a patient, significant other or a clinician that one or more of the prescribe, dispense or administer actions has occurred, resulting is a belief that the patient is, has, or will be using a particular medication.

MedicationStatement is an event resource from a FHIR workflow perspective - see Workflow Event

11.4.2 Boundaries and Relationships

The MedicationStatement resource is used to record a medications or substances that the patient reports as being taken, not taking, have taken in the past or may take in the future. It can also be used to record medication use that is derived from other records such as a MedicationRequest. The statement is not used to request or order a medication, supply or device. When requesting medicaation, supplies or devices when there is a patient focus or instructions regarding their use, a MedicationRequest, SupplyRequest or DeviceRequestDeviceRequest should be used instead

The Medication domain includes a number of related resources

MedicationRequest | An order for both supply of the medication and the instructions for administration of the medicine to a patient.
MedicationDispense | Provision of a supply of a medication with the intention that it is subsequently consumed by a patient (usually in response to a prescription).
MedicationAdministration | When a patient actually consumes a medicine, or it is otherwise administered to them
MedicationStatement | This is a record of a medication being taken by a patient or that a medication has been given to a patient, where the record is the result of a report from the patient or another clinician, or derived from supporting information (for example, Claim, Observation or MedicationRequest). A medication statement is not a part of the prescribe->dispense->administer sequence, but is a report that such a sequence (or at least a part of it) did take place, resulting in a belief that the patient has received a particular medication.

This resource is distinct from MedicationRequest, MedicationDispense and MedicationAdministration. Each of those resources refer to specific events - an individual order, an individual provisioning of medication or an individual dosing. MedicationStatement is a broader assertion covering a wider timespan and is independent of specific events. The existence of resource instances of any of the preceding three types may be used to infer a medication statement. However, medication statements can also be captured on the basis of other information, including an assertion by the patient or a care-giver, the results of a lab test, etc.

This resource is referenced by AdverseEvent and Goal

11.4.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
MedicationStatement | I | | DomainResource | Record of medication being taken by a patient + Reason not taken is only permitted if Taken is No
Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | External identifier
basedOn | ? | 0..* | Reference(MedicationRequest | CarePlan | ProcedureRequest | ReferralRequest) | Fulfils plan, proposal or order
partOf | ? | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Observation) | Part of referenced event
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter / Episode associated with MedicationStatement
status | ?!? | 1..1 | code | active | completed | entered-in-error | intended | stopped | on-hold MedicationStatementStatus (Required)
category | ? | 0..1 | CodeableConcept | Type of medication usage MedicationStatementCategory (Preferred)
medication[x] | ? | 1..1 | | What medication was taken SNOMED CT Medication Codes (Example)
medicationCodeableConcept | | | CodeableConcept
medicationReference | | | Reference(Medication)
effective[x] | ? | 0..1 | | The date/time or interval when the medication was taken
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
dateAsserted | ? | 0..1 | dateTime | When the statement was asserted?

informationSource | | 0..1 | Reference(Patient | Practitioner | RelatedPerson | Organization) | Person or organization that provided the information about the taking of this medication
subject | ? | 1..1 | Reference(Patient | Group) | Who is/was taking the medication
derivedFrom | | 0..* | Reference(Any) | Additional supporting information
taken | ?!? | 1..1 | code | y | n | unk | na MedicationStatementTaken (Required)
reasonNotTaken | I | 0..* | CodeableConcept | True if asserting medication was not given SNOMED CT Drugs not taken/completed Codes (Example)
reasonCode | | 0..* | CodeableConcept | Reason for why the medication is being/was taken Condition/Problem/Diagnosis Codes (Example)
reasonReference | | 0..* | Reference(Condition | Observation) | Condition or observation that supports why the medication is being/was taken
note | | 0..* | Annotation | Further information about the statement
dosage | | 0..* | Dosage | Details of how medication is/was taken or should be taken
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicationStatement xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External identifier --></identifier>
 <basedOn><!-- 0..* Reference(MedicationRequest|CarePlan|ProcedureRequest|
   ReferralRequest) Fulfils plan, proposal or order --></basedOn>
 <partOf><!-- 0..* Reference(MedicationAdministration|MedicationDispense|
   MedicationStatement|Procedure|Observation) Part of referenced event --></partOf>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter / Episode associated with MedicationStatement
    --></context>
 <status value="[code]"/><!-- 1..1 active | completed | entered-in-error | intended | stopped | on-hold -->
 <category><!-- 0..1 CodeableConcept Type of medication usage --></category>
 <medication[x]><!-- 1..1 CodeableConcept|Reference(Medication) What medication was taken --></medication[x]>
 <effective[x]><!-- 0..1 dateTime|Period The date/time or interval when the medication was taken --></effective[x]>
 <dateAsserted value="[dateTime]"/><!-- 0..1 When the statement was asserted? -->
 <informationSource><!-- 0..1 Reference(Patient|Practitioner|RelatedPerson|
   Organization) Person or organization that provided the information about the taking of this medication
    --></informationSource>
 <subject><!-- 1..1 Reference(Patient|Group) Who is/was taking  the medication --></subject>
 <derivedFrom><!-- 0..* Reference(Any) Additional supporting information --></derivedFrom>
 <taken value="[code]"/><!-- 1..1 y | n | unk | na -->
 <reasonNotTaken><!--  0..* CodeableConcept True if asserting medication was not given --></reasonNotTaken>
 <reasonCode><!-- 0..* CodeableConcept Reason for why the medication is being/was taken --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation) Condition or observation that supports why the medication
    is being/was taken --></reasonReference>
 <note><!-- 0..* Annotation Further information about the statement --></note>
 <dosage><!-- 0..* Dosage Details of how medication is/was taken or should be taken --></dosage>
</MedicationStatement>
```

JSON Template

```
{
  "resourceType" : "MedicationStatement",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External identifier
  "basedOn" : [{ Reference(MedicationRequest|CarePlan|ProcedureRequest|
   ReferralRequest) }], // Fulfils plan, proposal or order
  "partOf" : [{ Reference(MedicationAdministration|MedicationDispense|
   MedicationStatement|Procedure|Observation) }], // Part of referenced event
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter / Episode associated with MedicationStatement
  "status" : "<code>", // R!  active | completed | entered-in-error | intended | stopped | on-hold
  "category" : { CodeableConcept }, // Type of medication usage
  // medication[x]: What medication was taken. One of these 2:
  "medicationCodeableConcept" : { CodeableConcept },
  "medicationReference" : { Reference(Medication) },
  // effective[x]: The date/time or interval when the medication was taken. One of these 2:
  "effectiveDateTime" : "<dateTime>",
  "effectivePeriod" : { Period },
  "dateAsserted" : "<dateTime>", // When the statement was asserted?
  "informationSource" : { Reference(Patient|Practitioner|RelatedPerson|
   Organization) }, // Person or organization that provided the information about the taking of this medication
  "subject" : { Reference(Patient|Group) }, // R!  Who is/was taking  the medication
  "derivedFrom" : [{ Reference(Any) }], // Additional supporting information
  "taken" : "<code>", // R!  y | n | unk | na
  "reasonNotTaken" : [{ CodeableConcept }], // C? True if asserting medication was not given
  "reasonCode" : [{ CodeableConcept }], // Reason for why the medication is being/was taken
  "reasonReference" : [{ Reference(Condition|Observation) }], // Condition or observation that supports why the
    medication is being/was taken
  "note" : [{ Annotation }], // Further information about the statement
  "dosage" : [{ Dosage }] // Details of how medication is/was taken or should be taken
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicationStatement;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicationStatement.identifier [ Identifier ], ... ; # 0..* External identifier
```

```
  fhir:MedicationStatement.basedOn [ Reference(MedicationRequest|CarePlan|ProcedureRequest|ReferralRequest) ], ... ; #
    0..* Fulfils plan, proposal or order
  fhir:MedicationStatement.partOf [ Reference(MedicationAdministration|MedicationDispense|MedicationStatement|Procedure|
  Observation) ], ... ; # 0..* Part of referenced event
  fhir:MedicationStatement.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter / Episode associated with
    MedicationStatement
  fhir:MedicationStatement.status [ code ]; # 1..1 active | completed | entered-in-error | intended | stopped | on-hold
  fhir:MedicationStatement.category [ CodeableConcept ]; # 0..1 Type of medication usage
  # MedicationStatement.medication[x] : 1..1 What medication was taken. One of these 2
    fhir:MedicationStatement.medicationCodeableConcept [ CodeableConcept ]
    fhir:MedicationStatement.medicationReference [ Reference(Medication) ]
  # MedicationStatement.effective[x] : 0..1 The date/time or interval when the medication was taken. One of these 2
    fhir:MedicationStatement.effectiveDateTime [ dateTime ]
    fhir:MedicationStatement.effectivePeriod [ Period ]
  fhir:MedicationStatement.dateAsserted [ dateTime ]; # 0..1 When the statement was asserted?
  fhir:MedicationStatement.informationSource [ Reference(Patient|Practitioner|RelatedPerson|Organization) ]; # 0..1
    Person or organization that provided the information about the taking of this medication
  fhir:MedicationStatement.subject [ Reference(Patient|Group) ]; # 1..1 Who is/was taking  the medication
  fhir:MedicationStatement.derivedFrom [ Reference(Any) ], ... ; # 0..* Additional supporting information
  fhir:MedicationStatement.taken [ code ]; # 1..1 y | n | unk | na
  fhir:MedicationStatement.reasonNotTaken [ CodeableConcept ], ... ; # 0..* True if asserting medication was not given
  fhir:MedicationStatement.reasonCode [ CodeableConcept ], ... ; # 0..* Reason for why the medication is being/was taken
  fhir:MedicationStatement.reasonReference [ Reference(Condition|Observation) ], ... ; # 0..* Condition or observation
    that supports why the medication is being/was taken
  fhir:MedicationStatement.note [ Annotation ], ... ; # 0..* Further information about the statement
  fhir:MedicationStatement.dosage [ Dosage ], ... ; # 0..* Details of how medication is/was taken or should be taken
]
```

Changes since DSTU2

MedicationStatement
MedicationStatement.basedOn | Added Element
MedicationStatement.partOf | Added Element
MedicationStatement.context | Added Element
MedicationStatement.category | Added Element
MedicationStatement.informationSource | Add Reference(Organization)
MedicationStatement.subject | Renamed from patient to subject Add Reference(Group)
MedicationStatement.derivedFrom | Renamed from supportingInformation to derivedFrom
MedicationStatement.taken | Added Element
MedicationStatement.reasonCode | Added Element
MedicationStatement.reasonReference | Added Element
MedicationStatement.note | Max Cardinality changed from 1 to * Type changed from string to Annotation
MedicationStatement.dosage | Type changed from BackboneElement to Dosage
MedicationStatement.wasNotTaken | deleted
MedicationStatement.reasonForUse[x] | deleted
MedicationStatement.dosage.text | deleted
MedicationStatement.dosage.timing | deleted
MedicationStatement.dosage.asNeeded[x] | deleted
MedicationStatement.dosage.site[x] | deleted
MedicationStatement.dosage.route | deleted
MedicationStatement.dosage.method | deleted
MedicationStatement.dosage.quantity[x] | deleted
MedicationStatement.dosage.rate[x] | deleted
MedicationStatement.dosage.maxDosePerPeriod | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 7 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (1 errors).).

Structure

Name | Flags | Card. | Type | Description & Constraints
MedicationStatement | I | | DomainResource | Record of medication being taken by a patient + Reason not taken is only permitted if Taken is No
Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | External identifier
basedOn | ? | 0..* | Reference(MedicationRequest | CarePlan | ProcedureRequest | ReferralRequest) | Fulfils plan, proposal or order
partOf | ? | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Observation) | Part of referenced
event
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter / Episode associated with MedicationStatement
status | ?!? | 1..1 | code | active | completed | entered-in-error | intended | stopped | on-hold MedicationStatementStatus (Required)
category | ? | 0..1 | CodeableConcept | Type of medication usage MedicationStatementCategory (Preferred)
medication[x] | ? | 1..1 | | What medication was taken SNOMED CT Medication Codes (Example)
medicationCodeableConcept | | | CodeableConcept
medicationReference | | | Reference(Medication)
effective[x] | ? | 0..1 | | The date/time or interval when the medication was taken
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
dateAsserted | ? | 0..1 | dateTime | When the statement was asserted?

informationSource | | 0..1 | Reference(Patient | Practitioner | RelatedPerson | Organization) | Person or organization that provided the information about the taking of this medication
subject | ? | 1..1 | Reference(Patient | Group) | Who is/was taking the medication
derivedFrom | | 0..* | Reference(Any) | Additional supporting information
taken | ?!? | 1..1 | code | y | n | unk | na MedicationStatementTaken (Required)
reasonNotTaken | I | 0..* | CodeableConcept | True if asserting medication was not given SNOMED CT Drugs not taken/completed Codes (Example)
reasonCode | | 0..* | CodeableConcept | Reason for why the medication is being/was taken Condition/Problem/Diagnosis Codes (Example)
reasonReference | | 0..* | Reference(Condition | Observation) | Condition or observation that supports why the medication is being/was taken
note | | 0..* | Annotation | Further information about the statement
dosage | | 0..* | Dosage | Details of how medication is/was taken or should be taken
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicationStatement xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External identifier --></identifier>
 <basedOn><!-- 0..* Reference(MedicationRequest|CarePlan|ProcedureRequest|
   ReferralRequest) Fulfils plan, proposal or order --></basedOn>
 <partOf><!-- 0..* Reference(MedicationAdministration|MedicationDispense|
   MedicationStatement|Procedure|Observation) Part of referenced event --></partOf>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter / Episode associated with MedicationStatement
    --></context>
 <status value="[code]"/><!-- 1..1 active | completed | entered-in-error | intended | stopped | on-hold -->
 <category><!-- 0..1 CodeableConcept Type of medication usage --></category>
 <medication[x]><!-- 1..1 CodeableConcept|Reference(Medication) What medication was taken --></medication[x]>
 <effective[x]><!-- 0..1 dateTime|Period The date/time or interval when the medication was taken --></effective[x]>
 <dateAsserted value="[dateTime]"/><!-- 0..1 When the statement was asserted? -->
 <informationSource><!-- 0..1 Reference(Patient|Practitioner|RelatedPerson|
   Organization) Person or organization that provided the information about the taking of this medication
    --></informationSource>
 <subject><!-- 1..1 Reference(Patient|Group) Who is/was taking  the medication --></subject>
 <derivedFrom><!-- 0..* Reference(Any) Additional supporting information --></derivedFrom>
 <taken value="[code]"/><!-- 1..1 y | n | unk | na -->
 <reasonNotTaken><!--  0..* CodeableConcept True if asserting medication was not given --></reasonNotTaken>
 <reasonCode><!-- 0..* CodeableConcept Reason for why the medication is being/was taken --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation) Condition or observation that supports why the medication
    is being/was taken --></reasonReference>
 <note><!-- 0..* Annotation Further information about the statement --></note>
 <dosage><!-- 0..* Dosage Details of how medication is/was taken or should be taken --></dosage>
</MedicationStatement>
```

JSON Template

```
{
  "resourceType" : "MedicationStatement",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External identifier
  "basedOn" : [{ Reference(MedicationRequest|CarePlan|ProcedureRequest|
   ReferralRequest) }], // Fulfils plan, proposal or order
  "partOf" : [{ Reference(MedicationAdministration|MedicationDispense|
   MedicationStatement|Procedure|Observation) }], // Part of referenced event
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter / Episode associated with MedicationStatement
  "status" : "<code>", // R!  active | completed | entered-in-error | intended | stopped | on-hold
  "category" : { CodeableConcept }, // Type of medication usage
  // medication[x]: What medication was taken. One of these 2:
  "medicationCodeableConcept" : { CodeableConcept },
  "medicationReference" : { Reference(Medication) },
  // effective[x]: The date/time or interval when the medication was taken. One of these 2:
  "effectiveDateTime" : "<dateTime>",
  "effectivePeriod" : { Period },
  "dateAsserted" : "<dateTime>", // When the statement was asserted?
  "informationSource" : { Reference(Patient|Practitioner|RelatedPerson|
   Organization) }, // Person or organization that provided the information about the taking of this medication
  "subject" : { Reference(Patient|Group) }, // R!  Who is/was taking  the medication
  "derivedFrom" : [{ Reference(Any) }], // Additional supporting information
  "taken" : "<code>", // R!  y | n | unk | na
  "reasonNotTaken" : [{ CodeableConcept }], // C? True if asserting medication was not given
  "reasonCode" : [{ CodeableConcept }], // Reason for why the medication is being/was taken
  "reasonReference" : [{ Reference(Condition|Observation) }], // Condition or observation that supports why the
    medication is being/was taken
  "note" : [{ Annotation }], // Further information about the statement
  "dosage" : [{ Dosage }] // Details of how medication is/was taken or should be taken
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicationStatement;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicationStatement.identifier [ Identifier ], ... ; # 0..* External identifier
```

```
  fhir:MedicationStatement.basedOn [ Reference(MedicationRequest|CarePlan|ProcedureRequest|ReferralRequest) ], ... ; #
    0..* Fulfils plan, proposal or order
  fhir:MedicationStatement.partOf [ Reference(MedicationAdministration|MedicationDispense|MedicationStatement|Procedure|
  Observation) ], ... ; # 0..* Part of referenced event
  fhir:MedicationStatement.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter / Episode associated with
    MedicationStatement
  fhir:MedicationStatement.status [ code ]; # 1..1 active | completed | entered-in-error | intended | stopped | on-hold
  fhir:MedicationStatement.category [ CodeableConcept ]; # 0..1 Type of medication usage
  # MedicationStatement.medication[x] : 1..1 What medication was taken. One of these 2
    fhir:MedicationStatement.medicationCodeableConcept [ CodeableConcept ]
    fhir:MedicationStatement.medicationReference [ Reference(Medication) ]
  # MedicationStatement.effective[x] : 0..1 The date/time or interval when the medication was taken. One of these 2
    fhir:MedicationStatement.effectiveDateTime [ dateTime ]
    fhir:MedicationStatement.effectivePeriod [ Period ]
  fhir:MedicationStatement.dateAsserted [ dateTime ]; # 0..1 When the statement was asserted?
  fhir:MedicationStatement.informationSource [ Reference(Patient|Practitioner|RelatedPerson|Organization) ]; # 0..1
    Person or organization that provided the information about the taking of this medication
  fhir:MedicationStatement.subject [ Reference(Patient|Group) ]; # 1..1 Who is/was taking  the medication
  fhir:MedicationStatement.derivedFrom [ Reference(Any) ], ... ; # 0..* Additional supporting information
  fhir:MedicationStatement.taken [ code ]; # 1..1 y | n | unk | na
  fhir:MedicationStatement.reasonNotTaken [ CodeableConcept ], ... ; # 0..* True if asserting medication was not given
  fhir:MedicationStatement.reasonCode [ CodeableConcept ], ... ; # 0..* Reason for why the medication is being/was taken
  fhir:MedicationStatement.reasonReference [ Reference(Condition|Observation) ], ... ; # 0..* Condition or observation
    that supports why the medication is being/was taken
  fhir:MedicationStatement.note [ Annotation ], ... ; # 0..* Further information about the statement
  fhir:MedicationStatement.dosage [ Dosage ], ... ; # 0..* Details of how medication is/was taken or should be taken
]
```

Changes since DSTU2

MedicationStatement
MedicationStatement.basedOn | Added Element
MedicationStatement.partOf | Added Element
MedicationStatement.context | Added Element
MedicationStatement.category | Added Element
MedicationStatement.informationSource | Add Reference(Organization)
MedicationStatement.subject | Renamed from patient to subject Add Reference(Group)
MedicationStatement.derivedFrom | Renamed from supportingInformation to derivedFrom
MedicationStatement.taken | Added Element
MedicationStatement.reasonCode | Added Element
MedicationStatement.reasonReference | Added Element
MedicationStatement.note | Max Cardinality changed from 1 to * Type changed from string to Annotation
MedicationStatement.dosage | Type changed from BackboneElement to Dosage
MedicationStatement.wasNotTaken | deleted
MedicationStatement.reasonForUse[x] | deleted
MedicationStatement.dosage.text | deleted
MedicationStatement.dosage.timing | deleted
MedicationStatement.dosage.asNeeded[x] | deleted
MedicationStatement.dosage.site[x] | deleted
MedicationStatement.dosage.route | deleted
MedicationStatement.dosage.method | deleted
MedicationStatement.dosage.quantity[x] | deleted
MedicationStatement.dosage.rate[x] | deleted
MedicationStatement.dosage.maxDosePerPeriod | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 7 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (1 errors).).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

11.4.3.1 Terminology Bindings

Path | Definition | Type | Reference
MedicationStatement.status | A coded concept indicating the current status of a MedicationStatement. | Required | MedicationStatementStatus
MedicationStatement.category | A coded concept identifying where the medication included in the medicationstatement is expected to be consumed or administered | Preferred | MedicationStatementCategory
MedicationStatement.medication[x] | A coded concept identifying the substance or product being taken. | Example | SNOMED CT Medication Codes
MedicationStatement.taken | A coded concept identifying level of certainty if patient has taken or has not taken the medication | Required | MedicationStatementTaken
MedicationStatement.reasonNotTaken | A coded concept indicating the reason why the medication was not taken | Example | SNOMED CT Drugs not taken/completed Codes
MedicationStatement.reasonCode | A coded concept identifying why the medication is being taken. | Example | Condition/Problem/Diagnosis Codes

11.4.3.2 Constraints

* mst-1: Reason not taken is only permitted if Taken is No (expression : reasonNotTaken.exists().not() or (taken = 'n'))

11.4.3.3 How to determine if the patient has taken the medication

The MedicationStatement resource includes both a status and a taken code. The taken code conveys whether the medication was taken by the patient from the perspective of the information source. The status code reflects the current state of the practitioner's instructions to the patient whether the consumption of the medication should continue or not.

Note: Medication statements can be made about prescribed medications as well as non-prescribed (i.e. over the counter) medications.

If you desire to perform a query for all medication statements that "imply" that a medication has been taken, you will need to use both MedicationStatement.status and MedicationStatement.taken in your query. The following table is intended to provide guidance on the interpretation of these two attributes with respect to the MedicationStatement.

In the table below the "X" represents a valid status that can be present in combination with the Taken value.

Taken | Information Source | Active | Completed | Stopped | On Hold | Entered in Error | Intended | Interpretation or Meaning

N | Exists (e.g. Patient or RelatedPerson) | X | | | X | | X | Patient or related person states the medication is not currently being taken. Taken must = N. When status = Active, it means that although a statement was made that the patient isn't taking the medication, the practitioner still expects and instructs the patient to take the medication. When status = On Hold, it means that although a statement was made that the patient isn't taking the medication, the practitioner has suspended the medication, but intends for the patient to take the medication in the future. When status = Intended, it means that although a statement was made that the patient isn't taking the medication, the practitioner intends for the patient to take the medication in the future.

Y | Exists (e.g. Patient or RelatedPerson) | X | | | X | | X | Patient or related person states the medication is or will be taken. Taken must = Y. The status values can be any of the following: active, on hold, or intended.

UNK | No information source exists | X | | | X | | X | No assertion by patient or related person of whether the medication is being consumed. The MedicationStatement still exists because it can be derived from a MedicationRequest, but it is unknown whether the Patient is taking the medication as prescribed in the MedicationRequest.

NA | No information source exists | X | X | X | X | X | X | Patient reporting does not apply. For example, this can occur when MedicationStatements are derived from MedicationRequests that are administered by a practitioner. In this example, there is no need to ask for input from the patient or related person since the practitioner was responsible for the administration. Another example might be a MedicationStatement derived from an end-stated (stopped, completed, entered in error) MedicationRequest. In this example, there is no need to ask for input from the patient or related person since the MedicationRequest is no longer applicable.

11.4.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common

category | token | Returns statements of this category of medicationstatement | MedicationStatement.category

code | token | Return statements of this medication code | MedicationStatement.medication.as(CodeableConcept) | 4 Resources

context | reference | Returns statements for a specific context (episode or episode of Care). | MedicationStatement.context (EpisodeOfCare, Encounter)

effective | date | Date when patient was taking (or not taking) the medication | MedicationStatement.effective

identifier | token | Return statements with this external identifier | MedicationStatement.identifier | 3 Resources

medication | reference | Return statements of this medication reference | MedicationStatement.medication.as(Reference) (Medication) | 3 Resources

part-of | reference | Returns statements that are part of another event. | MedicationStatement.partOf (MedicationDispense, Observation, MedicationAdministration, Procedure, MedicationStatement)

patient | reference | Returns statements for a specific patient. | MedicationStatement.subject (Patient) | 3 Resources

source | reference | Who or where the information in the statement came from | MedicationStatement.informationSource (Practitioner, Organization, Patient, RelatedPerson)

status | token | Return statements that match the given status | MedicationStatement.status | 3 Resources

subject | reference | The identity of a patient, animal or group to list statements for | MedicationStatement.subject (Group, Patient)

11.4.5 Resource MedicationStatement - Examples

URL: https://hl7.org/fhir/STU3/medicationstatement-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.4.5 Resource MedicationStatement - Examples

Pharmacy Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner, RelatedPerson

The following are examples of Medication Statement resources:

* Patient reports they are taking Tylenol PM and has the bottle

* Patient reports they are not taking Tylenol PM

* Patient reports they took Amoxycillin in the past but are not taking it now

* Patient reports they previous said they took Amoxycillin but really didn't

* Care Giver reports that Patient took Amoxycillin

* Patient reports that they intend to use Mometasone in the future

* Patient reports they are taking the "little pink pill for water retention

Example List:

Patient reports they are taking Tylenol PM and has the bottle | example001 | XML | JSON | Turtle

Patient reports they are not taking Tylenol PM | example002 | XML | JSON | Turtle

Patient reports they were taking Amoxycillin in the past but are not taking it now | example004 | XML | JSON | Turtle

Patient reports they previous said they took Amoxycillin but really didn't | example005 | XML | JSON | Turtle
Care Giver reports that the Patient took Amoxicillin | example006 | XML | JSON | Turtle
Patient reports they intend to use Mometasone in the future | example007 | XML | JSON | Turtle
Patient reports that they are taking the "little pink pill" | example003 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.4.6 Resource MedicationStatement - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/medicationstatement-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.4.6 Resource MedicationStatement - Detailed Descriptions

Pharmacy Work Group | Maturity Level: 3 | Trial Use | Compartments: Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the MedicationStatement resource.

MedicationStatement

Definition | A record of a medication that is being consumed by a patient. A MedicationStatement may indicate that the patient may be taking the medication now, or has taken the medication in the past or will be taking the medication in the future. The source of this information can be the patient, significant other (such as a family member or spouse), or a clinician. A common scenario where this information is captured is during the history taking process during a patient visit or stay. The medication information may come from sources such as the patient's memory, from a prescription bottle, or from a list of medications the patient, clinician or other party maintains The primary difference between a medication statement and a medication administration is that the medication administration has complete administration information and is based on actual administration information from the person who administered the medication. A medication statement is often, if not always, less specific. There is no required date/time when the medication was administered, in fact we only know that a source has reported the patient is taking this medication, where details such as time, quantity, or rate or even medication product may be incomplete or missing or less precise. As stated earlier, the medication statement information may come from the patient's memory, from a prescription bottle or from a list of medications the patient, clinician or other party maintains. Medication administration is more formal and is not missing detailed information.

Control | 1..1

Comments | When interpreting a medicationStatement, the value of the status and NotTaken needed to be considered: MedicationStatement.status + MedicationStatement.wasNotTaken Status=Active + NotTaken=T = Not currently taking Status=Completed + NotTaken=T = Not taken in the past Status=Intended + NotTaken=T = No intention of taking Status=Active + NotTaken=F = Taking, but not as prescribed Status=Active + NotTaken=F = Taking Status=Intended +NotTaken= F = Will be taking (not started) Status=Completed + NotTaken=F = Taken in past Status=In Error + NotTaken=N/A = In Error.

Invariants | Defined on this element mst-1: Reason not taken is only permitted if Taken is No (expression : reasonNotTaken.exists().not() or (taken = 'n'), xpath: not(exists(f:reasonNotTaken)) or f:taken/@value='n')

MedicationStatement.identifier

Definition | External identifier - FHIR will generate its own internal identifiers (probably URLs) which do not need to be explicitly managed by the resource. The identifier here is one that would be used by another non-FHIR system - for example an automated medication pump would provide a record each time it operated; an administration while the patient was off the ward might be made with a different system and entered after the event. Particularly important if these records have to be updated.

Note | This is a business identifer, not a resource identifier (see discussion)

Control | 0..*

Type | Identifier

Summary | true

MedicationStatement.basedOn

Definition | A plan, proposal or order that is fulfilled in whole or in part by this event.

Control | 0..*

Type | Reference(MedicationRequest | CarePlan | ProcedureRequest | ReferralRequest)

Requirements | Allows tracing of authorization for the event and tracking whether proposals/recommendations were acted upon.

Summary | true

MedicationStatement.partOf

Definition | A larger event of which this particular event is a component or step.

Control | 0..*

Type | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Observation)

Requirements | This should not be used when indicating which resource a MedicationStatement has been derived from. If that is the use case, then MedicationStatement.derivedFrom should be used.

Summary | true

MedicationStatement.context

Definition | The encounter or episode of care that establishes the context for this MedicationStatement.

Control | 0..1

Type | Reference(Encounter | EpisodeOfCare)

Summary | true

MedicationStatement.status

Definition | A code representing the patient or other source's judgment about the state of the medication used that this statement is about. Generally this will be active or completed.

Control | 1..1

Terminology Binding | MedicationStatementStatus (Required)

Type | code

Is Modifier | true

Summary | true

Comments | MedicationStatement is a statement at a point in time. The status is only representative at the point when it was asserted. The value set for MedicationStatement.status contains codes that assert the status of the use of the medication by the patient (for example, stopped or on hold) as well as codes that assert the status of the medication statement itself (for example, entered in error). This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
MedicationStatement.category
Definition | Indicates where type of medication statement and where the medication is expected to be consumed or administered.
Control | 0..1
Terminology Binding | MedicationStatementCategory (Preferred)
Type | CodeableConcept
Summary | true
MedicationStatement.medication[x]
Definition | Identifies the medication being administered. This is either a link to a resource representing the details of the medication or a simple attribute carrying a code that identifies the medication from a known list of medications.
Control | 1..1
Terminology Binding | SNOMED CT Medication Codes (Example)
Type | CodeableConcept|Reference(Medication)
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | If only a code is specified, then it needs to be a code for a specific product. If more information is required, then the use of the medication resource is recommended. For example if you require form or lot number, then you must reference the Medication resource. .
MedicationStatement.effective[x]
Definition | The interval of time during which it is being asserted that the patient was taking the medication (or was not taking, when the wasNotGiven element is true).
Control | 0..1
Type | dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | This attribute reflects the period over which the patient consumed the medication and is expected to be populated on the majority of Medication Statements. If the medication is still being taken at the time the statement is recorded, the "end" date will be omitted.
MedicationStatement.dateAsserted
Definition | The date when the medication statement was asserted by the information source.
Control | 0..1
Type | dateTime
Summary | true
MedicationStatement.informationSource
Definition | The person or organization that provided the information about the taking of this medication. Note: Use derivedFrom when a MedicationStatement is derived from other resources, e.g Claim or MedicationRequest.
Control | 0..1
Type | Reference(Patient | Practitioner | RelatedPerson | Organization)
MedicationStatement.subject
Definition | The person, animal or group who is/was taking the medication.
Control | 1..1
Type | Reference(Patient | Group)
Summary | true
MedicationStatement.derivedFrom
Definition | Allows linking the MedicationStatement to the underlying MedicationRequest, or to other information that supports or is used to derive the MedicationStatement.
Control | 0..*
Type | Reference(Any)
Comments | Likely references would be to MedicationRequest, MedicationDispense, Claim, Observation or QuestionnaireAnswers. The most common use cases for deriving a MedicationStatement comes from creating a MedicationStatement from a MedicationRequest or from a lab observation or a claim. it should be noted that the amount of information that is available varies from the type resource that you derive the MedicationStatement from.
MedicationStatement.taken
Definition | Indicator of the certainty of whether the medication was taken by the patient.
Control | 1..1
Terminology Binding | MedicationStatementTaken (Required)
Type | code
Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because it indicates that the medication was not taken.
MedicationStatement.reasonNotTaken
Definition | A code indicating why the medication was not taken.
Control | 0..*
Terminology Binding | SNOMED CT Drugs not taken/completed Codes (Example)
Type | CodeableConcept
Invariants | Affect this element mst-1: Reason not taken is only permitted if Taken is No (expression : reasonNotTaken.exists().not() or (taken = 'n'), xpath: not(exists(f:reasonNotTaken)) or f:taken/@value='n')
MedicationStatement.reasonCode
Definition | A reason for why the medication is being/was taken.
Control | 0..*
Terminology Binding | Condition/Problem/Diagnosis Codes (Example)
Type | CodeableConcept

Comments | This could be a diagnosis code. If a full condition record exists or additional detail is needed, use reasonForUseReference.
MedicationStatement.reasonReference
Definition | Condition or observation that supports why the medication is being/was taken.
Control | 0..*
Type | Reference(Condition | Observation)
Comments | This is a reference to a condition that is the reason why the medication is being/was taken. If only a code exists, use reasonForUseCode.
MedicationStatement.note
Definition | Provides extra information about the medication statement that is not conveyed by the other attributes.
Control | 0..*
Type | Annotation
MedicationStatement.dosage
Definition | Indicates how the medication is/was or should be taken by the patient.
Control | 0..*
Type | Dosage
Comments | The dates included in the dosage on a Medication Statement reflect the dates for a given dose. For example, "from November 1, 2016 to November 3, 2016, take one tablet daily and from November 4, 2016 to November 7, 2016, take two tablets daily." It is expected that this specificity may only be populated where the patient brings in their labeled container or where the Medication Statement is derived from a MedicationRequest.

11.4.7 Resource MedicationStatement - Mappings

URL: https://hl7.org/fhir/STU3/medicationstatement-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.4.7 Resource MedicationStatement - Mappings

Pharmacy Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner, RelatedPerson

Mappings:

RIM Mapping

HL7 v2 Mapping

Workflow Mapping

Mappings for the medicationstatement resource.

11.4.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

MedicationStatement | SubstanceAdministration
identifier | .id
basedOn | .outboundRelationship[typeCode=FLFS].target[classCode=SBADM or PROC or PCPR or OBS, moodCode=RQO orPLAN or PRP]
partOf | .outboundRelationship[typeCode=COMP]/target[classCode=SPLY or SBADM or PROC or OBS,moodCode=EVN]
context | .inboundRelationship[typeCode=COMP].source[classCode=ENC, moodCode=EVN, code="type of encounter or episode"]
status | .statusCode
category | .inboundRelationship[typeCode=COMP].source[classCode=OBS, moodCode=EVN, code="type of medication usage"].value
medication[x] | .participation[typeCode=CSM].role[classCode=ADMM or MANU]
effective[x] | .effectiveTime
dateAsserted | .participation[typeCode=AUT].time
informationSource | .participation[typeCode=INF].role[classCode=PAT, or codes for Practioner or Related Person (if PAT is the informer, then syntax for self-reported =true)
subject | .participation[typeCode=SBJ].role[classCode=PAT]
derivedFrom | .outboundRelationship[typeCode=SPRT]/target[classCode=ACT,moodCode=EVN]
taken | .actionNegationInd
reasonNotTaken | .outboundRelationship[typeCode=RSON]/target[classCode=OBS,moodCode=EVN, code="reason not taken"].value
reasonCode | .reasonCode
reasonReference | .outboundRelationship[typeCode=RSON]/target[classCode=OBS,moodCode=EVN, code="reason for use"].value
note | .inboundRelationship[typeCode=SUBJ]/source[classCode=OBS,moodCode=EVN,code="annotation"].value
dosage | refer dosageInstruction mapping

11.4.7.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

MedicationStatement
identifier
basedOn
partOf
context
status
category
medication[x]
effective[x]
dateAsserted
informationSource
subject | PID-3-Patient ID List

derivedFrom
taken
reasonNotTaken
reasonCode
reasonReference
note
dosage

11.4.7.3 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

MedicationStatement | ..Event
identifier | …identifer
basedOn | …basedOn
partOf | …part of
context | …context
status | …status
category
medication[x] | …code
effective[x] | …occurrence[x]
dateAsserted
informationSource
subject | …subject
derivedFrom
taken | …notDone
reasonNotTaken
reasonCode | …reasoneCodeableConcept
reasonReference | …reasonReference
note | …note
dosage

11.4.8 Resource MedicationStatement - Design Notes

URL: https://hl7.org/fhir/STU3/medicationstatement-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.4.8 Resource MedicationStatement - Design Notes

to do

11.4.9 Resource MedicationStatement - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/medicationstatement-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.4.9 Resource MedicationStatement - Extensions & Profiles

Pharmacy Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the MedicationStatement resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

11.4.9.1 Profiles

No Profiles defined for this resource

11.4.9.2 Extensions

medicationstatement-Prescriber | Prescriber | for MedicationStatement HL7 Extensions

Extensions that reference this resource:

condition-dueTo | dueTo | for Condition HL7 Extensions
condition-occurredFollowing | occurredFollowing | for Condition HL7 Extensions
procedure-causedBy | causedBy | for Procedure HL7 Extensions

11.4.9.3 Search Extensions

No Search Extensions defined for this resource

11.4.10 MedicationStatement HL7 Extensions

URL: https://hl7.org/fhir/STU3/medicationstatement-extensions.html

Defines common extensions used with or related to the MedicationStatement resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.4.10 MedicationStatement HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

11.4.10.1 Content

Extensions:
medicationstatement-Prescriber | Prescriber : The healthcare professional responsible for authorizing the initial prescription.

11.4.11 Resource MedicationStatement - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/medicationstatement-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.4.11 Resource MedicationStatement - Detailed Descriptions

Pharmacy Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner, RelatedPerson

R2 : R3 Converson maps for MedicationStatement.

Functional status for this map: 7 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (1 errors). (see documentation)

11.4.11.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/MedicationStatement2to3" = "R2 to R3 Conversions for MedicationStatement"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/MedicationStatement" alias MedicationStatementR2 as source
uses "http://hl7.org/fhir/StructureDefinition/MedicationStatement" alias MedicationStatement as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  MedicationStatement extends DomainResource
  input src : MedicationStatementR2 as source
  input tgt : MedicationStatement as target

  "MedicationStatement-identifier" : for src.identifier make tgt.identifier
  "MedicationStatement-subject" : for src.patient make tgt.subject
  "MedicationStatement-informationSource" : for src.informationSource make tgt.informationSource
  "MedicationStatement-dateAsserted" : for src.dateAsserted make tgt.dateAsserted
  "MedicationStatement-status" : for src.status make tgt.status
  "MedicationStatement-wasNotTaken" : for src.wasNotTaken as vt where $this = true make tgt.taken = "n"
  "MedicationStatement-wasNotTaken" : for src.wasNotTaken as vt where $this = false make tgt.taken = "y"
  "MedicationStatement-reasonNotTaken" : for src.reasonNotTaken make tgt.reasonNotTaken
  "MedicationStatement-reasonForUse" : for src.reasonForUse where $this is CodeableConcept make tgt.reasonCode
  "MedicationStatement-reasonForUse" : for src.reasonForUse where $this is Reference make tgt.reasonReference
  "MedicationStatement-effective" : for src.effective make tgt.effective
  "MedicationStatement-note" : for src.note make tgt.note
  "MedicationStatement-supportingInformation" : for src.supportingInformation make tgt.derivedFrom
  "MedicationStatement-medication" : for src.medication make tgt.medication

  "MedicationStatement-dosage" : for src.dosage as vs0 make tgt.dosage as vt0 then mstDosage(vs0, vt0)
endgroup

group mstDosage extends Element // BackboneElement?
  input src as source
  input tgt as target

  "DosageInstruction-text" : for src.text make tgt.text
  "DosageInstruction-timing" : for src.timing make tgt.timing
  "DosageInstruction-asNeeded" : for src.asNeeded make tgt.asNeeded
  "DosageInstruction-site" : for src.site where $this is CodeableConcept make tgt.site // todo: handle Reference?
  "DosageInstruction-route" : for src.route make tgt.route
  "DosageInstruction-method" : for src.method make tgt.method
  "DosageInstruction-quantity" : for src.quantity make tgt.dose
  "DosageInstruction-rate" : for src.rate make tgt.rate
  "DosageInstruction-maxDosePerPeriod" : for src.maxDosePerPeriod make tgt.maxDosePerPeriod
endgroup
```

11.4.11.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/MedicationStatement3to2" = "R3 to R2 Conversion for MedicationStatement"

uses "http://hl7.org/fhir/StructureDefinition/MedicationStatement" alias MedicationStatement as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/MedicationStatement" alias MedicationStatementR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  MedicationStatement extends DomainResource
  input src : MedicationStatement as source
  input tgt : MedicationStatementR2 as target

  "MedicationStatement-identifier" : for src.identifier make tgt.identifier
```

```
  "MedicationStatement-subject" : for src.subject make tgt.patient
  "MedicationStatement-informationSource" : for src.informationSource make tgt.informationSource
  "MedicationStatement-dateAsserted" : for src.dateAsserted make tgt.dateAsserted
  "MedicationStatement-status" : for src.status check src.status != 'stopped' and status != 'on-hold' make tgt.status
  "MedicationStatement-wasNotTaken" : for src.taken as vt where $this = 'n' make tgt.wasNotTaken = true
  "MedicationStatement-wasNotTaken" : for src.taken as vt where $this = 'y' make tgt.wasNotTaken = false
  "MedicationStatement-reasonNotTaken" : for src.reasonNotTaken make tgt.reasonNotTaken
  "MedicationStatement-reasonForUse1" : for src.reasonCode make tgt.reasonForUse
  "MedicationStatement-reasonForUse2" : for src.reasonReference make tgt.reasonForUse
  "MedicationStatement-effective" : for src.effective make tgt.effective
  "MedicationStatement-note" : for src.note make tgt.note
  "MedicationStatement-supportingInformation" : for src.derivedFrom make tgt.supportingInformation
  "MedicationStatement-medication" : for src.medication make tgt.medication

  "MedicationStatement-dosage" : for src.dosage as vs0 make tgt.dosage as vt0 then mstDosage(vs0, vt0)
endgroup

group mstDosage extends Element // BackboneElement?
  input src as source
  input tgt as target

  "DosageInstruction-text" : for src.text make tgt.text
  "DosageInstruction-timing" : for src.timing make tgt.timing
  "DosageInstruction-asNeeded" : for src.asNeeded make tgt.asNeeded
  "DosageInstruction-site" : for src.site make tgt.site // todo: handle Reference?
  "DosageInstruction-route" : for src.route make tgt.route
  "DosageInstruction-method" : for src.method make tgt.method
  "DosageInstruction-quantity" : for src.dose make tgt.quantity
  "DosageInstruction-rate" : for src.rate make tgt.rate
  "DosageInstruction-maxDosePerPeriod" : for src.maxDosePerPeriod make tgt.maxDosePerPeriod
endgroup
```

11.4.11.3 R3 Validation Errors

medicationstatement-example | ERROR @ MedicationStatement: Profile http://hl7.org/fhir/StructureDefinition/MedicationStatement, Element 'MedicationStatement.taken': minimum required = 1, but only found 0 (src = InstanceValidator)

11.5 Resource Medication - Content

URL: https://hl7.org/fhir/STU3/medication.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Pharmacy Work Group | Maturity Level: 3 | Trial Use | Compartments: Not linked to any defined compartments

This resource is primarily used for the identification and definition of a medication. It covers the ingredients and the packaging for a medication.

11.5.1 Scope and Usage

Representing medications in the majority of healthcare settings is a matter of identifying an item from a list and then conveying a reference for the item selected either into a patient related resource or to other applications. Additional information about the medication is frequently provided for human verification, but a full representation of the details of composition and efficacy of the medicine is conveyed by referring to drug dictionaries by means of the codes they define. There are some occasions where it is necessary to identify slightly more detail, such as when dispensing a package containing a particular medicine requires identification both of the medicine and the package at once. There are also some occasions (e.g. custom formulations) where the composition of a medicine must be represented. In these cases the ingredients of the medicine have to be specified together with the amount contained, though the medication resource does not provide full details.

The Medication resource allows for medications to be characterized by the form of the drug and the ingredient (or ingredients), as well as how it is packaged. The medication will include the ingredient(s) and their strength(s) and the package can include the amount (for example, number of tablets, volume, etc.) that is contained in a particular container (for example, 100 capsules of Amoxicillin 500mg per bottle).

The medication resource can be used to describe a compounded (aka extemporaneous or magistral) product that is manufactured by the pharmacy at the time of dispensing. In this case there will be multiple ingredients which are typically base chemicals (for example, hydrocortisone powder) and there may be other ingredients that are manufactured products (for example, Glaxal Base).

If medication includes a package, further details about the composition can be provided. A package has a container (vacuum packed box, jar, etc.) and a list of the products or other packages that are in the package.

This resource is referenced by ActivityDefinition, AdverseEvent, CarePlan, Flag, Group, MedicationAdministration, MedicationDispense, MedicationRequest, MedicationStatement, Procedure, SupplyDelivery and SupplyRequest

11.5.2 Resource Content

* Structure

* UML

* XML

* JSON
* Turtle
* R2 Diff
* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Medication | | | DomainResource | Definition of a Medication Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
code | ? | 0..1 | CodeableConcept | Codes that identify this medication SNOMED CT Medication Codes (Example)
status | ? | 0..1 | code | active | inactive | entered-in-error MedicationStatus (Required)
isBrand | ? | 0..1 | boolean | True if a brand
isOverTheCounter | ? | 0..1 | boolean | True if medication does not require a prescription
manufacturer | ? | 0..1 | Reference(Organization) | Manufacturer of the item
form | | 0..1 | CodeableConcept | powder | tablets | capsule + SNOMED CT Form Codes (Example)
ingredient | | 0..* | BackboneElement | Active or inactive ingredient
item[x] | | 1..1 | | The product contained
itemCodeableConcept | | | CodeableConcept
itemReference | | | Reference(Substance | Medication)
isActive | | 0..1 | boolean | Active ingredient indicator
amount | | 0..1 | Ratio | Quantity of ingredient present
package | | 0..1 | BackboneElement | Details about packaged medications
container | | 0..1 | CodeableConcept | E.g. box, vial, blister-pack MedicationContainer (Example)
content | | 0..* | BackboneElement | What is in the package
item[x] | | 1..1 | | The item in the package
itemCodeableConcept | | | CodeableConcept
itemReference | | | Reference(Medication)
amount | | 0..1 | SimpleQuantity | Quantity present in the package
batch | | 0..* | BackboneElement | Identifies a single production run
lotNumber | | 0..1 | string | Identifier assigned to batch
expirationDate | | 0..1 | dateTime | When batch will expire
image | | 0..* | Attachment | Picture of the medication
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Medication xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <code><!-- 0..1 CodeableConcept Codes that identify this medication --></code>
 <status value="[code]"/><!-- 0..1 active | inactive | entered-in-error -->
 <isBrand value="[boolean]"/><!-- 0..1 True if a brand -->
 <isOverTheCounter value="[boolean]"/><!-- 0..1 True if medication does not require a prescription -->
 <manufacturer><!-- 0..1 Reference(Organization) Manufacturer of the item --></manufacturer>
 <form><!-- 0..1 CodeableConcept powder | tablets | capsule + --></form>
 <ingredient>  <!-- 0..* Active or inactive ingredient -->
  <item[x]><!-- 1..1 CodeableConcept|Reference(Substance|Medication) The product contained --></item[x]>
  <isActive value="[boolean]"/><!-- 0..1 Active ingredient indicator -->
  <amount><!-- 0..1 Ratio Quantity of ingredient present --></amount>
 </ingredient>
 <package>  <!-- 0..1 Details about packaged medications -->
  <container><!-- 0..1 CodeableConcept E.g. box, vial, blister-pack --></container>
  <content>  <!-- 0..* What is  in the package -->
   <item[x]><!-- 1..1 CodeableConcept|Reference(Medication) The item in the package --></item[x]>
   <amount><!-- 0..1 Quantity(SimpleQuantity) Quantity present in the package --></amount>
  </content>
  <batch>  <!-- 0..* Identifies a single production run -->
   <lotNumber value="[string]"/><!-- 0..1 Identifier assigned to batch -->
   <expirationDate value="[dateTime]"/><!-- 0..1 When batch will expire -->
  </batch>
 </package>
 <image><!-- 0..* Attachment Picture of the medication --></image>
</Medication>
```

JSON Template

```
{
  "resourceType" : "Medication",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "code" : { CodeableConcept }, // Codes that identify this medication
  "status" : "<code>", // active | inactive | entered-in-error
  "isBrand" : <boolean>, // True if a brand
  "isOverTheCounter" : <boolean>, // True if medication does not require a prescription
  "manufacturer" : { Reference(Organization) }, // Manufacturer of the item
  "form" : { CodeableConcept }, // powder | tablets | capsule +
  "ingredient" : [{ // Active or inactive ingredient
    // item[x]: The product contained. One of these 2:
    "itemCodeableConcept" : { CodeableConcept },
    "itemReference" : { Reference(Substance|Medication) },
```

```
    "isActive" : <boolean>, // Active ingredient indicator
    "amount" : { Ratio } // Quantity of ingredient present
  }],
  "package" : { // Details about packaged medications
    "container" : { CodeableConcept }, // E.g. box, vial, blister-pack
    "content" : [{ // What is  in the package
      // item[x]: The item in the package. One of these 2:
      "itemCodeableConcept" : { CodeableConcept },
      "itemReference" : { Reference(Medication) },
      "amount" : { Quantity(SimpleQuantity) } // Quantity present in the package
    }],
    "batch" : [{ // Identifies a single production run
      "lotNumber" : "<string>", // Identifier assigned to batch
      "expirationDate" : "<dateTime>" // When batch will expire
    }]
  },
  "image" : [{ Attachment }] // Picture of the medication
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Medication;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Medication.code [ CodeableConcept ]; # 0..1 Codes that identify this medication
  fhir:Medication.status [ code ]; # 0..1 active | inactive | entered-in-error
  fhir:Medication.isBrand [ boolean ]; # 0..1 True if a brand
  fhir:Medication.isOverTheCounter [ boolean ]; # 0..1 True if medication does not require a prescription
  fhir:Medication.manufacturer [ Reference(Organization) ]; # 0..1 Manufacturer of the item
  fhir:Medication.form [ CodeableConcept ]; # 0..1 powder | tablets | capsule +
  fhir:Medication.ingredient [ # 0..* Active or inactive ingredient
    # Medication.ingredient.item[x] : 1..1 The product contained. One of these 2
      fhir:Medication.ingredient.itemCodeableConcept [ CodeableConcept ]
      fhir:Medication.ingredient.itemReference [ Reference(Substance|Medication) ]
    fhir:Medication.ingredient.isActive [ boolean ]; # 0..1 Active ingredient indicator
    fhir:Medication.ingredient.amount [ Ratio ]; # 0..1 Quantity of ingredient present
  ], ...;
  fhir:Medication.package [ # 0..1 Details about packaged medications
    fhir:Medication.package.container [ CodeableConcept ]; # 0..1 E.g. box, vial, blister-pack
    fhir:Medication.package.content [ # 0..* What is  in the package
      # Medication.package.content.item[x] : 1..1 The item in the package. One of these 2
        fhir:Medication.package.content.itemCodeableConcept [ CodeableConcept ]
        fhir:Medication.package.content.itemReference [ Reference(Medication) ]
      fhir:Medication.package.content.amount [ Quantity(SimpleQuantity) ]; # 0..1 Quantity present in the package
    ], ...;
    fhir:Medication.package.batch [ # 0..* Identifies a single production run
      fhir:Medication.package.batch.lotNumber [ string ]; # 0..1 Identifier assigned to batch
      fhir:Medication.package.batch.expirationDate [ dateTime ]; # 0..1 When batch will expire
    ], ...;
  ];
  fhir:Medication.image [ Attachment ], ... ; # 0..* Picture of the medication
]
```

Changes since DSTU2

Medication

Medication.status | Added Element

Medication.isOverTheCounter | Added Element

Medication.form | Added Element

Medication.ingredient | Added Element

Medication.ingredient.item[x] | Added Element

Medication.ingredient.isActive | Added Element

Medication.ingredient.amount | Added Element

Medication.package.content.item[x] | Renamed from item to item[x] Add CodeableConcept

Medication.package.batch | Added Element

Medication.package.batch.lotNumber | Added Element

Medication.package.batch.expirationDate | Added Element

Medication.image | Added Element

Medication.product | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 23 tests that all execute ok. 16 fail round-trip testing and 1 r3 resources are invalid (1 errors).).

Structure

Name | Flags | Card. | Type | Description & Constraints

Medication | | | DomainResource | Definition of a Medication Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

code | ? | 0..1 | CodeableConcept | Codes that identify this medication SNOMED CT Medication Codes (Example)

status | ? | 0..1 | code | active | inactive | entered-in-error MedicationStatus (Required)
isBrand | ? | 0..1 | boolean | True if a brand
isOverTheCounter | ? | 0..1 | boolean | True if medication does not require a prescription
manufacturer | ? | 0..1 | Reference(Organization) | Manufacturer of the item
form | | 0..1 | CodeableConcept | powder | tablets | capsule + SNOMED CT Form Codes (Example)
ingredient | | 0..* | BackboneElement | Active or inactive ingredient
item[x] | | 1..1 | | The product contained
itemCodeableConcept | | | CodeableConcept
itemReference | | | Reference(Substance | Medication)
isActive | | 0..1 | boolean | Active ingredient indicator
amount | | 0..1 | Ratio | Quantity of ingredient present
package | | 0..1 | BackboneElement | Details about packaged medications
container | | 0..1 | CodeableConcept | E.g. box, vial, blister-pack MedicationContainer (Example)
content | | 0..* | BackboneElement | What is in the package
item[x] | | 1..1 | | The item in the package
itemCodeableConcept | | | CodeableConcept
itemReference | | | Reference(Medication)
amount | | 0..1 | SimpleQuantity | Quantity present in the package
batch | | 0..* | BackboneElement | Identifies a single production run
lotNumber | | 0..1 | string | Identifier assigned to batch
expirationDate | | 0..1 | dateTime | When batch will expire
image | | 0..* | Attachment | Picture of the medication
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Medication xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <code><!-- 0..1 CodeableConcept Codes that identify this medication --></code>
 <status value="[code]"/><!-- 0..1 active | inactive | entered-in-error -->
 <isBrand value="[boolean]"/><!-- 0..1 True if a brand -->
 <isOverTheCounter value="[boolean]"/><!-- 0..1 True if medication does not require a prescription -->
 <manufacturer><!-- 0..1 Reference(Organization) Manufacturer of the item --></manufacturer>
 <form><!-- 0..1 CodeableConcept powder | tablets | capsule + --></form>
 <ingredient>  <!-- 0..* Active or inactive ingredient -->
  <item[x]><!-- 1..1 CodeableConcept|Reference(Substance|Medication) The product contained --></item[x]>
  <isActive value="[boolean]"/><!-- 0..1 Active ingredient indicator -->
  <amount><!-- 0..1 Ratio Quantity of ingredient present --></amount>
 </ingredient>
 <package>  <!-- 0..1 Details about packaged medications -->
  <container><!-- 0..1 CodeableConcept E.g. box, vial, blister-pack --></container>
  <content>  <!-- 0..* What is  in the package -->
   <item[x]><!-- 1..1 CodeableConcept|Reference(Medication) The item in the package --></item[x]>
   <amount><!-- 0..1 Quantity(SimpleQuantity) Quantity present in the package --></amount>
  </content>
  <batch>  <!-- 0..* Identifies a single production run -->
   <lotNumber value="[string]"/><!-- 0..1 Identifier assigned to batch -->
   <expirationDate value="[dateTime]"/><!-- 0..1 When batch will expire -->
  </batch>
 </package>
 <image><!-- 0..* Attachment Picture of the medication --></image>
</Medication>
```

JSON Template

```
{
  "resourceType" : "Medication",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "code" : { CodeableConcept }, // Codes that identify this medication
  "status" : "<code>", // active | inactive | entered-in-error
  "isBrand" : <boolean>, // True if a brand
  "isOverTheCounter" : <boolean>, // True if medication does not require a prescription
  "manufacturer" : { Reference(Organization) }, // Manufacturer of the item
  "form" : { CodeableConcept }, // powder | tablets | capsule +
  "ingredient" : [{ // Active or inactive ingredient
    // item[x]: The product contained. One of these 2:
    "itemCodeableConcept" : { CodeableConcept },
    "itemReference" : { Reference(Substance|Medication) },
    "isActive" : <boolean>, // Active ingredient indicator
    "amount" : { Ratio } // Quantity of ingredient present
  }],
  "package" : { // Details about packaged medications
    "container" : { CodeableConcept }, // E.g. box, vial, blister-pack
    "content" : [{ // What is  in the package
      // item[x]: The item in the package. One of these 2:
      "itemCodeableConcept" : { CodeableConcept },
      "itemReference" : { Reference(Medication) },
      "amount" : { Quantity(SimpleQuantity) } // Quantity present in the package
    }],
    "batch" : [{ // Identifies a single production run
      "lotNumber" : "<string>", // Identifier assigned to batch
      "expirationDate" : "<dateTime>" // When batch will expire
```

```
    }]
  },
  "image" : [{ Attachment }] // Picture of the medication
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Medication;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Medication.code [ CodeableConcept ]; # 0..1 Codes that identify this medication
  fhir:Medication.status [ code ]; # 0..1 active | inactive | entered-in-error
  fhir:Medication.isBrand [ boolean ]; # 0..1 True if a brand
  fhir:Medication.isOverTheCounter [ boolean ]; # 0..1 True if medication does not require a prescription
  fhir:Medication.manufacturer [ Reference(Organization) ]; # 0..1 Manufacturer of the item
  fhir:Medication.form [ CodeableConcept ]; # 0..1 powder | tablets | capsule +
  fhir:Medication.ingredient [ # 0..* Active or inactive ingredient
    # Medication.ingredient.item[x] : 1..1 The product contained. One of these 2
      fhir:Medication.ingredient.itemCodeableConcept [ CodeableConcept ]
      fhir:Medication.ingredient.itemReference [ Reference(Substance|Medication) ]
    fhir:Medication.ingredient.isActive [ boolean ]; # 0..1 Active ingredient indicator
    fhir:Medication.ingredient.amount [ Ratio ]; # 0..1 Quantity of ingredient present
  ], ...;
  fhir:Medication.package [ # 0..1 Details about packaged medications
    fhir:Medication.package.container [ CodeableConcept ]; # 0..1 E.g. box, vial, blister-pack
    fhir:Medication.package.content [ # 0..* What is  in the package
      # Medication.package.content.item[x] : 1..1 The item in the package. One of these 2
        fhir:Medication.package.content.itemCodeableConcept [ CodeableConcept ]
        fhir:Medication.package.content.itemReference [ Reference(Medication) ]
      fhir:Medication.package.content.amount [ Quantity(SimpleQuantity) ]; # 0..1 Quantity present in the package
    ], ...;
    fhir:Medication.package.batch [ # 0..* Identifies a single production run
      fhir:Medication.package.batch.lotNumber [ string ]; # 0..1 Identifier assigned to batch
      fhir:Medication.package.batch.expirationDate [ dateTime ]; # 0..1 When batch will expire
    ], ...;
  ];
  fhir:Medication.image [ Attachment ], ... ; # 0..* Picture of the medication
]
```

Changes since DSTU2

Medication
Medication.status | Added Element
Medication.isOverTheCounter | Added Element
Medication.form | Added Element
Medication.ingredient | Added Element
Medication.ingredient.item[x] | Added Element
Medication.ingredient.isActive | Added Element
Medication.ingredient.amount | Added Element
Medication.package.content.item[x] | Renamed from item to item[x] Add CodeableConcept
Medication.package.batch | Added Element
Medication.package.batch.lotNumber | Added Element
Medication.package.batch.expirationDate | Added Element
Medication.image | Added Element
Medication.product | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 23 tests that all execute ok. 16 fail round-trip testing and 1 r3 resources are invalid (1 errors).).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

11.5.2.1 Terminology Bindings

Path | Definition | Type | Reference
Medication.code | A coded concept that defines the type of a medication | Example | SNOMED CT Medication Codes
Medication.status | A coded concept defining if the medication is in active use | Required | MedicationStatus
Medication.form | A coded concept defining the form of a medication | Example | SNOMED CT Form Codes
Medication.package.container | A coded concept defining the kind of container a medication package is packaged in | Example | MedicationContainer

11.5.3 Best practices for using 'Medication'

Medication does not have a status. Pharmacy is evaluating formulary use cases. Feedback is encouraged to the Pharmacy working group committee.

11.5.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
code | token | Codes that identify this medication | Medication.code | 4 Resources
container | token | E.g. box, vial, blister-pack | Medication.package.container
form | token | powder | tablets | capsule + | Medication.form
ingredient | reference | The product contained | Medication.ingredient.item.as(Reference) (Medication, Substance)
ingredient-code | token | The product contained | Medication.ingredient.item.as(CodeableConcept)
manufacturer | reference | Manufacturer of the item | Medication.manufacturer (Organization)
over-the-counter | token | True if medication does not require a prescription | Medication.isOverTheCounter
package-item | reference | The item in the package | Medication.package.content.item.as(Reference) (Medication)
package-item-code | token | The item in the package | Medication.package.content.item.as(CodeableConcept)
status | token | active | inactive | entered-in-error | Medication.status

11.5.5 Resource Medication - Examples

URL: https://hl7.org/fhir/STU3/medication-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.5.5 Resource Medication - Examples

Pharmacy Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

The following are examples of Medication resources where the medication is a manufactured product:

* IV injection (inpatient order) - Manufactured Product - Vancomycin

* IV Injection - Manufactured Product - Piperacillin

* IV Chemotherapy Example - Manufactured Product - Alemtuzumab

* Oral Chemotherapy Example - Manufactured Product - Myleran

* Eye Drop Example - Manufactured Product - Timoptic

* IV Chemotherapy Example - Manufactured Product - Adcetris

* Subcutaneous injection Example - Manufactured Product - Novolog

* As needed Oral tablet - Manufactured Product - Percocet

* Over the Counter Product - Manufactured Product - Tylenol PM

* Oral Chemotherapy Example - Manufactured Product - Xeloda - includes a reference to a substance in ingredient element

The following are examples of Medication resources where the medication is a generic formulation (i.e., represents an ingredient or set of ingredients plus dosage form)

* Generic Formulation - Oxycodone

* Generic Formulation - Prednisone 5mg tablet

* Generic Formulation - Nystatin 100,000u/ml

* Generic Formulation - Lorazepam 2mg/ml injection

* Generic Formulation - Alprazolam 0.25mg tablet

* Generic Formulation - Mometasone Furorate 0.05mg/Actuat

* Generic Formulation - Fully populated example of a Medication - Chlorthalidone

* Generic Formulation - Oral Chemotherapy Example - Capecitabine – includes a reference to a substance in the Ingredient element

* Generic Formulation - Azithromycin 250mg oral capsule - includes a reference to a substance in the Ingredient element

The following are examples of Medication resources where the medication is a compounded (aka extemporaneous or magistral) product

* Compounded Preparation - KCl in D5W Admixture - includes a reference to a substance in the ingredient element

* Compounded Preparation - TPN Solution - includes a reference to a substance in the ingredient element

* Compounded Preparation - HC 1%, Salicylic Acid 5% in Glaxal Base - includes a reference to a substance in the ingredient element

The following is an example where the medication resource is minimally populated:

* Name Element Only populated - Amoxicillin 250mg/5ml Suspension

Example List:

Name element only populated - Amoxicillin 250mg/5ml Suspension | medicationexample1 | XML | JSON | Turtle
Oral Chemotherapy Example - Manufactured Product - Xeloda - includes a reference to a substance in ingredient element | medexample015 | XML | JSON | Turtle
IV injection (inpatient order) - Manufactured Product - Vancomycin | med0301 | XML | JSON | Turtle
IV Injection with a Rate Example - Manufactured Product - Piperacillin | med0302 | XML | JSON | Turtle

IV Chemotherapy Example - Manufactured Product - Alemtuzumab | med0303 | XML | JSON | Turtle
Oral Chemotherapy Example - Manufactured Product - Myleran | med0304 | XML | JSON | Turtle
Eye Drop Example - Manufactured Product - Timoptic | med0305 | XML | JSON | Turtle
IV Chemotherapy Example - Manufactured Product - Adcetris | med0306 | XML | JSON | Turtle
Subcutaneous injection Example - Manufactured Product - Novolog | med0307 | XML | JSON | Turtle
As needed Oral tablet - Manufactured Product - Percocet | med0308 | XML | JSON | Turtle
As needed dosage range - Manufactured Product - Tylenol No 1 Caplet | med0309 | XML | JSON | Turtle
Generic Formulation - Oxycodone | med0310 | XML | JSON | Turtle
Generic Formulation - Prednisone 5mg tablet | med0311 | XML | JSON | Turtle
Generic Formulation - Nystatin 100,000 u/ml suspension | med0312 | XML | JSON | Turtle
Generic Formulation - Lorazepam 2mg/ml injection | med0313 | XML | JSON | Turtle
Generic Formulation - Alprazolam 0.25mg tablet | med0314 | XML | JSON | Turtle
Generic Formulation - Mometasone Furoate 0.05mg/Actuat | med0315 | XML | JSON | Turtle
Generic Formulation - Fully populated example of a Medication - Chlorthalidone | med0316 | XML | JSON | Turtle
Compounded Preparation - KCl in D5W Admixture - includes a reference to a substance in the ingredient element | med0317 | XML | JSON | Turtle
Compounded Preparation - TPN Solution - includes a reference to a substance in the ingredient element | med0318 | XML | JSON | Turtle
Compounded Preparation - HC 1%, Salicylic Acid 5% in Glaxal Base - includes a reference to a substance in the ingredient element | med0319 | XML | JSON | Turtle
Generic Formulation - Azithromycin 250mg oral capsule - includes a reference to a substance in the Ingredient element | med0320 | XML | JSON | Turtle
Generic Formulation - Oral Chemotherapy Example - Capecitabine – includes a reference to a substance in the Ingredient element | med0321 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.5.6 Resource Medication - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/medication-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.5.6 Resource Medication - Detailed Descriptions

Pharmacy Work Group | Maturity Level: 3 | Trial Use | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the Medication resource.

Medication
Definition | This resource is primarily used for the identification and definition of a medication. It covers the ingredients and the packaging for a medication.
Control | 1..1
Medication.code
Definition | A code (or set of codes) that specify this medication, or a textual description if no code is available. Usage note: This could be a standard medication code such as a code from RxNorm, SNOMED CT, IDMP etc. It could also be a national or local formulary code, optionally with translations to other code systems.
Control | 0..1
Terminology Binding | SNOMED CT Medication Codes (Example)
Type | CodeableConcept
Summary | true
Comments | Depending on the context of use, the code that was actually selected by the user (prescriber, dispenser, etc.) will have the coding.userSelected set to true. As described in the coding datatype: "A coding may be marked as a "userSelected" if a user selected the particular coded value in a user interface (e.g. the user selects an item in a pick-list). If a user selected coding exists, it is the preferred choice for performing translations etc.
Medication.status
Definition | A code to indicate if the medication is in active use.
Control | 0..1
Terminology Binding | MedicationStatus (Required)
Type | code
Summary | true
Comments | This status is not intended to specify if a medication is part of a formulary.
Medication.isBrand
Definition | Set to true if the item is attributable to a specific manufacturer.
Control | 0..1
Type | boolean
Summary | true
Medication.isOverTheCounter
Definition | Set to true if the medication can be obtained without an order from a prescriber.
Control | 0..1
Type | boolean
Summary | true
Medication.manufacturer
Definition | Describes the details of the manufacturer of the medication product. This is not intended to represent the distributor of a medication product.
Control | 0..1

Type | Reference(Organization)
Summary | true
Medication.form
Definition | Describes the form of the item. Powder; tablets; capsule.
Control | 0..1
Terminology Binding | SNOMED CT Form Codes (Example)
Type | CodeableConcept
Comments | When Medication is referenced from MedicationRequest, this is the ordered form. When Medication is referenced within MedicationDispense, this is the dispensed form. When Medication is referenced within MedicationAdministration, this is administered form.
Medication.ingredient
Definition | Identifies a particular constituent of interest in the product.
Control | 0..*
Comments | The ingredients need not be a complete list. If an ingredient is not specified, this does not indicate whether an ingredient is present or absent. If an ingredient is specified it does not mean that all ingredients are specified. It is possible to specify both inactive and active ingredients.
Medication.ingredient.item[x]
Definition | The actual ingredient - either a substance (simple ingredient) or another medication.
Control | 1..1
Type | CodeableConcept|Reference(Substance | Medication)
[x] Note | See Choice of Data Types for further information about how to use [x]
Medication.ingredient.isActive
Definition | Indication of whether this ingredient affects the therapeutic action of the drug.
Control | 0..1
Type | boolean
Requirements | True indicates that the ingredient affects the therapeutic action of the drug (i.e. active). False indicates that the ingredient does not affect the therapeutic action of the drug (i.e. inactive).
Medication.ingredient.amount
Definition | Specifies how many (or how much) of the items there are in this Medication. For example, 250 mg per tablet. This is expressed as a ratio where the numerator is 250mg and the denominator is 1 tablet.
Control | 0..1
Type | Ratio
Medication.package
Definition | Information that only applies to packages (not products).
Control | 0..1
Medication.package.container
Definition | The kind of container that this package comes as.
Control | 0..1
Terminology Binding | MedicationContainer (Example)
Type | CodeableConcept
Medication.package.content
Definition | A set of components that go to make up the described item.
Control | 0..*
Medication.package.content.item[x]
Definition | Identifies one of the items in the package.
Control | 1..1
Type | CodeableConcept|Reference(Medication)
[x] Note | See Choice of Data Types for further information about how to use [x]
Medication.package.content.amount
Definition | The amount of the product that is in the package.
Control | 0..1
Type | SimpleQuantity
Medication.package.batch
Definition | Information about a group of medication produced or packaged from one production run.
Control | 0..*
Medication.package.batch.lotNumber
Definition | The assigned lot number of a batch of the specified product.
Control | 0..1
Type | string
Medication.package.batch.expirationDate
Definition | When this specific batch of product will expire.
Control | 0..1
Type | dateTime
Medication.image
Definition | Photo(s) or graphic representation(s) of the medication.
Control | 0..*
Type | Attachment
Requirements | Note that "image" is only applicable when the medication.code represents a physical item (e.g. Amoxil 250 mg capsule) and not an abstract item (e.g. amoxicillin 250 mg).

11.5.7 Resource Medication - Mappings

URL: https://hl7.org/fhir/STU3/medication-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.5.7 Resource Medication - Mappings

Pharmacy Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Mappings:

RIM Mapping

HL7 v2 Mapping

Mapping to NCPDP SCRIPT 10.6

Mappings for the medication resource.

11.5.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

Medication | ManufacturedProduct[classCode=ADMM]
code | .code
status | .statusCode
isBrand | true if .player.scopingRole[typeCode=MANU] exists
isOverTheCounter | NA
manufacturer | .player.scopingRole[typeCode=MANU].scoper
form | .formCode
ingredient | .scopesRole[typeCode=INGR]
item[x] | .player
isActive | NA
amount | .quantity
package | .player[classCode=CONT]
container | .code
content | .scopesRole[typeCode=CONT]
item[x] | .player
amount | .quantity
batch | participation[typeCode=CSM].role[classCode=INST].scopedRole.scoper[classCode=MMAT].id
lotNumber | .id
expirationDate | participation[typeCode=CSM].role[classCode=INST].scopedRole.scoper[classCode=MMAT].expirationTime
image | .participation[typeCode=SUBJ].role[classCode=MANU],source[classCode=OBS,moodCode=EVN,code="type of image"].value(ED.image)

11.5.7.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

Medication
code | RXO-1.1-Requested Give Code.code / RXE-2.1-Give Code.code / RXD-2.1-Dispense/Give Code.code / RXG-4.1-Give Code.code /RXA-5.1-Administered Code.code / RXC-2.1 Component Code
status
isBrand
isOverTheCounter
manufacturer | RXD-20-Substance Manufacturer Name / RXG-21-Substance Manufacturer Name / RXA-17-Substance Manufacturer Name
form | RXO-5-Requested Dosage Form / RXE-6-Give Dosage Form / RXD-6-Actual Dosage Form / RXG-8-Give Dosage Form / RXA-8-Administered Dosage Form
ingredient
item[x] | RXC-2-Component Code if medication: RXO-1-Requested Give Code / RXE-2-Give Code / RXD-2-Dispense/Give Code / RXG-4-Give Code / RXA-5-Administered Code
isActive
amount | RXC-3-Component Amount & RXC-4-Component Units if medication: RXO-2-Requested Give Amount - Minimum & RXO-4-Requested Give Units / RXO-3-Requested Give Amount - Maximum & RXO-4-Requested Give Units / RXO-11-Requested Dispense Amount & RXO-12-Requested Dispense Units / RXE-3-Give Amount - Minimum & RXE-5-Give Units / RXE-4-Give Amount - Maximum & RXE-5-Give Units / RXE-10-Dispense Amount & RXE-10-Dispense Units
package
container
content
item[x]
amount
batch
lotNumber | RXA-15 Substance Lot Number / RXG-19 Substance Lot Number
expirationDate | RXA-16 Substance Expiration Date / RXG-20 Substance Expiration Date
image

11.5.7.3 Mappings for Mapping to NCPDP SCRIPT 10.6 (http://ncpdp.org/SCRIPT10_6)

Medication | NewRx/MedicationPrescribed -or- RxFill/MedicationDispensed -or- RxHistoryResponse/MedicationDispensed -or- RxHistoryResponse/MedicationPrescribed
code | coding.code = //element(*,MedicationType)/DrugCoded/ProductCode coding.system = //element(*,MedicationType)/DrugCoded/ProductCodeQualifier coding.display = //element(*,MedicationType)/DrugDescription
status
isBrand | no mapping
isOverTheCounter
manufacturer | no mapping

form | coding.code = //element(*,DrugCodedType)/FormCode coding.system = //element(*,DrugCodedType)/FormSourceCode
ingredient
item[x] | coding.code = //element(*,MedicationType)/DrugCoded/ProductCode coding.system =
//element(*,MedicationType)/DrugCoded/ProductCodeQualifier coding.display = //element(*,MedicationType)/DrugDescription
isActive
amount | //element(*,DrugCodedType)/Strength
package | no mapping
container | no mapping
content | no mapping
item[x] | no mapping
amount | no mapping
batch | no mapping
lotNumber | no mapping
expirationDate | no mapping
image

11.5.8 Resource Medication - Design Notes

URL: https://hl7.org/fhir/STU3/medication-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.5.8 Resource Medication - Design Notes

to do

11.5.9 Resource Medication - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/medication-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.5.9 Resource Medication - Extensions & Profiles

Pharmacy Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Medication resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

11.5.9.1 Profiles

No Profiles defined for this resource

11.5.9.2 Extensions

medication-usualRoute | usualRoute | for Medication HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource

11.5.9.3 Search Extensions

No Search Extensions defined for this resource

11.5.10 Medication HL7 Extensions

URL: https://hl7.org/fhir/STU3/medication-extensions.html

Defines common extensions used with or related to the Medication resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.5.10 Medication HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

11.5.10.1 Content

Extensions:
medication-usualRoute | usualRoute : The typical route into the body for this medication.

11.5.11 Resource Medication - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/medication-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.5.11 Resource Medication - Detailed Descriptions

Pharmacy Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

R2 : R3 Converson maps for Medication.

Functional status for this map: 23 tests that all execute ok. 16 fail round-trip testing and 1 r3 resources are invalid (1 errors). (see documentation)

11.5.11.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Medication2to3" = "R2 to R3 Conversions for Medication"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Medication" alias MedicationR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Medication" alias Medication as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  Medication extends DomainResource
  input src : MedicationR2 as source
  input tgt : Medication as target

  "Medication-code" : for src.code make tgt.code
  "Medication-isBrand" : for src.isBrand make tgt.isBrand
  "Medication-manufacturer" : for src.manufacturer make tgt.manufacturer

  "Medication-product" : for src.product as vs0 then medProduct(vs0, tgt)
  "Medication-package" : for src.package as vs0 make tgt.package as vt0 then medPackage(vs0, vt0)
endgroup

group medProduct extends BackboneElement
  input src as source
  input tgt as target

  "Medication.product-form" : for src.form make tgt.form
  "Medication.product-ingredient" : for src.ingredient as vs0 make tgt.ingredient as vt0 then medIngredient(vs0, vt0)
  "Medication.product-batch" : for src.batch as vs0 make tgt.package as pkg, pkg.batch as vt0 then medBatch(vs0, vt0)
endgroup

group medIngredient extends BackboneElement
  input src as source
  input tgt as target

  "Medication.product.batch-item" : for src.item make tgt.item
  "Medication.product.batch-amount" : for src.amount make tgt.amount
endgroup

group medBatch extends BackboneElement
  input src as source
  input tgt as target

  "Medication.product.batch-lotNumber" : for src.lotNumber make tgt.lotNumber
  "Medication.product.batch-expirationDate" : for src.expirationDate make tgt.expirationDate
endgroup

group medPackage extends BackboneElement
  input src as source
  input tgt as target

  "Medication.package-container" : for src.container make tgt.container
  "Medication.package-content" : for src.content as vs0 make tgt.content as vt0 then medContent(vs0, vt0)
endgroup

group medContent extends BackboneElement
  input src as source
  input tgt as target

  "Medication.package.content-item" : for src.item make tgt.item
  "Medication.package.content-amount" : for src.amount make tgt.amount
endgroup
```

11.5.11.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Medication3to2" = "R3 to R2 Conversion for Medication"

uses "http://hl7.org/fhir/StructureDefinition/Medication" alias Medication as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Medication" alias MedicationR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  Medication extends DomainResource
  input src : Medication as source
  input tgt : MedicationR2 as target

  "Medication-code" : for src.code make tgt.code
  "Medication-isBrand" : for src.isBrand make tgt.isBrand
```

```
  "Medication-manufacturer" : for src.manufacturer make tgt.manufacturer

  "Medication.product-form" : for src.form as v make tgt.product as vt0, vt0.form as v
  "Medication.product-ingredient" : for src.ingredient as vs0 make tgt.product as vt0, vt0.ingredient as vt1 then
    medIngredient(vs0, vt1)
  "Medication-package" : for src.package as vs0 make tgt.package as vt0 then medPackage(vs0, vt0, tgt)
endgroup

group medProduct extends BackboneElement
  input src as source
  input tgt as target

  "Medication.product-form" : for src.form make tgt.form
  "Medication.product-ingredient" : for src.ingredient as vs0 make tgt.ingredient as vt0 then medIngredient(vs0, vt0)
endgroup

group medIngredient extends BackboneElement
  input src as source
  input tgt as target

  "Medication.product.batch-item" : for src.item where $this is Reference make tgt.item
  "Medication.product.batch-amount" : for src.amount make tgt.amount
endgroup

group medBatch extends BackboneElement
  input src as source
  input tgt as target

  "Medication.product.batch-lotNumber" : for src.lotNumber make tgt.lotNumber
  "Medication.product.batch-expirationDate" : for src.expirationDate make tgt.expirationDate
endgroup

group medPackage extends BackboneElement
  input src as source
  input tgt as target
  input res as target

  "Medication.package-container" : for src.container make tgt.container
  "Medication.package-content" : for src.content as vs0 make tgt.content as vt0 then medContent(vs0, vt0)
  "Medication.product-batch" : for src.batch as vs0 make res.product as vt1, vt1.batch as vt0 then medBatch(vs0, vt0)
endgroup

group medContent extends BackboneElement
  input src as source
  input tgt as target

  "Medication.package.content-item" : for src.item where $this is Reference make tgt.item
  "Medication.package.content-amount" : for src.amount make tgt.amount
endgroup
```

11.5.11.3 R3 Validation Errors

medexample012 | WARNING @ Medication.code.coding.display: value should not start or finish with whitespace (src = InstanceValidator)

11.6 Resource Immunization - Content

URL: https://hl7.org/fhir/STU3/immunization.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Public Health and Emergency Response Work Group | Maturity Level: 3 | Trial Use | Compartments: Patient, Practitioner

Describes the event of a patient being administered a vaccination or a record of a vaccination as reported by a patient, a clinician or another party and may include vaccine reaction information and what vaccination protocol was followed.

11.6.1 Scope and Usage

The Immunization resource is intended to cover the recording of current and historical administration of vaccines to patients across all healthcare disciplines in all care settings and all regions. This includes immunization of both humans and animals, but does not include the administration of non-vaccine agents, even those that may have or claim to have immunological effects.

Additionally, the Immunization resource is expected to cover key concepts related to the creation, revision and querying of a patient's immunization history. This resource - through consultation with the PHER work group - is believed to meet key use cases and information requirements as defined in the existing HL7 v2 immunization implementation guide, HL7 v3 POIZ domain and Immunization Domain Analysis Model.

11.6.2 Boundaries and Relationships

This resource references the following resources:

* Patient

* Practitioner

* Organization

* Location

* Observation

* Encounter

Administration of vaccines is intended to be handled using the Immunization resource. MedicationAdministration is intended for tracking the administration of non-vaccine medications. Some systems treat immunizations in the same way as any other medication administration. Such systems SHOULD use an immunization resource to represent these. If systems need to use a MedicationAdministration resource to capture vaccinations for workflow or other reasons, they SHOULD also expose an equivalent Immunization instance.

AllergyIntolerance and Immunization.reaction

Immunization.reaction may be an indication of an allergy or intolerance. If this is deemed to be the case, a separate AllergyIntolerance resource instance should be created to indicate it, as most systems will not query against past Immunization.reaction.

This resource is referenced by AdverseEvent, ChargeItem and ImmunizationRecommendation

11.6.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Immunization | I | | DomainResource | Immunization event information + If immunization was administered (notGiven=false) then explanation.reasonNotGiven SHALL be absent. + If immunization was not administered (notGiven=true) then there SHALL be no reaction nor explanation.reason present Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | Business identifier
status | ?!? | 1..1 | code | completed | entered-in-error Immunization Status Codes (Required)
notGiven | ?!? | 1..1 | boolean | Flag for whether immunization was given
vaccineCode | | 1..1 | CodeableConcept | Vaccine product administered Vaccine Administered Value Set (Example)
patient | | 1..1 | Reference(Patient) | Who was immunized
encounter | | 0..1 | Reference(Encounter) | Encounter administered as part of
date | | 0..1 | dateTime | Vaccination administration date
primarySource | | 1..1 | boolean | Indicates context the data was recorded in
reportOrigin | | 0..1 | CodeableConcept | Indicates the source of a secondarily reported record Immunization Origin Codes (Example)
location | | 0..1 | Reference(Location) | Where vaccination occurred
manufacturer | | 0..1 | Reference(Organization) | Vaccine manufacturer
lotNumber | | 0..1 | string | Vaccine lot number
expirationDate | | 0..1 | date | Vaccine expiration date
site | | 0..1 | CodeableConcept | Body site vaccine was administered Codes for Immunization Site of Administration (Example)
route | | 0..1 | CodeableConcept | How vaccine entered body Immunization Route Codes (Example)
doseQuantity | | 0..1 | SimpleQuantity | Amount of vaccine administered
practitioner | ? | 0..* | BackboneElement | Who performed event
role | ? | 0..1 | CodeableConcept | What type of performance was done Immunization Role Codes (Extensible)
actor | ? | 1..1 | Reference(Practitioner) | Individual who was performing
note | ? | 0..* | Annotation | Vaccination notes
explanation | | 0..1 | BackboneElement | Administration/non-administration reasons
reason | | 0..* | CodeableConcept | Why immunization occurred Immunization Reason Codes (Example)
reasonNotGiven | | 0..* | CodeableConcept | Why immunization did not occur Immunization Reasons for Not Immunizing Codes (Example)
reaction | | 0..* | BackboneElement | Details of a reaction that follows immunization
date | | 0..1 | dateTime | When reaction started
detail | | 0..1 | Reference(Observation) | Additional information on reaction
reported | | 0..1 | boolean | Indicates self-reported reaction
vaccinationProtocol | | 0..* | BackboneElement | What protocol was followed
doseSequence | | 0..1 | positiveInt | Dose number within series
description | | 0..1 | string | Details of vaccine protocol
authority | | 0..1 | Reference(Organization) | Who is responsible for protocol
series | | 0..1 | string | Name of vaccine series
seriesDoses | | 0..1 | positiveInt | Recommended number of doses for immunity
targetDisease | | 1..* | CodeableConcept | Disease immunized against Vaccination Protocol Dose Target Codes (Example)
doseStatus | | 1..1 | CodeableConcept | Indicates if dose counts towards immunity Vaccination Protocol Dose Status codes (Example)
doseStatusReason | | 0..1 | CodeableConcept | Why dose does (not) count Vaccination Protocol Dose Status Reason codes (Example)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Immunization xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <status value="[code]"/><!-- 1..1 completed | entered-in-error -->
 <notGiven value="[boolean]"/><!-- 1..1 Flag for whether immunization was given -->
 <vaccineCode><!-- 1..1 CodeableConcept Vaccine product administered --></vaccineCode>
 <patient><!-- 1..1 Reference(Patient) Who was immunized --></patient>
 <encounter><!-- 0..1 Reference(Encounter) Encounter administered as part of --></encounter>
 <date value="[dateTime]"/><!-- 0..1 Vaccination administration date -->
 <primarySource value="[boolean]"/><!-- 1..1 Indicates context the data was recorded in -->
 <reportOrigin><!-- 0..1 CodeableConcept Indicates the source of a secondarily reported record --></reportOrigin>
 <location><!-- 0..1 Reference(Location) Where vaccination occurred --></location>
 <manufacturer><!-- 0..1 Reference(Organization) Vaccine manufacturer --></manufacturer>
 <lotNumber value="[string]"/><!-- 0..1 Vaccine lot number -->
 <expirationDate value="[date]"/><!-- 0..1 Vaccine expiration date -->
 <site><!-- 0..1 CodeableConcept Body site vaccine  was administered --></site>
 <route><!-- 0..1 CodeableConcept How vaccine entered body --></route>
 <doseQuantity><!-- 0..1 Quantity(SimpleQuantity) Amount of vaccine administered --></doseQuantity>
 <practitioner>  <!-- 0..* Who performed event -->
  <role><!-- 0..1 CodeableConcept What type of performance was done --></role>
  <actor><!-- 1..1 Reference(Practitioner) Individual who was performing --></actor>
 </practitioner>
 <note><!-- 0..* Annotation Vaccination notes --></note>
 <explanation>  <!-- 0..1 Administration/non-administration reasons -->
  <reason><!-- 0..* CodeableConcept Why immunization occurred --></reason>
  <reasonNotGiven><!-- 0..* CodeableConcept Why immunization did not occur --></reasonNotGiven>
 </explanation>
 <reaction>  <!-- 0..* Details of a reaction that follows immunization -->
  <date value="[dateTime]"/><!-- 0..1 When reaction started -->
  <detail><!-- 0..1 Reference(Observation) Additional information on reaction --></detail>
  <reported value="[boolean]"/><!-- 0..1 Indicates self-reported reaction -->
 </reaction>
 <vaccinationProtocol>  <!-- 0..* What protocol was followed -->
  <doseSequence value="[positiveInt]"/><!-- 0..1 Dose number within series -->
  <description value="[string]"/><!-- 0..1 Details of vaccine protocol -->
  <authority><!-- 0..1 Reference(Organization) Who is responsible for protocol --></authority>
  <series value="[string]"/><!-- 0..1 Name of vaccine series -->
  <seriesDoses value="[positiveInt]"/><!-- 0..1 Recommended number of doses for immunity -->
  <targetDisease><!-- 1..* CodeableConcept Disease immunized against --></targetDisease>
  <doseStatus><!-- 1..1 CodeableConcept Indicates if dose counts towards immunity --></doseStatus>
  <doseStatusReason><!-- 0..1 CodeableConcept Why dose does (not) count --></doseStatusReason>
 </vaccinationProtocol>
</Immunization>
```

JSON Template

```
{
  "resourceType" : "Immunization",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "status" : "<code>", // R!  completed | entered-in-error
  "notGiven" : <boolean>, // R!  Flag for whether immunization was given
  "vaccineCode" : { CodeableConcept }, // R!  Vaccine product administered
  "patient" : { Reference(Patient) }, // R!  Who was immunized
  "encounter" : { Reference(Encounter) }, // Encounter administered as part of
  "date" : "<dateTime>", // Vaccination administration date
  "primarySource" : <boolean>, // R!  Indicates context the data was recorded in
  "reportOrigin" : { CodeableConcept }, // Indicates the source of a secondarily reported record
  "location" : { Reference(Location) }, // Where vaccination occurred
  "manufacturer" : { Reference(Organization) }, // Vaccine manufacturer
  "lotNumber" : "<string>", // Vaccine lot number
  "expirationDate" : "<date>", // Vaccine expiration date
  "site" : { CodeableConcept }, // Body site vaccine  was administered
  "route" : { CodeableConcept }, // How vaccine entered body
  "doseQuantity" : { Quantity(SimpleQuantity) }, // Amount of vaccine administered
  "practitioner" : [{ // Who performed event
    "role" : { CodeableConcept }, // What type of performance was done
    "actor" : { Reference(Practitioner) } // R!  Individual who was performing
  }],
  "note" : [{ Annotation }], // Vaccination notes
  "explanation" : { // Administration/non-administration reasons
    "reason" : [{ CodeableConcept }], // Why immunization occurred
    "reasonNotGiven" : [{ CodeableConcept }] // Why immunization did not occur
  },
  "reaction" : [{ // Details of a reaction that follows immunization
    "date" : "<dateTime>", // When reaction started
    "detail" : { Reference(Observation) }, // Additional information on reaction
    "reported" : <boolean> // Indicates self-reported reaction
  }],
  "vaccinationProtocol" : [{ // What protocol was followed
    "doseSequence" : "<positiveInt>", // Dose number within series
    "description" : "<string>", // Details of vaccine protocol
    "authority" : { Reference(Organization) }, // Who is responsible for protocol
    "series" : "<string>", // Name of vaccine series
    "seriesDoses" : "<positiveInt>", // Recommended number of doses for immunity
    "targetDisease" : [{ CodeableConcept }], // R!  Disease immunized against
```

```
    "doseStatus" : { CodeableConcept }, // R!  Indicates if dose counts towards immunity
    "doseStatusReason" : { CodeableConcept } // Why dose does (not) count
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Immunization;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Immunization.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:Immunization.status [ code ]; # 1..1 completed | entered-in-error
  fhir:Immunization.notGiven [ boolean ]; # 1..1 Flag for whether immunization was given
  fhir:Immunization.vaccineCode [ CodeableConcept ]; # 1..1 Vaccine product administered
  fhir:Immunization.patient [ Reference(Patient) ]; # 1..1 Who was immunized
  fhir:Immunization.encounter [ Reference(Encounter) ]; # 0..1 Encounter administered as part of
  fhir:Immunization.date [ dateTime ]; # 0..1 Vaccination administration date
  fhir:Immunization.primarySource [ boolean ]; # 1..1 Indicates context the data was recorded in
  fhir:Immunization.reportOrigin [ CodeableConcept ]; # 0..1 Indicates the source of a secondarily reported record
  fhir:Immunization.location [ Reference(Location) ]; # 0..1 Where vaccination occurred
  fhir:Immunization.manufacturer [ Reference(Organization) ]; # 0..1 Vaccine manufacturer
  fhir:Immunization.lotNumber [ string ]; # 0..1 Vaccine lot number
  fhir:Immunization.expirationDate [ date ]; # 0..1 Vaccine expiration date
  fhir:Immunization.site [ CodeableConcept ]; # 0..1 Body site vaccine  was administered
  fhir:Immunization.route [ CodeableConcept ]; # 0..1 How vaccine entered body
  fhir:Immunization.doseQuantity [ Quantity(SimpleQuantity) ]; # 0..1 Amount of vaccine administered
  fhir:Immunization.practitioner [ # 0..* Who performed event
    fhir:Immunization.practitioner.role [ CodeableConcept ]; # 0..1 What type of performance was done
    fhir:Immunization.practitioner.actor [ Reference(Practitioner) ]; # 1..1 Individual who was performing
  ], ...;
  fhir:Immunization.note [ Annotation ], ... ; # 0..* Vaccination notes
  fhir:Immunization.explanation [ # 0..1 Administration/non-administration reasons
    fhir:Immunization.explanation.reason [ CodeableConcept ], ... ; # 0..* Why immunization occurred
    fhir:Immunization.explanation.reasonNotGiven [ CodeableConcept ], ... ; # 0..* Why immunization did not occur
  ];
  fhir:Immunization.reaction [ # 0..* Details of a reaction that follows immunization
    fhir:Immunization.reaction.date [ dateTime ]; # 0..1 When reaction started
    fhir:Immunization.reaction.detail [ Reference(Observation) ]; # 0..1 Additional information on reaction
    fhir:Immunization.reaction.reported [ boolean ]; # 0..1 Indicates self-reported reaction
  ], ...;
  fhir:Immunization.vaccinationProtocol [ # 0..* What protocol was followed
    fhir:Immunization.vaccinationProtocol.doseSequence [ positiveInt ]; # 0..1 Dose number within series
    fhir:Immunization.vaccinationProtocol.description [ string ]; # 0..1 Details of vaccine protocol
    fhir:Immunization.vaccinationProtocol.authority [ Reference(Organization) ]; # 0..1 Who is responsible for protocol
    fhir:Immunization.vaccinationProtocol.series [ string ]; # 0..1 Name of vaccine series
    fhir:Immunization.vaccinationProtocol.seriesDoses [ positiveInt ]; # 0..1 Recommended number of doses for immunity
    fhir:Immunization.vaccinationProtocol.targetDisease [ CodeableConcept ], ... ; # 1..* Disease immunized against
    fhir:Immunization.vaccinationProtocol.doseStatus [ CodeableConcept ]; # 1..1 Indicates if dose counts towards
    immunity
    fhir:Immunization.vaccinationProtocol.doseStatusReason [ CodeableConcept ]; # 0..1 Why dose does (not) count
  ], ...;
]
```

Changes since DSTU2

Immunization

Immunization.status | Change value set from http://hl7.org/fhir/ValueSet/medication-admin-status to http://hl7.org/fhir/ValueSet/immunization-status

Immunization.notGiven | Renamed from wasNotGiven to notGiven

Immunization.primarySource | Added Element

Immunization.reportOrigin | Added Element

Immunization.practitioner | Added Element

Immunization.practitioner.role | Added Element

Immunization.practitioner.actor | Added Element

Immunization.vaccinationProtocol.doseSequence | Min Cardinality changed from 1 to 0

Immunization.reported | deleted

Immunization.performer | deleted

Immunization.requester | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 2 tests that all execute ok. 2 fail round-trip testing and 2 r3 resources are invalid (2 errors).).

Structure

Name | Flags | Card. | Type | Description & Constraints

Immunization | I | | DomainResource | Immunization event information + If immunization was administered (notGiven=false) then explanation.reasonNotGiven SHALL be absent. + If immunization was not administered (notGiven=true) then there SHALL be no reaction nor explanation.reason present Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | | 0..* | Identifier | Business identifier

status | ?!? | 1..1 | code | completed | entered-in-error Immunization Status Codes (Required)

notGiven | ?!? | 1..1 | boolean | Flag for whether immunization was given
vaccineCode | | 1..1 | CodeableConcept | Vaccine product administered Vaccine Administered Value Set (Example)
patient | | 1..1 | Reference(Patient) | Who was immunized
encounter | | 0..1 | Reference(Encounter) | Encounter administered as part of
date | | 0..1 | dateTime | Vaccination administration date
primarySource | | 1..1 | boolean | Indicates context the data was recorded in
reportOrigin | | 0..1 | CodeableConcept | Indicates the source of a secondarily reported record Immunization Origin Codes (Example)
location | | 0..1 | Reference(Location) | Where vaccination occurred
manufacturer | | 0..1 | Reference(Organization) | Vaccine manufacturer
lotNumber | | 0..1 | string | Vaccine lot number
expirationDate | | 0..1 | date | Vaccine expiration date
site | | 0..1 | CodeableConcept | Body site vaccine was administered Codes for Immunization Site of Administration (Example)
route | | 0..1 | CodeableConcept | How vaccine entered body Immunization Route Codes (Example)
doseQuantity | | 0..1 | SimpleQuantity | Amount of vaccine administered
practitioner | ? | 0..* | BackboneElement | Who performed event
role | ? | 0..1 | CodeableConcept | What type of performance was done Immunization Role Codes (Extensible)
actor | ? | 1..1 | Reference(Practitioner) | Individual who was performing
note | ? | 0..* | Annotation | Vaccination notes
explanation | | 0..1 | BackboneElement | Administration/non-administration reasons
reason | | 0..* | CodeableConcept | Why immunization occurred Immunization Reason Codes (Example)
reasonNotGiven | | 0..* | CodeableConcept | Why immunization did not occur Immunization Reasons for Not Immunizing Codes (Example)
reaction | | 0..* | BackboneElement | Details of a reaction that follows immunization
date | | 0..1 | dateTime | When reaction started
detail | | 0..1 | Reference(Observation) | Additional information on reaction
reported | | 0..1 | boolean | Indicates self-reported reaction
vaccinationProtocol | | 0..* | BackboneElement | What protocol was followed
doseSequence | | 0..1 | positiveInt | Dose number within series
description | | 0..1 | string | Details of vaccine protocol
authority | | 0..1 | Reference(Organization) | Who is responsible for protocol
series | | 0..1 | string | Name of vaccine series
seriesDoses | | 0..1 | positiveInt | Recommended number of doses for immunity
targetDisease | | 1..* | CodeableConcept | Disease immunized against Vaccination Protocol Dose Target Codes (Example)
doseStatus | | 1..1 | CodeableConcept | Indicates if dose counts towards immunity Vaccination Protocol Dose Status codes (Example)
doseStatusReason | | 0..1 | CodeableConcept | Why dose does (not) count Vaccination Protocol Dose Status Reason codes (Example)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Immunization xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <status value="[code]"/><!-- 1..1 completed | entered-in-error -->
 <notGiven value="[boolean]"/><!-- 1..1 Flag for whether immunization was given -->
 <vaccineCode><!-- 1..1 CodeableConcept Vaccine product administered --></vaccineCode>
 <patient><!-- 1..1 Reference(Patient) Who was immunized --></patient>
 <encounter><!-- 0..1 Reference(Encounter) Encounter administered as part of --></encounter>
 <date value="[dateTime]"/><!-- 0..1 Vaccination administration date -->
 <primarySource value="[boolean]"/><!-- 1..1 Indicates context the data was recorded in -->
 <reportOrigin><!-- 0..1 CodeableConcept Indicates the source of a secondarily reported record --></reportOrigin>
 <location><!-- 0..1 Reference(Location) Where vaccination occurred --></location>
 <manufacturer><!-- 0..1 Reference(Organization) Vaccine manufacturer --></manufacturer>
 <lotNumber value="[string]"/><!-- 0..1 Vaccine lot number -->
 <expirationDate value="[date]"/><!-- 0..1 Vaccine expiration date -->
 <site><!-- 0..1 CodeableConcept Body site vaccine  was administered --></site>
 <route><!-- 0..1 CodeableConcept How vaccine entered body --></route>
 <doseQuantity><!-- 0..1 Quantity(SimpleQuantity) Amount of vaccine administered --></doseQuantity>
 <practitioner>  <!-- 0..* Who performed event -->
  <role><!-- 0..1 CodeableConcept What type of performance was done --></role>
  <actor><!-- 1..1 Reference(Practitioner) Individual who was performing --></actor>
 </practitioner>
 <note><!-- 0..* Annotation Vaccination notes --></note>
 <explanation>  <!-- 0..1 Administration/non-administration reasons -->
  <reason><!-- 0..* CodeableConcept Why immunization occurred --></reason>
  <reasonNotGiven><!-- 0..* CodeableConcept Why immunization did not occur --></reasonNotGiven>
 </explanation>
 <reaction>  <!-- 0..* Details of a reaction that follows immunization -->
  <date value="[dateTime]"/><!-- 0..1 When reaction started -->
  <detail><!-- 0..1 Reference(Observation) Additional information on reaction --></detail>
  <reported value="[boolean]"/><!-- 0..1 Indicates self-reported reaction -->
 </reaction>
 <vaccinationProtocol>  <!-- 0..* What protocol was followed -->
  <doseSequence value="[positiveInt]"/><!-- 0..1 Dose number within series -->
  <description value="[string]"/><!-- 0..1 Details of vaccine protocol -->
  <authority><!-- 0..1 Reference(Organization) Who is responsible for protocol --></authority>
  <series value="[string]"/><!-- 0..1 Name of vaccine series -->
  <seriesDoses value="[positiveInt]"/><!-- 0..1 Recommended number of doses for immunity -->
  <targetDisease><!-- 1..* CodeableConcept Disease immunized against --></targetDisease>
  <doseStatus><!-- 1..1 CodeableConcept Indicates if dose counts towards immunity --></doseStatus>
  <doseStatusReason><!-- 0..1 CodeableConcept Why dose does (not) count --></doseStatusReason>
```

```
 </vaccinationProtocol>
</Immunization>
```

JSON Template

```
{
  "resourceType" : "Immunization",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "status" : "<code>", // R!  completed | entered-in-error
  "notGiven" : <boolean>, // R!  Flag for whether immunization was given
  "vaccineCode" : { CodeableConcept }, // R!  Vaccine product administered
  "patient" : { Reference(Patient) }, // R!  Who was immunized
  "encounter" : { Reference(Encounter) }, // Encounter administered as part of
  "date" : "<dateTime>", // Vaccination administration date
  "primarySource" : <boolean>, // R!  Indicates context the data was recorded in
  "reportOrigin" : { CodeableConcept }, // Indicates the source of a secondarily reported record
  "location" : { Reference(Location) }, // Where vaccination occurred
  "manufacturer" : { Reference(Organization) }, // Vaccine manufacturer
  "lotNumber" : "<string>", // Vaccine lot number
  "expirationDate" : "<date>", // Vaccine expiration date
  "site" : { CodeableConcept }, // Body site vaccine  was administered
  "route" : { CodeableConcept }, // How vaccine entered body
  "doseQuantity" : { Quantity(SimpleQuantity) }, // Amount of vaccine administered
  "practitioner" : [{ // Who performed event
    "role" : { CodeableConcept }, // What type of performance was done
    "actor" : { Reference(Practitioner) } // R!  Individual who was performing
  }],
  "note" : [{ Annotation }], // Vaccination notes
  "explanation" : { // Administration/non-administration reasons
    "reason" : [{ CodeableConcept }], // Why immunization occurred
    "reasonNotGiven" : [{ CodeableConcept }] // Why immunization did not occur
  },
  "reaction" : [{ // Details of a reaction that follows immunization
    "date" : "<dateTime>", // When reaction started
    "detail" : { Reference(Observation) }, // Additional information on reaction
    "reported" : <boolean> // Indicates self-reported reaction
  }],
  "vaccinationProtocol" : [{ // What protocol was followed
    "doseSequence" : "<positiveInt>", // Dose number within series
    "description" : "<string>", // Details of vaccine protocol
    "authority" : { Reference(Organization) }, // Who is responsible for protocol
    "series" : "<string>", // Name of vaccine series
    "seriesDoses" : "<positiveInt>", // Recommended number of doses for immunity
    "targetDisease" : [{ CodeableConcept }], // R!  Disease immunized against
    "doseStatus" : { CodeableConcept }, // R!  Indicates if dose counts towards immunity
    "doseStatusReason" : { CodeableConcept } // Why dose does (not) count
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Immunization;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Immunization.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:Immunization.status [ code ]; # 1..1 completed | entered-in-error
  fhir:Immunization.notGiven [ boolean ]; # 1..1 Flag for whether immunization was given
  fhir:Immunization.vaccineCode [ CodeableConcept ]; # 1..1 Vaccine product administered
  fhir:Immunization.patient [ Reference(Patient) ]; # 1..1 Who was immunized
  fhir:Immunization.encounter [ Reference(Encounter) ]; # 0..1 Encounter administered as part of
  fhir:Immunization.date [ dateTime ]; # 0..1 Vaccination administration date
  fhir:Immunization.primarySource [ boolean ]; # 1..1 Indicates context the data was recorded in
  fhir:Immunization.reportOrigin [ CodeableConcept ]; # 0..1 Indicates the source of a secondarily reported record
  fhir:Immunization.location [ Reference(Location) ]; # 0..1 Where vaccination occurred
  fhir:Immunization.manufacturer [ Reference(Organization) ]; # 0..1 Vaccine manufacturer
  fhir:Immunization.lotNumber [ string ]; # 0..1 Vaccine lot number
  fhir:Immunization.expirationDate [ date ]; # 0..1 Vaccine expiration date
  fhir:Immunization.site [ CodeableConcept ]; # 0..1 Body site vaccine  was administered
  fhir:Immunization.route [ CodeableConcept ]; # 0..1 How vaccine entered body
  fhir:Immunization.doseQuantity [ Quantity(SimpleQuantity) ]; # 0..1 Amount of vaccine administered
  fhir:Immunization.practitioner [ # 0..* Who performed event
    fhir:Immunization.practitioner.role [ CodeableConcept ]; # 0..1 What type of performance was done
    fhir:Immunization.practitioner.actor [ Reference(Practitioner) ]; # 1..1 Individual who was performing
  ], ...;
  fhir:Immunization.note [ Annotation ], ... ; # 0..* Vaccination notes
  fhir:Immunization.explanation [ # 0..1 Administration/non-administration reasons
    fhir:Immunization.explanation.reason [ CodeableConcept ], ... ; # 0..* Why immunization occurred
    fhir:Immunization.explanation.reasonNotGiven [ CodeableConcept ], ... ; # 0..* Why immunization did not occur
  ];
  fhir:Immunization.reaction [ # 0..* Details of a reaction that follows immunization
    fhir:Immunization.reaction.date [ dateTime ]; # 0..1 When reaction started
    fhir:Immunization.reaction.detail [ Reference(Observation) ]; # 0..1 Additional information on reaction
    fhir:Immunization.reaction.reported [ boolean ]; # 0..1 Indicates self-reported reaction
  ], ...;
```

```
  fhir:Immunization.vaccinationProtocol [ # 0..* What protocol was followed
    fhir:Immunization.vaccinationProtocol.doseSequence [ positiveInt ]; # 0..1 Dose number within series
    fhir:Immunization.vaccinationProtocol.description [ string ]; # 0..1 Details of vaccine protocol
    fhir:Immunization.vaccinationProtocol.authority [ Reference(Organization) ]; # 0..1 Who is responsible for protocol
    fhir:Immunization.vaccinationProtocol.series [ string ]; # 0..1 Name of vaccine series
    fhir:Immunization.vaccinationProtocol.seriesDoses [ positiveInt ]; # 0..1 Recommended number of doses for immunity
    fhir:Immunization.vaccinationProtocol.targetDisease [ CodeableConcept ], ... ; # 1..* Disease immunized against
    fhir:Immunization.vaccinationProtocol.doseStatus [ CodeableConcept ]; # 1..1 Indicates if dose counts towards
    immunity
    fhir:Immunization.vaccinationProtocol.doseStatusReason [ CodeableConcept ]; # 0..1 Why dose does (not) count
  ], ...;
]
```

Changes since DSTU2

Immunization
Immunization.status | Change value set from http://hl7.org/fhir/ValueSet/medication-admin-status to http://hl7.org/fhir/ValueSet/immunization-status
Immunization.notGiven | Renamed from wasNotGiven to notGiven
Immunization.primarySource | Added Element
Immunization.reportOrigin | Added Element
Immunization.practitioner | Added Element
Immunization.practitioner.role | Added Element
Immunization.practitioner.actor | Added Element
Immunization.vaccinationProtocol.doseSequence | Min Cardinality changed from 1 to 0
Immunization.reported | deleted
Immunization.performer | deleted
Immunization.requester | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 2 tests that all execute ok. 2 fail round-trip testing and 2 r3 resources are invalid (2 errors).).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

11.6.3.1 Terminology Bindings

Path | Definition | Type | Reference
Immunization.status | A set of codes indicating the current status of an Immunization | Required | Immunization Status Codes
Immunization.vaccineCode | The code for vaccine product administered | Example | Vaccine Administered Value Set
Immunization.reportOrigin | The source of the data for a record which is not from a primary source. | Example | Immunization Origin Codes
Immunization.site | The site at which the vaccine was administered | Example | Codes for Immunization Site of Administration
Immunization.route | The route by which the vaccine was administered | Example | Immunization Route Codes
Immunization.practitioner.role | The role a practitioner plays in the immunization event | Extensible | Immunization Role Codes
Immunization.explanation.reason | The reason why a vaccine was administered | Example | Immunization Reason Codes
Immunization.explanation.reasonNotGiven | The reason why a vaccine was not administered | Example | Immunization Reasons for Not Immunizing Codes
Immunization.vaccinationProtocol.targetDisease | The disease target of the vaccination protocol | Example | Vaccination Protocol Dose Target Codes
Immunization.vaccinationProtocol.doseStatus | The status of the vaccination protocol (i.e. should this count) | Example | Vaccination Protocol Dose Status codes
Immunization.vaccinationProtocol.doseStatusReason | The reason for the determining if a vaccination should count or why vaccination should not count. | Example | Vaccination Protocol Dose Status Reason codes

11.6.3.2 Constraints

* imm-1: If immunization was not administered (notGiven=true) then there SHALL be no reaction nor explanation.reason present (expression : (notGiven = true).not() or (reaction.empty() and explanation.reason.empty()))

* imm-2: If immunization was administered (notGiven=false) then explanation.reasonNotGiven SHALL be absent. (expression : (notGiven = true) or explanation.reasonNotGiven.empty())

11.6.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
date | date | Vaccination (non)-Administration Date | Immunization.date | 18 Resources
dose-sequence | number | Dose number within series | Immunization.vaccinationProtocol.doseSequence
identifier | token | Business identifier | Immunization.identifier | 26 Resources
location | reference | The service delivery location or facility in which the vaccine was / was to be administered | Immunization.location (Location)
lot-number | string | Vaccine Lot Number | Immunization.lotNumber
manufacturer | reference | Vaccine Manufacturer | Immunization.manufacturer (Organization)
notgiven | token | Administrations which were not given | Immunization.notGiven
patient | reference | The patient for the vaccination record | Immunization.patient (Patient) | 31 Resources
practitioner | reference | The practitioner who played a role in the vaccination | Immunization.practitioner.actor (Practitioner)
reaction | reference | Additional information on reaction | Immunization.reaction.detail (Observation)
reaction-date | date | When reaction started | Immunization.reaction.date

reason | token | Why immunization occurred | Immunization.explanation.reason
reason-not-given | token | Explanation of reason vaccination was not administered | Immunization.explanation.reasonNotGiven
status | token | Immunization event status | Immunization.status
vaccine-code | token | Vaccine Product Administered | Immunization.vaccineCode

11.6.5 Resource Immunization - Examples

URL: https://hl7.org/fhir/STU3/immunization-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.6.5 Resource Immunization - Examples

Public Health and Emergency Response Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner

Example List:

Example of Immunization | example | XML | JSON | Turtle
Not Given Immunization Example | notGiven | XML | JSON | Turtle
Example of a Historical Example | historical | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.6.6 Resource Immunization - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/immunization-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.6.6 Resource Immunization - Detailed Descriptions

Public Health and Emergency Response Work Group | Maturity Level: 3 | Trial Use | Compartments: Patient, Practitioner

Detailed Descriptions for the elements in the Immunization resource.

Immunization
Definition | Describes the event of a patient being administered a vaccination or a record of a vaccination as reported by a patient, a clinician or another party and may include vaccine reaction information and what vaccination protocol was followed.
Control | 1..1
Invariants | Defined on this element imm-1: If immunization was not administered (notGiven=true) then there SHALL be no reaction nor explanation.reason present (expression : (notGiven = true).not() or (reaction.empty() and explanation.reason.empty()), xpath: not(f:notGiven/@value=true() and (count(f:reaction) > 0 or exists(f:explanation/f:reason)))) imm-2: If immunization was administered (notGiven=false) then explanation.reasonNotGiven SHALL be absent. (expression : (notGiven = true) or explanation.reasonNotGiven.empty(), xpath: not(f:notGiven/@value=false() and exists(f:explanation/f:reasonNotGiven)))
Immunization.identifier
Definition | A unique identifier assigned to this immunization record.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Immunization.status
Definition | Indicates the current status of the vaccination event.
Control | 1..1
Terminology Binding | Immunization Status Codes (Required)
Type | code
Is Modifier | true
Summary | true
Comments | Will generally be set to show that the immunization has been completed. This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
Immunization.notGiven
Definition | Indicates if the vaccination was or was not given.
Control | 1..1
Type | boolean
Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because it indicates that an immunization didn't happen.
Immunization.vaccineCode
Definition | Vaccine that was administered or was to be administered.
Control | 1..1
Terminology Binding | Vaccine Administered Value Set (Example)
Type | CodeableConcept
Immunization.patient
Definition | The patient who either received or did not receive the immunization.
Control | 1..1
Type | Reference(Patient)
Immunization.encounter

Definition | The visit or admission or other contact between patient and health care provider the immunization was performed as part of.
Control | 0..1
Type | Reference(Encounter)
Immunization.date
Definition | Date vaccine administered or was to be administered.
Control | 0..1
Type | dateTime
Comments | When immunizations are given a specific date and time should always be known. When immunizations are patient reported, a specific date might not be known. Although partial dates are allowed, an adult patient might not be able to recall the year a childhood immunization was given.
Immunization.primarySource
Definition | An indication that the content of the record is based on information from the person who administered the vaccine. This reflects the context under which the data was originally recorded.
Control | 1..1
Type | boolean
Comments | Reflects the “reliability” of the content.
Immunization.reportOrigin
Definition | The source of the data when the report of the immunization event is not based on information from the person who administered the vaccine.
Control | 0..1
Terminology Binding | Immunization Origin Codes (Example)
Type | CodeableConcept
Comments | Should not be populated if primarySource = True, will not be required even if primarySource = False.
Immunization.location
Definition | The service delivery location where the vaccine administration occurred.
Control | 0..1
Type | Reference(Location)
Immunization.manufacturer
Definition | Name of vaccine manufacturer.
Control | 0..1
Type | Reference(Organization)
Immunization.lotNumber
Definition | Lot number of the vaccine product.
Control | 0..1
Type | string
Immunization.expirationDate
Definition | Date vaccine batch expires.
Control | 0..1
Type | date
Immunization.site
Definition | Body site where vaccine was administered.
Control | 0..1
Terminology Binding | Codes for Immunization Site of Administration (Example)
Type | CodeableConcept
Immunization.route
Definition | The path by which the vaccine product is taken into the body.
Control | 0..1
Terminology Binding | Immunization Route Codes (Example)
Type | CodeableConcept
Immunization.doseQuantity
Definition | The quantity of vaccine product that was administered.
Control | 0..1
Type | SimpleQuantity
Immunization.practitioner
Definition | Indicates who or what performed the event.
Control | 0..*
Summary | true
Immunization.practitioner.role
Definition | Describes the type of performance (e.g. ordering provider, administering provider, etc.).
Control | 0..1
Terminology Binding | Immunization Role Codes (Extensible)
Type | CodeableConcept
Summary | true
Immunization.practitioner.actor
Definition | The device, practitioner, etc. who performed the action.
Control | 1..1
Type | Reference(Practitioner)
Summary | true
Immunization.note
Definition | Extra information about the immunization that is not conveyed by the other attributes.
Control | 0..*
Type | Annotation
Summary | true

Immunization.explanation
Definition | Reasons why a vaccine was or was not administered.
Control | 0..1
Immunization.explanation.reason
Definition | Reasons why a vaccine was administered.
Control | 0..*
Terminology Binding | Immunization Reason Codes (Example)
Type | CodeableConcept
Immunization.explanation.reasonNotGiven
Definition | Reason why a vaccine was not administered.
Control | 0..*
Terminology Binding | Immunization Reasons for Not Immunizing Codes (Example)
Type | CodeableConcept
Immunization.reaction
Definition | Categorical data indicating that an adverse event is associated in time to an immunization.
Control | 0..*
Comments | A reaction may be an indication of an allergy or intolerance and, if this is determined to be the case, it should be recorded as a new AllergyIntolerance resource instance as most systems will not query against past Immunization.reaction elements.
Immunization.reaction.date
Definition | Date of reaction to the immunization.
Control | 0..1
Type | dateTime
Immunization.reaction.detail
Definition | Details of the reaction.
Control | 0..1
Type | Reference(Observation)
Immunization.reaction.reported
Definition | Self-reported indicator.
Control | 0..1
Type | boolean
Immunization.vaccinationProtocol
Definition | Contains information about the protocol(s) under which the vaccine was administered.
Control | 0..*
Immunization.vaccinationProtocol.doseSequence
Definition | Nominal position in a series.
Control | 0..1
Type | positiveInt
Immunization.vaccinationProtocol.description
Definition | Contains the description about the protocol under which the vaccine was administered.
Control | 0..1
Type | string
Immunization.vaccinationProtocol.authority
Definition | Indicates the authority who published the protocol. E.g. ACIP.
Control | 0..1
Type | Reference(Organization)
Immunization.vaccinationProtocol.series
Definition | One possible path to achieve presumed immunity against a disease - within the context of an authority.
Control | 0..1
Type | string
Immunization.vaccinationProtocol.seriesDoses
Definition | The recommended number of doses to achieve immunity.
Control | 0..1
Type | positiveInt
Immunization.vaccinationProtocol.targetDisease
Definition | The targeted disease.
Control | 1..*
Terminology Binding | Vaccination Protocol Dose Target Codes (Example)
Type | CodeableConcept
Immunization.vaccinationProtocol.doseStatus
Definition | Indicates if the immunization event should "count" against the protocol.
Control | 1..1
Terminology Binding | Vaccination Protocol Dose Status codes (Example)
Type | CodeableConcept
Comments | May need to differentiate between status declarations by a provider vs. a CDS engine.
Immunization.vaccinationProtocol.doseStatusReason
Definition | Provides an explanation as to why an immunization event should or should not count against the protocol.
Control | 0..1
Terminology Binding | Vaccination Protocol Dose Status Reason codes (Example)
Type | CodeableConcept

11.6.7 Resource Immunization - Mappings

URL: https://hl7.org/fhir/STU3/immunization-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.6.7 Resource Immunization - Mappings

Public Health and Emergency Response Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner

Mappings:

RIM Mapping

CDA (R2)

HL7 v2 Mapping

Workflow Mapping

Mappings for the immunization resource.

11.6.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

Immunization | SubstanceAdministration
identifier | .id
status | statusCode
notGiven | [actionNegationInd=true].reasonCode
vaccineCode | .code
patient | .partipication[ttypeCode=].role
encounter | component->EncounterEvent
date | .effectiveTime
primarySource | immunization.uncertaintycode (if primary source=false, uncertainty=U)
reportOrigin | .participation[typeCode=INF].role[classCode=PAT] (this syntax for self-reported) .participation[typeCode=INF].role[classCode=LIC] (this syntax for health care professional) .participation[typeCode=INF].role[classCode=PRS] (this syntax for family member)
location | .participation[typeCode=LOC].COCT_MT240000UV
manufacturer | .participation[typeCode=CSM].role[classCode=INST].scopedRole.scoper[classCode=ORG]
lotNumber | .participation[typeCode=CSM].role[classCode=INST].scopedRole.scoper[classCode=MMAT].id
expirationDate | .participation[typeCode=CSM].role[classCode=INST].scopedRole.scoper[classCode=MMAT].expirationTime
site | observation.targetSiteCode
route | .routeCode
doseQuantity | .doseQuantity
practitioner | .participation[typeCode=PRF].role[scoper.determinerCode=INSTANCE]
role | .participation.functionCode
actor | .player
note | note
explanation | n/a
reason | [actionNegationInd=false].reasonCode
reasonNotGiven | [actionNegationInd=true].reasonCode
reaction | Observation[classCode=obs].code
date | .effectiveTime
detail | .value
reported | .participation[typeCode=INF].role[classCode=PAT] (this syntax for self-reported=true)
vaccinationProtocol | outboundRelationship[typeCode=INST].target[classCode=SBADM]
doseSequence | outboundRelationship[typeCode=INST].target[classCode=SBADM].inboundRelationship[typeCode=COMP].sequenceNumber
description | outboundRelationship[typeCode=INST].target[classCode=SBADM].inboundRelationship[typeCode=COMP].source[classCode=SBADM].text
authority | outboundRelationship[typeCode=INST].target[classCode=SBADM].inboundRelationship[typeCode=COMP].source[classCode=SBADM].participation[typeCode=AUT].COCT_MT090200UV
series | outboundRelationship[typeCode=INST].target[classCode=SBADM]
seriesDoses | outboundRelationship[typeCode=INST].target[classCode=SBADM].inboundRelationship[typeCode=COMP].source[classCode=SBADM].text
targetDisease | outboundRelationship.target[classCode=OBS].code
doseStatus | .participation[typeCode=CSM].role[classCode=INST].scopedRole.scoper[classCode=MMAT].playedRole[classCode=INGR].scopedRole.scoper[classCode=MMAT].playedRole[classCode=HLTHCHRT].participation[typeCode=SBJ].observation[classCode=OBS].value
doseStatusReason | .participation[typeCode=CSM].role[classCode=INST].scopedRole.scoper[classCode=MMAT].playedRole[classCode=INGR].scopedRole.scoper[classCode=MMAT].playedRole[classCode=HLTHCHRT].participation[typeCode=SBJ].observation[classCode=OBS].reasonCode

11.6.7.2 Mappings for CDA (R2) (http://hl7.org/v3/cda)

Immunization
identifier | ClinicalDocument/component/StructuredBody/component/section/entry/substanceAdministration/id
status
notGiven
vaccineCode | ClinicalDocument/component/StructuredBody/component/section/entry/substanceAdministration/consumable/manfacturedProduct/manufacturedMaterial/realmCode/code
patient
encounter
date | ClinicalDocument/component/StructuredBody/component/section/entry/substanceAdministration/effectiveTime/value
primarySource

reportOrigin
location
manufacturer | ClinicalDocument/component/StructuredBody/component/section/entry/substanceAdministration/consumable/manfacturedProduct/manufacuturerOrganization/name
lotNumber | ClinicalDocument/component/StructuredBody/component/section/entry/substanceAdministration/consumable/manfacturedProduct/manufacturedMaterial/lotNumberText
expirationDate
site | ClinicalDocument/component/StructuredBody/component/section/entry/substanceAdministration/approachSiteCode/code
route | ClinicalDocument/component/StructuredBody/component/section/entry/substanceAdministration/routeCode/code
doseQuantity
practitioner
role
actor
note
explanation
reason
reasonNotGiven
reaction
date
detail
reported
vaccinationProtocol
doseSequence
description
authority
series
seriesDoses
targetDisease
doseStatus
doseStatusReason

11.6.7.3 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

Immunization | VXU_V04
identifier
status
notGiven
vaccineCode | RXA-5
patient | PID-3
encounter | PV1-19
date | RXA-3
primarySource | RXA-9
reportOrigin | RXA-9
location | RXA-27 (or RXA-11, deprecated as of v2.7)
manufacturer | RXA-17
lotNumber | RXA-15
expirationDate | RXA-16
site | RXR-2
route | RXR-1
doseQuantity | RXA-6 / RXA-7.1
practitioner | ORC-12 / RXA-10
role
actor
note
explanation
reason
reasonNotGiven | RXA-18
reaction | OBX-3
date | OBX-14 (ideally this would be reported in an IAM segment, but IAM is not part of the HL7 v2 VXU message - most likely would appear in OBX segments if at all)
detail | OBX-5
reported | (no such concept seems to exist for allergy/adverse reaction in HL7 v2)
vaccinationProtocol | (HL7 v2 doesn't seem to provide for this)
doseSequence
description
authority
series
seriesDoses
targetDisease
doseStatus
doseStatusReason

11.6.7.4 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

Immunization
identifier | Event.identifier
status | Event.status
notGiven | Event.notDone
vaccineCode | Event.code
patient | Event.subject
encounter | Event.context
date | Event.occurrence.occurrenceDateTime
primarySource
reportOrigin
location
manufacturer
lotNumber
expirationDate
site
route
doseQuantity
practitioner | Event.performer
role | Event.performer.role
actor | Event.performer.actor
note | Event.note
explanation
reason | Event.reasonCodeableConcept
reasonNotGiven | Event.notDoneReason
reaction
date
detail
reported
vaccinationProtocol
doseSequence
description
authority
series
seriesDoses
targetDisease
doseStatus
doseStatusReason

11.6.8 Resource Immunization - Design Notes

URL: https://hl7.org/fhir/STU3/immunization-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.6.8 Resource Immunization - Design Notes

to do

11.6.9 Resource Immunization - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/immunization-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.6.9 Resource Immunization - Extensions & Profiles

Public Health and Emergency Response Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Immunization resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

11.6.9.1 Profiles

No Profiles defined for this resource

11.6.9.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

condition-dueTo | dueTo | for Condition HL7 Extensions
condition-occurredFollowing | occurredFollowing | for Condition HL7 Extensions
procedure-causedBy | causedBy | for Procedure HL7 Extensions

11.6.9.3 Search Extensions

No Search Extensions defined for this resource

11.6.10 Immunization HL7 Extensions

URL: https://hl7.org/fhir/STU3/immunization-extensions.html

Defines common extensions used with or related to the Immunization resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.6.10 Immunization HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

11.6.10.1 Content

11.6.11 Resource Immunization - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/immunization-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.6.11 Resource Immunization - Detailed Descriptions

Public Health and Emergency Response Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner

R2 : R3 Converson maps for Immunization.

Functional status for this map: 2 tests that all execute ok. 2 fail round-trip testing and 2 r3 resources are invalid (2 errors). (see documentation)

11.6.11.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Immunization2to3" = "R2 to R3 Conversions for Immunization"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Immunization" alias ImmunizationR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Immunization" alias Immunization as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  Immunization extends DomainResource
  input src : ImmunizationR2 as source
  input tgt : Immunization as target

  "Immunization-identifier" : for src.identifier make tgt.identifier
  "Immunization-status" : for src.status where $this = 'completed' or $this = 'entered-in-error' make tgt.status
  "Immunization-date" : for src.date make tgt.date
  "Immunization-vaccineCode" : for src.vaccineCode make tgt.vaccineCode
  "Immunization-patient" : for src.patient make tgt.patient
  "Immunization-wasNotGiven" : for src.wasNotGiven make tgt.notGiven
  "Immunization-performer" : for src.performer as v make tgt.practitioner as p, p.role =
    cc("http://hl7.org/fhir/v2/0443", "AP"), p.actor = v
  "Immunization-requester" : for src.requester as v make tgt.practitioner as p, p.role =
    cc("http://hl7.org/fhir/v2/0443", "OP"), p.actor = v
  "Immunization-encounter" : for src.encounter make tgt.encounter
  "Immunization-manufacturer" : for src.manufacturer make tgt.manufacturer
  "Immunization-location" : for src.location make tgt.location
  "Immunization-lotNumber" : for src.lotNumber make tgt.lotNumber
  "Immunization-expirationDate" : for src.expirationDate make tgt.expirationDate
  "Immunization-site" : for src.site make tgt.site
  "Immunization-route" : for src.route make tgt.route
  "Immunization-doseQuantity" : for src.doseQuantity make tgt.doseQuantity
  "Immunization-note" : for src.note make tgt.note
  "Immunization-explanation" : for src.explanation as vs0 make tgt.explanation as vt0 then immExplanation(vs0, vt0)
  "Immunization-reaction" : for src.reaction as vs0 make tgt.reaction as vt0 then immReaction(vs0, vt0)
  "Immunization-vaccinationProtocol" : for src.vaccinationProtocol as vs0 make tgt.vaccinationProtocol as vt0 then
    immProtocol(vs0, vt0)
endgroup

group immExplanation extends BackboneElement
  input src as source
  input tgt as target

  "Immunization.explanation-reason" : for src.reason make tgt.reason
  "Immunization.explanation-reasonNotGiven" : for src.reasonNotGiven make tgt.reasonNotGiven
endgroup

group immReaction extends BackboneElement
  input src as source
  input tgt as target

  "Immunization.reaction-date" : for src.date make tgt.date
  "Immunization.reaction-detail" : for src.detail make tgt.detail
```

```
  "Immunization.reaction-reported" : for src.reported make tgt.reported
endgroup

group immProtocol extends BackboneElement
  input src as source
  input tgt as target

  "Immunization.vaccinationProtocol-doseSequence" : for src.doseSequence make tgt.doseSequence
  "Immunization.vaccinationProtocol-description" : for src.description make tgt.description
  "Immunization.vaccinationProtocol-authority" : for src.authority make tgt.authority
  "Immunization.vaccinationProtocol-series" : for src.series make tgt.series
  "Immunization.vaccinationProtocol-seriesDoses" : for src.seriesDoses make tgt.seriesDoses
  "Immunization.vaccinationProtocol-targetDisease" : for src.targetDisease make tgt.targetDisease
  "Immunization.vaccinationProtocol-doseStatus" : for src.doseStatus make tgt.doseStatus
  "Immunization.vaccinationProtocol-doseStatusReason" : for src.doseStatusReason make tgt.doseStatusReason
endgroup
```

11.6.11.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Immunization3to2" = "R3 to R2 Conversion for Immunization"

uses "http://hl7.org/fhir/StructureDefinition/Immunization" alias Immunization as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Immunization" alias ImmunizationR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  Immunization extends DomainResource
  input src : Immunization as source
  input tgt : ImmunizationR2 as target

  "Immunization-identifier" : for src.identifier make tgt.identifier
  "Immunization-status" : for src.status make tgt.status
  "Immunization-date" : for src.date make tgt.date
  "Immunization-vaccineCode" : for src.vaccineCode make tgt.vaccineCode
  "Immunization-patient" : for src.patient make tgt.patient
  "Immunization-wasNotGiven" : for src.notGiven make tgt.wasNotGiven
  // can't translate reported
  "Immunization-performer" : for src.practitioner as v where v.role.coding.code.contains("AP") then {
    "Immunization-performer-inner" : for v.actor make tgt.performer
  }
  "Immunization-requester" : for src.practitioner as v where v.role.coding.code.contains("OP") then {
    "Immunization-requester-inner" : for v.actor make tgt.requester
  }
  "Immunization-encounter" : for src.encounter make tgt.encounter
  "Immunization-manufacturer" : for src.manufacturer make tgt.manufacturer
  "Immunization-location" : for src.location make tgt.location
  "Immunization-lotNumber" : for src.lotNumber make tgt.lotNumber
  "Immunization-expirationDate" : for src.expirationDate make tgt.expirationDate
  "Immunization-site" : for src.site make tgt.site
  "Immunization-route" : for src.route make tgt.route
  "Immunization-doseQuantity" : for src.doseQuantity make tgt.doseQuantity
  "Immunization-note" : for src.note make tgt.note
  "Immunization-explanation" : for src.explanation as vs0 make tgt.explanation as vt0 then immExplanation(vs0, vt0)
  "Immunization-reaction" : for src.reaction as vs0 make tgt.reaction as vt0 then immReaction(vs0, vt0)
  "Immunization-vaccinationProtocol" : for src.vaccinationProtocol as vs0 make tgt.vaccinationProtocol as vt0 then
    immProtocol(vs0, vt0)
endgroup

group immExplanation extends BackboneElement
  input src as source
  input tgt as target

  "Immunization.explanation-reason" : for src.reason make tgt.reason
  "Immunization.explanation-reasonNotGiven" : for src.reasonNotGiven make tgt.reasonNotGiven
endgroup

group immReaction extends BackboneElement
  input src as source
  input tgt as target

  "Immunization.reaction-date" : for src.date make tgt.date
  "Immunization.reaction-detail" : for src.detail make tgt.detail
  "Immunization.reaction-reported" : for src.reported make tgt.reported
endgroup

group immProtocol extends BackboneElement
  input src as source
  input tgt as target

  "Immunization.vaccinationProtocol-doseSequence" : for src.doseSequence make tgt.doseSequence
  "Immunization.vaccinationProtocol-description" : for src.description make tgt.description
  "Immunization.vaccinationProtocol-authority" : for src.authority make tgt.authority
  "Immunization.vaccinationProtocol-series" : for src.series make tgt.series
  "Immunization.vaccinationProtocol-seriesDoses" : for src.seriesDoses make tgt.seriesDoses
  "Immunization.vaccinationProtocol-targetDisease" : for src.targetDisease make tgt.targetDisease
  "Immunization.vaccinationProtocol-doseStatus" : for src.doseStatus make tgt.doseStatus
  "Immunization.vaccinationProtocol-doseStatusReason" : for src.doseStatusReason make tgt.doseStatusReason
endgroup
```

11.6.11.3 R3 Validation Errors

example | ERROR @ Immunization: Profile http://hl7.org/fhir/StructureDefinition/Immunization, Element 'Immunization.primarySource': minimum required = 1, but only found 0 (src = InstanceValidator)

notGiven | ERROR @ Immunization: Profile http://hl7.org/fhir/StructureDefinition/Immunization, Element 'Immunization.primarySource': minimum required = 1, but only found 0 (src = InstanceValidator)

11.7 Resource ImmunizationRecommendation - Content

URL: https://hl7.org/fhir/STU3/immunizationrecommendation.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Public Health and Emergency Response Work Group | Maturity Level: 1 | Trial Use | Compartments: Patient

A patient's point-in-time immunization and recommendation (i.e. forecasting a patient's immunization eligibility according to a published schedule) with optional supporting justification.

11.7.1 Scope and Usage

The ImmunizationRecommendation resource is intended to cover communication of a specified patient's immunization recommendations and status across all healthcare disciplines in all care settings and all regions.

Additionally, the ImmunizationRecommendation resource is expected to cover key concepts related to the querying of a patient's immunization recommendations and status. This resource - through consultation with the PHER work group - is believed to meet key use cases and information requirements as defined in the existing HL7 v3 POIZ domain and Immunization Domain Analysis Model.

11.7.2 Boundaries and Relationships

This resource references the following resources:

* Allergy/Intolerance

* Patient

* Organization

* Immunization

* Observation

One of the considerations for this resource is if it is better for this resource to be a profile of the CarePlan resource, or if it is more appropriate for this to be a separate resource due to the number of immunization profile-specific data elements.

This resource is referenced by DiagnosticReport and Observation

11.7.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints

ImmunizationRecommendation | | | DomainResource | Guidance or advice relating to an immunization Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..* | Identifier | Business identifier

patient | ? | 1..1 | Reference(Patient) | Who this profile is for

recommendation | ?I | 1..* | BackboneElement | Vaccine administration recommendations + One of vaccineCode or targetDisease SHALL be present

date | ? | 1..1 | dateTime | Date recommendation created

vaccineCode | ? | 0..1 | CodeableConcept | Vaccine recommendation applies to Vaccine Administered Value Set (Example)

targetDisease | ? | 0..1 | CodeableConcept | Disease to be immunized against Immunization Recommendation Target Disease Codes (Example)

doseNumber | ? | 0..1 | positiveInt | Recommended dose number

forecastStatus | ? | 1..1 | CodeableConcept | Vaccine administration status Immunization Recommendation Status Codes (Example)

dateCriterion | | 0..* | BackboneElement | Dates governing proposed immunization

code | | 1..1 | CodeableConcept | Type of date Immunization Recommendation Date Criterion Codes (Example)

value | | 1..1 | dateTime | Recommended date

protocol | | 0..1 | BackboneElement | Protocol used by recommendation

doseSequence | | 0..1 | positiveInt | Dose number within sequence

description | | 0..1 | string | Protocol details

authority | | 0..1 | Reference(Organization) | Who is responsible for protocol

series | | 0..1 | string | Name of vaccination series

supportingImmunization | | 0..* | Reference(Immunization) | Past immunizations supporting recommendation
supportingPatientInformation | | 0..* | Reference(Observation | AllergyIntolerance) | Patient observations supporting recommendation
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ImmunizationRecommendation xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <patient><!-- 1..1 Reference(Patient) Who this profile is for --></patient>
 <recommendation>  <!-- 1..* Vaccine administration recommendations -->
  <date value="[dateTime]"/><!-- 1..1 Date recommendation created -->
  <vaccineCode><!-- 0..1 CodeableConcept Vaccine recommendation applies to --></vaccineCode>
  <targetDisease><!-- 0..1 CodeableConcept Disease to be immunized against --></targetDisease>
  <doseNumber value="[positiveInt]"/><!-- 0..1 Recommended dose number -->
  <forecastStatus><!-- 1..1 CodeableConcept Vaccine administration status --></forecastStatus>
  <dateCriterion>  <!-- 0..* Dates governing proposed immunization -->
   <code><!-- 1..1 CodeableConcept Type of date --></code>
   <value value="[dateTime]"/><!-- 1..1 Recommended date -->
  </dateCriterion>
  <protocol>  <!-- 0..1 Protocol used by recommendation -->
   <doseSequence value="[positiveInt]"/><!-- 0..1 Dose number within sequence -->
   <description value="[string]"/><!-- 0..1 Protocol details -->
   <authority><!-- 0..1 Reference(Organization) Who is responsible for protocol --></authority>
   <series value="[string]"/><!-- 0..1 Name of vaccination series -->
  </protocol>
  <supportingImmunization><!-- 0..* Reference(Immunization) Past immunizations supporting recommendation
    --></supportingImmunization>
  <supportingPatientInformation><!-- 0..* Reference(Observation|
    AllergyIntolerance) Patient observations supporting recommendation --></supportingPatientInformation>
 </recommendation>
</ImmunizationRecommendation>
```

JSON Template

```
{
  "resourceType" : "ImmunizationRecommendation",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "patient" : { Reference(Patient) }, // R!  Who this profile is for
  "recommendation" : [{ // R!  Vaccine administration recommendations
    "date" : "<dateTime>", // R!  Date recommendation created
    "vaccineCode" : { CodeableConcept }, // Vaccine recommendation applies to
    "targetDisease" : { CodeableConcept }, // Disease to be immunized against
    "doseNumber" : "<positiveInt>", // Recommended dose number
    "forecastStatus" : { CodeableConcept }, // R!  Vaccine administration status
    "dateCriterion" : [{ // Dates governing proposed immunization
      "code" : { CodeableConcept }, // R!  Type of date
      "value" : "<dateTime>" // R!  Recommended date
    }],
    "protocol" : { // Protocol used by recommendation
      "doseSequence" : "<positiveInt>", // Dose number within sequence
      "description" : "<string>", // Protocol details
      "authority" : { Reference(Organization) }, // Who is responsible for protocol
      "series" : "<string>" // Name of vaccination series
    },
    "supportingImmunization" : [{ Reference(Immunization) }], // Past immunizations supporting recommendation
    "supportingPatientInformation" : [{ Reference(Observation|
    AllergyIntolerance) }] // Patient observations supporting recommendation
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ImmunizationRecommendation;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ImmunizationRecommendation.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:ImmunizationRecommendation.patient [ Reference(Patient) ]; # 1..1 Who this profile is for
  fhir:ImmunizationRecommendation.recommendation [ # 1..* Vaccine administration recommendations
    fhir:ImmunizationRecommendation.recommendation.date [ dateTime ]; # 1..1 Date recommendation created
    fhir:ImmunizationRecommendation.recommendation.vaccineCode [ CodeableConcept ]; # 0..1 Vaccine recommendation
    applies to
    fhir:ImmunizationRecommendation.recommendation.targetDisease [ CodeableConcept ]; # 0..1 Disease to be immunized
    against
    fhir:ImmunizationRecommendation.recommendation.doseNumber [ positiveInt ]; # 0..1 Recommended dose number
    fhir:ImmunizationRecommendation.recommendation.forecastStatus [ CodeableConcept ]; # 1..1 Vaccine administration
    status
    fhir:ImmunizationRecommendation.recommendation.dateCriterion [ # 0..* Dates governing proposed immunization
      fhir:ImmunizationRecommendation.recommendation.dateCriterion.code [ CodeableConcept ]; # 1..1 Type of date
      fhir:ImmunizationRecommendation.recommendation.dateCriterion.value [ dateTime ]; # 1..1 Recommended date
    ], ...;
```

```
    fhir:ImmunizationRecommendation.recommendation.protocol [ # 0..1 Protocol used by recommendation
      fhir:ImmunizationRecommendation.recommendation.protocol.doseSequence [ positiveInt ]; # 0..1 Dose number within
    sequence
      fhir:ImmunizationRecommendation.recommendation.protocol.description [ string ]; # 0..1 Protocol details
      fhir:ImmunizationRecommendation.recommendation.protocol.authority [ Reference(Organization) ]; # 0..1 Who is
    responsible for protocol
      fhir:ImmunizationRecommendation.recommendation.protocol.series [ string ]; # 0..1 Name of vaccination series
    ];
    fhir:ImmunizationRecommendation.recommendation.supportingImmunization [ Reference(Immunization) ], ... ; # 0..* Past
     immunizations supporting recommendation
    fhir:ImmunizationRecommendation.recommendation.supportingPatientInformation [
    Reference(Observation|AllergyIntolerance) ], ... ; # 0..* Patient observations supporting recommendation
  ], ...;
]
```

Changes since DSTU2

ImmunizationRecommendation

ImmunizationRecommendation.recommendation.vaccineCode | Min Cardinality changed from 1 to 0

ImmunizationRecommendation.recommendation.targetDisease | Added Element

ImmunizationRecommendation.recommendation.protocol.doseSequence | Type changed from integer to positiveInt

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints

ImmunizationRecommendation | | | DomainResource | Guidance or advice relating to an immunization Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..* | Identifier | Business identifier

patient | ? | 1..1 | Reference(Patient) | Who this profile is for

recommendation | ?I | 1..* | BackboneElement | Vaccine administration recommendations + One of vaccineCode or targetDisease SHALL be present

date | ? | 1..1 | dateTime | Date recommendation created

vaccineCode | ? | 0..1 | CodeableConcept | Vaccine recommendation applies to Vaccine Administered Value Set (Example)

targetDisease | ? | 0..1 | CodeableConcept | Disease to be immunized against Immunization Recommendation Target Disease Codes (Example)

doseNumber | ? | 0..1 | positiveInt | Recommended dose number

forecastStatus | ? | 1..1 | CodeableConcept | Vaccine administration status Immunization Recommendation Status Codes (Example)

dateCriterion | | 0..* | BackboneElement | Dates governing proposed immunization

code | | 1..1 | CodeableConcept | Type of date Immunization Recommendation Date Criterion Codes (Example)

value | | 1..1 | dateTime | Recommended date

protocol | | 0..1 | BackboneElement | Protocol used by recommendation

doseSequence | | 0..1 | positiveInt | Dose number within sequence

description | | 0..1 | string | Protocol details

authority | | 0..1 | Reference(Organization) | Who is responsible for protocol

series | | 0..1 | string | Name of vaccination series

supportingImmunization | | 0..* | Reference(Immunization) | Past immunizations supporting recommendation

supportingPatientInformation | | 0..* | Reference(Observation | AllergyIntolerance) | Patient observations supporting recommendation

Documentation for this format

UML Diagram (Legend)

XML Template

```
<ImmunizationRecommendation xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <patient><!-- 1..1 Reference(Patient) Who this profile is for --></patient>
 <recommendation>  <!-- 1..* Vaccine administration recommendations -->
  <date value="[dateTime]"/><!-- 1..1 Date recommendation created -->
  <vaccineCode><!-- 0..1 CodeableConcept Vaccine recommendation applies to --></vaccineCode>
  <targetDisease><!-- 0..1 CodeableConcept Disease to be immunized against --></targetDisease>
  <doseNumber value="[positiveInt]"/><!-- 0..1 Recommended dose number -->
  <forecastStatus><!-- 1..1 CodeableConcept Vaccine administration status --></forecastStatus>
  <dateCriterion>  <!-- 0..* Dates governing proposed immunization -->
   <code><!-- 1..1 CodeableConcept Type of date --></code>
   <value value="[dateTime]"/><!-- 1..1 Recommended date -->
  </dateCriterion>
  <protocol>  <!-- 0..1 Protocol used by recommendation -->
   <doseSequence value="[positiveInt]"/><!-- 0..1 Dose number within sequence -->
   <description value="[string]"/><!-- 0..1 Protocol details -->
   <authority><!-- 0..1 Reference(Organization) Who is responsible for protocol --></authority>
   <series value="[string]"/><!-- 0..1 Name of vaccination series -->
  </protocol>
  <supportingImmunization><!-- 0..* Reference(Immunization) Past immunizations supporting recommendation
    --></supportingImmunization>
  <supportingPatientInformation><!-- 0..* Reference(Observation|
    AllergyIntolerance) Patient observations supporting recommendation --></supportingPatientInformation>
 </recommendation>
</ImmunizationRecommendation>
```

JSON Template

```
{
  "resourceType" : "ImmunizationRecommendation",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "patient" : { Reference(Patient) }, // R!  Who this profile is for
  "recommendation" : [{ // R!  Vaccine administration recommendations
    "date" : "<dateTime>", // R!  Date recommendation created
    "vaccineCode" : { CodeableConcept }, // Vaccine recommendation applies to
    "targetDisease" : { CodeableConcept }, // Disease to be immunized against
    "doseNumber" : "<positiveInt>", // Recommended dose number
    "forecastStatus" : { CodeableConcept }, // R!  Vaccine administration status
    "dateCriterion" : [{ // Dates governing proposed immunization
      "code" : { CodeableConcept }, // R!  Type of date
      "value" : "<dateTime>" // R!  Recommended date
    }],
    "protocol" : { // Protocol used by recommendation
      "doseSequence" : "<positiveInt>", // Dose number within sequence
      "description" : "<string>", // Protocol details
      "authority" : { Reference(Organization) }, // Who is responsible for protocol
      "series" : "<string>" // Name of vaccination series
    },
    "supportingImmunization" : [{ Reference(Immunization) }], // Past immunizations supporting recommendation
    "supportingPatientInformation" : [{ Reference(Observation|
    AllergyIntolerance) }] // Patient observations supporting recommendation
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ImmunizationRecommendation;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ImmunizationRecommendation.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:ImmunizationRecommendation.patient [ Reference(Patient) ]; # 1..1 Who this profile is for
  fhir:ImmunizationRecommendation.recommendation [ # 1..* Vaccine administration recommendations
    fhir:ImmunizationRecommendation.recommendation.date [ dateTime ]; # 1..1 Date recommendation created
    fhir:ImmunizationRecommendation.recommendation.vaccineCode [ CodeableConcept ]; # 0..1 Vaccine recommendation
    applies to
    fhir:ImmunizationRecommendation.recommendation.targetDisease [ CodeableConcept ]; # 0..1 Disease to be immunized
    against
    fhir:ImmunizationRecommendation.recommendation.doseNumber [ positiveInt ]; # 0..1 Recommended dose number
    fhir:ImmunizationRecommendation.recommendation.forecastStatus [ CodeableConcept ]; # 1..1 Vaccine administration
    status
    fhir:ImmunizationRecommendation.recommendation.dateCriterion [ # 0..* Dates governing proposed immunization
      fhir:ImmunizationRecommendation.recommendation.dateCriterion.code [ CodeableConcept ]; # 1..1 Type of date
      fhir:ImmunizationRecommendation.recommendation.dateCriterion.value [ dateTime ]; # 1..1 Recommended date
    ], ...;
    fhir:ImmunizationRecommendation.recommendation.protocol [ # 0..1 Protocol used by recommendation
      fhir:ImmunizationRecommendation.recommendation.protocol.doseSequence [ positiveInt ]; # 0..1 Dose number within
    sequence
      fhir:ImmunizationRecommendation.recommendation.protocol.description [ string ]; # 0..1 Protocol details
      fhir:ImmunizationRecommendation.recommendation.protocol.authority [ Reference(Organization) ]; # 0..1 Who is
    responsible for protocol
      fhir:ImmunizationRecommendation.recommendation.protocol.series [ string ]; # 0..1 Name of vaccination series
    ];
    fhir:ImmunizationRecommendation.recommendation.supportingImmunization [ Reference(Immunization) ], ... ; # 0..* Past
     immunizations supporting recommendation
    fhir:ImmunizationRecommendation.recommendation.supportingPatientInformation [
    Reference(Observation|AllergyIntolerance) ], ... ; # 0..* Patient observations supporting recommendation
  ], ...;
]
```

Changes since DSTU2

ImmunizationRecommendation

ImmunizationRecommendation.recommendation.vaccineCode | Min Cardinality changed from 1 to 0

ImmunizationRecommendation.recommendation.targetDisease | Added Element

ImmunizationRecommendation.recommendation.protocol.doseSequence | Type changed from integer to positiveInt

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

11.7.3.1 Terminology Bindings

Path | Definition | Type | Reference

ImmunizationRecommendation.recommendation.vaccineCode | The type of vaccine administered | Example | Vaccine Administered Value Set

ImmunizationRecommendation.recommendation.targetDisease | The disease that the recommended vaccination targets | Example | Immunization Recommendation Target Disease Codes

ImmunizationRecommendation.recommendation.forecastStatus | The patient's status with respect to a vaccination protocol | Example | Immunization Recommendation Status Codes
ImmunizationRecommendation.recommendation.dateCriterion.code | Classifies date criterion with respect to conveying information about a patient's vaccination status (e.g. due date, latest to give date, etc.) | Example | Immunization Recommendation Date Criterion Codes

11.7.3.2 Constraints

* imr-1: On ImmunizationRecommendation.recommendation: One of vaccineCode or targetDisease SHALL be present (expression on ImmunizationRecommendation.recommendation: vaccineCode.exists() or targetDisease.exists())

11.7.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
date | date | Date recommendation created | ImmunizationRecommendation.recommendation.date
dose-number | number | Recommended dose number | ImmunizationRecommendation.recommendation.doseNumber
dose-sequence | number | Dose number within sequence | ImmunizationRecommendation.recommendation.protocol.doseSequence
identifier | token | Business identifier | ImmunizationRecommendation.identifier
information | reference | Patient observations supporting recommendation | ImmunizationRecommendation.recommendation.supportingPatientInformation (AllergyIntolerance, Observation)
patient | reference | Who this profile is for | ImmunizationRecommendation.patient (Patient)
status | token | Vaccine administration status | ImmunizationRecommendation.recommendation.forecastStatus
support | reference | Past immunizations supporting recommendation | ImmunizationRecommendation.recommendation.supportingImmunization (Immunization)
target-disease | token | Disease to be immunized against | ImmunizationRecommendation.recommendation.targetDisease
vaccine-type | token | Vaccine recommendation applies to | ImmunizationRecommendation.recommendation.vaccineCode

11.7.5 Resource ImmunizationRecommendation - Examples

URL: https://hl7.org/fhir/STU3/immunizationrecommendation-examples.html
This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.7.5 Resource ImmunizationRecommendation - Examples

Public Health and Emergency Response Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient

Example List:

Example of immunizationrecommendation using a vaccine code | example | XML | JSON | Turtle
Example of immunizationrecommendation using a target disease | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.7.6 Resource ImmunizationRecommendation - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/immunizationrecommendation-definitions.html
This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.7.6 Resource ImmunizationRecommendation - Detailed Descriptions

Public Health and Emergency Response Work Group | Maturity Level: 1 | Trial Use | Compartments: Patient

Detailed Descriptions for the elements in the ImmunizationRecommendation resource.

ImmunizationRecommendation
Definition | A patient's point-in-time immunization and recommendation (i.e. forecasting a patient's immunization eligibility according to a published schedule) with optional supporting justification.
Control | 1..1
ImmunizationRecommendation.identifier
Definition | A unique identifier assigned to this particular recommendation record.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Summary | true
ImmunizationRecommendation.patient
Definition | The patient the recommendations are for.
Control | 1..1
Type | Reference(Patient)
Summary | true
ImmunizationRecommendation.recommendation
Definition | Vaccine administration recommendations.
Control | 1..*

Summary | true
Invariants | Defined on this element imr-1: One of vaccineCode or targetDisease SHALL be present (expression : vaccineCode.exists() or targetDisease.exists(), xpath: exists(f:vaccineCode) or exists(f:targetDisease))
ImmunizationRecommendation.recommendation.date
Definition | The date the immunization recommendation was created.
Control | 1..1
Type | dateTime
Summary | true
ImmunizationRecommendation.recommendation.vaccineCode
Definition | Vaccine that pertains to the recommendation.
Control | 0..1
Terminology Binding | Vaccine Administered Value Set (Example)
Type | CodeableConcept
Summary | true
ImmunizationRecommendation.recommendation.targetDisease
Definition | The targeted disease for the recommendation.
Control | 0..1
Terminology Binding | Immunization Recommendation Target Disease Codes (Example)
Type | CodeableConcept
Summary | true
ImmunizationRecommendation.recommendation.doseNumber
Definition | The next recommended dose number (e.g. dose 2 is the next recommended dose).
Control | 0..1
Type | positiveInt
Summary | true
Comments | May need other dose concepts such as administered vs. valid.
ImmunizationRecommendation.recommendation.forecastStatus
Definition | Vaccine administration status.
Control | 1..1
Terminology Binding | Immunization Recommendation Status Codes (Example)
Type | CodeableConcept
Summary | true
ImmunizationRecommendation.recommendation.dateCriterion
Definition | Vaccine date recommendations. For example, earliest date to administer, latest date to administer, etc.
Control | 0..*
ImmunizationRecommendation.recommendation.dateCriterion.code
Definition | Date classification of recommendation. For example, earliest date to give, latest date to give, etc.
Control | 1..1
Terminology Binding | Immunization Recommendation Date Criterion Codes (Example)
Type | CodeableConcept
ImmunizationRecommendation.recommendation.dateCriterion.value
Definition | The date whose meaning is specified by dateCriterion.code.
Control | 1..1
Type | dateTime
ImmunizationRecommendation.recommendation.protocol
Definition | Contains information about the protocol under which the vaccine was administered.
Control | 0..1
ImmunizationRecommendation.recommendation.protocol.doseSequence
Definition | Indicates the nominal position in a series of the next dose. This is the recommended dose number as per a specified protocol.
Control | 0..1
Type | positiveInt
ImmunizationRecommendation.recommendation.protocol.description
Definition | Contains the description about the protocol under which the vaccine was administered.
Control | 0..1
Type | string
ImmunizationRecommendation.recommendation.protocol.authority
Definition | Indicates the authority who published the protocol. For example, ACIP.
Control | 0..1
Type | Reference(Organization)
ImmunizationRecommendation.recommendation.protocol.series
Definition | One possible path to achieve presumed immunity against a disease - within the context of an authority.
Control | 0..1
Type | string
ImmunizationRecommendation.recommendation.supportingImmunization
Definition | Immunization event history that supports the status and recommendation.
Control | 0..*
Type | Reference(Immunization)
ImmunizationRecommendation.recommendation.supportingPatientInformation
Definition | Patient Information that supports the status and recommendation. This includes patient observations, adverse reactions and allergy/intolerance information.
Control | 0..*
Type | Reference(Observation | AllergyIntolerance)

11.7.7 Resource ImmunizationRecommendation - Mappings

URL: https://hl7.org/fhir/STU3/immunizationrecommendation-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.7.7 Resource ImmunizationRecommendation - Mappings

Public Health and Emergency Response Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient

Mappings:

RIM Mapping

HL7 v2 Mapping

Mappings for the immunizationrecommendation resource.

11.7.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

ImmunizationRecommendation | SubstanceAdministration[moodCode=RMD]
identifier | .id
patient | .partipication[typeCode=SUBJ].role
recommendation | .outboundRelationship[typeCode='COMP'].target[classCode=SBADM, moodCode=RMD]
date | .partipication[typeCode=AUTH].time
vaccineCode | .partipication[typeCode=CSM].role[classCode=ADMM].code
targetDisease | .partipication[typeCode=CSM].role[classCode=ADMM].code
doseNumber | .inboundRelationship[typeCode=COMP, .source[classCode=SBADM, moodCode=INT]].sequenceNumber
forecastStatus | .inboundRelationship[typeCode=SUBJ] .source[classCode=OBS, moodCode=EVN, code="Immunization Forecast Status"].value
dateCriterion | .outboundRelationship[typeCode='LIMIT'].target[classCode=OBS, moodCode=EVN, isCriterionInd=true]
code | .code
value | .value
protocol | .inboundRelationship[typeCode=INST].source[classCode=SBADM, moodCode=DEF]
doseSequence | .inboundRelationship[typeCode=COMP, .source[classCode=SBADM, moodCode=DEF]].sequenceNumber
description | .text
authority | .participation[typeCode=AUT]
series | .title
supportingImmunization | .outboundRelationship[typeCode=RSON].target[classCode=SBADM, moodCode=EVN, code="immunization"]
supportingPatientInformation | .outboundRelationship[typeCode=RSON].target[classCode=OBS, moodCode=EVN]

11.7.7.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

ImmunizationRecommendation | VXU_V04
identifier
patient | PID-3
recommendation
date | OBX-15
vaccineCode | OBX-5 : OBX-3 e { 30956-7 | 38890-0 }
targetDisease | OBX-5 : OBX-3 e { 30956-7 | 38890-0 }
doseNumber | OBX-5 : OBX-3 = 30973-2
forecastStatus | OBX-5 : OBX-3 = 59783-1
dateCriterion | OBX-3: 30980-7 = Next dose recommended, 59777-3 = Latest next dose should be given, 59778-1 = Dose is overdue
code | OBX-3
value | OBX-5
protocol
doseSequence | OBX-5 : OBX-3 = 30973-2
description | OBX-5 : OBX-3 = 30956-7
authority | OBX-5 : OBX-3 = 59779-9
series | RXA-5-administered code
supportingImmunization | ? No clear way to associate in HL7 v2 or IG
supportingPatientInformation | OBX If OBX is associated with (RXA : RXA-5-administered code = '998^No vaccine administered^CVX'), could contain the reason the vaccine was not administered

11.7.8 Resource ImmunizationRecommendation - Design Notes

URL: https://hl7.org/fhir/STU3/immunizationrecommendation-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.7.8 Resource ImmunizationRecommendation - Design Notes

to do

11.7.9 Resource ImmunizationRecommendation - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/immunizationrecommendation-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.7.9 Resource ImmunizationRecommendation - Extensions & Profiles

Public Health and Emergency Response Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the ImmunizationRecommendation resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

11.7.9.1 Profiles

No Profiles defined for this resource

11.7.9.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

procedure-causedBy | causedBy | for Procedure HL7 Extensions

11.7.9.3 Search Extensions

No Search Extensions defined for this resource

11.7.10 ImmunizationRecommendation HL7 Extensions

URL: https://hl7.org/fhir/STU3/immunizationrecommendation-extensions.html

Defines common extensions used with or related to the ImmunizationRecommendation resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.7.10 ImmunizationRecommendation HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

11.7.10.1 Content

11.7.11 Resource ImmunizationRecommendation - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/immunizationrecommendation-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.7.11 Resource ImmunizationRecommendation - Detailed Descriptions

Public Health and Emergency Response Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient

R2 : R3 Converson maps for ImmunizationRecommendation.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

11.7.11.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/ImmunizationRecommendation2to3" = "R2 to R3 Conversions for
    ImmunizationRecommendation"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/ImmunizationRecommendation" alias ImmunizationRecommendationR2 as
    source
uses "http://hl7.org/fhir/StructureDefinition/ImmunizationRecommendation" alias ImmunizationRecommendation as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  ImmunizationRecommendation extends DomainResource
  input src : ImmunizationRecommendationR2 as source
  input tgt : ImmunizationRecommendation as target

  "ImmunizationRecommendation-identifier" : for src.identifier make tgt.identifier
  "ImmunizationRecommendation-patient" : for src.patient make tgt.patient
  "ImmunizationRecommendation-recommendation" : for src.recommendation as vs0 make tgt.recommendation as vt0 then
    imrRecommendation(vs0, vt0)
endgroup

group imrRecommendation extends BackboneElement
  input src as source
  input tgt as target

  "ImmunizationRecommendation.recommendation-date" : for src.date make tgt.date
  "ImmunizationRecommendation.recommendation-vaccineCode" : for src.vaccineCode make tgt.vaccineCode
  "ImmunizationRecommendation.recommendation-doseNumber" : for src.doseNumber make tgt.doseNumber
```

```
  "ImmunizationRecommendation.recommendation-forecastStatus" : for src.forecastStatus make tgt.forecastStatus
  "ImmunizationRecommendation.recommendation-dateCriterion" : for src.dateCriterion as vs make tgt.dateCriterion as vt
    then imrDateCriterion(vs, vt)
  "ImmunizationRecommendation.recommendation-protocol" : for src.protocol as vs make tgt.protocol as vt then
    imrProtocol(vs, vt)
  "ImmunizationRecommendation.recommendation-supportingImmunization" : for src.supportingImmunization make
    tgt.supportingImmunization
  "ImmunizationRecommendation.recommendation-supportingPatientInformation" : for src.supportingPatientInformation make
    tgt.supportingPatientInformation
endgroup

group imrDateCriterion extends BackboneElement
  input src as source
  input tgt as target

  "ImmunizationRecommendation.recommendation.dateCriterion-date" : for src.code make tgt.code
  "ImmunizationRecommendation.recommendation.dateCriterion-date" : for src.value make tgt.value
endgroup

group imrProtocol extends BackboneElement
  input src as source
  input tgt as target

  "ImmunizationRecommendation.recommendation.protocol-doseSequence" : for src.doseSequence make tgt.doseSequence
  "ImmunizationRecommendation.recommendation.protocol-description" : for src.description make tgt.description
  "ImmunizationRecommendation.recommendation.protocol-authority" : for src.authority make tgt.authority
  "ImmunizationRecommendation.recommendation.protocol-series" : for src.series make tgt.series
endgroup
```

11.7.11.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/ImmunizationRecommendation3to2" = "R3 to R2 Conversion for
    ImmunizationRecommendation"

uses "http://hl7.org/fhir/StructureDefinition/ImmunizationRecommendation" alias ImmunizationRecommendation as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/ImmunizationRecommendation" alias ImmunizationRecommendationR2 as
    target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  ImmunizationRecommendation extends DomainResource
  input src : ImmunizationRecommendation as source
  input tgt : ImmunizationRecommendationR2 as target

  "ImmunizationRecommendation-identifier" : for src.identifier make tgt.identifier
  "ImmunizationRecommendation-patient" : for src.patient make tgt.patient
  "ImmunizationRecommendation-recommendation" : for src.recommendation as vs0 make tgt.recommendation as vt0 then
    imrRecommendation(vs0, vt0)
endgroup

group imrRecommendation extends BackboneElement
  input src as source
  input tgt as target

  "ImmunizationRecommendation.recommendation-date" : for src.date make tgt.date
  "ImmunizationRecommendation.recommendation-vaccineCode" : for src.vaccineCode make tgt.vaccineCode
  "ImmunizationRecommendation.recommendation-doseNumber" : for src.doseNumber make tgt.doseNumber
  "ImmunizationRecommendation.recommendation-forecastStatus" : for src.forecastStatus make tgt.forecastStatus
  "ImmunizationRecommendation.recommendation-dateCriterion" : for src.dateCriterion as vs make tgt.dateCriterion as vt
    then imrDateCriterion(vs, vt)
  "ImmunizationRecommendation.recommendation-protocol" : for src.protocol as vs make tgt.protocol as vt then
    imrProtocol(vs, vt)
  "ImmunizationRecommendation.recommendation-supportingImmunization" : for src.supportingImmunization make
    tgt.supportingImmunization
  "ImmunizationRecommendation.recommendation-supportingPatientInformation" : for src.supportingPatientInformation make
    tgt.supportingPatientInformation
endgroup

group imrDateCriterion extends BackboneElement
  input src as source
  input tgt as target

  "ImmunizationRecommendation.recommendation.dateCriterion-date" : for src.code make tgt.code
  "ImmunizationRecommendation.recommendation.dateCriterion-date" : for src.value make tgt.value
endgroup

group imrProtocol extends BackboneElement
  input src as source
  input tgt as target

  "ImmunizationRecommendation.recommendation.protocol-doseSequence" : for src.doseSequence make tgt.doseSequence
  "ImmunizationRecommendation.recommendation.protocol-description" : for src.description make tgt.description
  "ImmunizationRecommendation.recommendation.protocol-authority" : for src.authority make tgt.authority
  "ImmunizationRecommendation.recommendation.protocol-series" : for src.series make tgt.series
endgroup
```

11.7.11.3 R3 Validation Errors

11.8 Dosage

URL: https://hl7.org/fhir/STU3/dosage.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Pharmacy Work Group | Maturity Level: 3 | Ballot Status: Trial Use

The Dosage structure defines general dosage instruction information typically represented in medication requests, medication dispenses and medication statements.

11.8.0.1 Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints

Dosage | ? | | Element | How the medication is/was taken or should be taken Elements defined in Ancestors: id, extension

sequence | ? | 0..1 | integer | The order of the dosage instructions

text | ? | 0..1 | string | Free text dosage instructions e.g. SIG

additionalInstruction | ? | 0..* | CodeableConcept | Supplemental instruction - e.g. "with meals" SNOMED CT Additional Dosage Instructions (Example)

patientInstruction | ? | 0..1 | string | Patient or consumer oriented instructions

timing | ? | 0..1 | Timing | When medication should be administered

asNeeded[x] | ? | 0..1 | | Take "as needed" (for x) SNOMED CT Medication As Needed Reason Codes (Example)

asNeededBoolean | | | boolean

asNeededCodeableConcept | | | CodeableConcept

site | ? | 0..1 | CodeableConcept | Body site to administer to SNOMED CT Anatomical Structure for Administration Site Codes (Example)

route | ? | 0..1 | CodeableConcept | How drug should enter body SNOMED CT Route Codes (Example)

method | ? | 0..1 | CodeableConcept | Technique for administering medication SNOMED CT Administration Method Codes (Example)

dose[x] | ? | 0..1 | | Amount of medication per dose

doseRange | | | Range

doseQuantity | | | SimpleQuantity

maxDosePerPeriod | ? | 0..1 | Ratio | Upper limit on medication per unit of time

maxDosePerAdministration | ? | 0..1 | SimpleQuantity | Upper limit on medication per administration

maxDosePerLifetime | ? | 0..1 | SimpleQuantity | Upper limit on medication per lifetime of the patient

rate[x] | ? | 0..1 | | Amount of medication per unit of time

rateRatio | | | Ratio

rateRange | | | Range

rateQuantity | | | SimpleQuantity

Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <sequence value="[integer]"/><!-- 0..1 The order of the dosage instructions -->
 <text value="[string]"/><!-- 0..1 Free text dosage instructions e.g. SIG -->
 <additionalInstruction><!-- 0..* CodeableConcept Supplemental instruction - e.g. "with meals"
    --></additionalInstruction>
 <patientInstruction value="[string]"/><!-- 0..1 Patient or consumer oriented instructions -->
 <timing><!-- 0..1 Timing When medication should be administered --></timing>
 <asNeeded[x]><!-- 0..1 boolean|CodeableConcept Take "as needed" (for x) --></asNeeded[x]>
 <site><!-- 0..1 CodeableConcept Body site to administer to --></site>
 <route><!-- 0..1 CodeableConcept How drug should enter body --></route>
 <method><!-- 0..1 CodeableConcept Technique for administering medication --></method>
 <dose[x]><!-- 0..1 Range|Quantity(SimpleQuantity) Amount of medication per dose --></dose[x]>
 <maxDosePerPeriod><!-- 0..1 Ratio Upper limit on medication per unit of time --></maxDosePerPeriod>
 <maxDosePerAdministration><!-- 0..1 Quantity(SimpleQuantity) Upper limit on medication per administration
    --></maxDosePerAdministration>
 <maxDosePerLifetime><!-- 0..1 Quantity(SimpleQuantity) Upper limit on medication per lifetime of the patient
    --></maxDosePerLifetime>
 <rate[x]><!-- 0..1 Ratio|Range|Quantity(SimpleQuantity) Amount of medication per unit of time --></rate[x]>
</[name]>
```

JSON Template

```
{
  "resourceType" : "Dosage",
  // from Element: extension
  "sequence" : <integer>, // The order of the dosage instructions
```

```
  "text" : "<string>", // Free text dosage instructions e.g. SIG
  "additionalInstruction" : [{ CodeableConcept }], // Supplemental instruction - e.g. "with meals"
  "patientInstruction" : "<string>", // Patient or consumer oriented instructions
  "timing" : { Timing }, // When medication should be administered
  // asNeeded[x]: Take "as needed" (for x). One of these 2:
  "asNeededBoolean" : <boolean>,
  "asNeededCodeableConcept" : { CodeableConcept },
  "site" : { CodeableConcept }, // Body site to administer to
  "route" : { CodeableConcept }, // How drug should enter body
  "method" : { CodeableConcept }, // Technique for administering medication
  // dose[x]: Amount of medication per dose. One of these 2:
  "doseRange" : { Range },
  "doseQuantity" : { Quantity(SimpleQuantity) },
  "maxDosePerPeriod" : { Ratio }, // Upper limit on medication per unit of time
  "maxDosePerAdministration" : { Quantity(SimpleQuantity) }, // Upper limit on medication per administration
  "maxDosePerLifetime" : { Quantity(SimpleQuantity) }, // Upper limit on medication per lifetime of the patient
  // rate[x]: Amount of medication per unit of time. One of these 3:
  "rateRatio" : { Ratio }
  "rateRange" : { Range }
  "rateQuantity" : { Quantity(SimpleQuantity) }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Dosage.sequence [ integer ]; # 0..1 The order of the dosage instructions
  fhir:Dosage.text [ string ]; # 0..1 Free text dosage instructions e.g. SIG
  fhir:Dosage.additionalInstruction [ CodeableConcept ], ... ; # 0..* Supplemental instruction - e.g. "with meals"
  fhir:Dosage.patientInstruction [ string ]; # 0..1 Patient or consumer oriented instructions
  fhir:Dosage.timing [ Timing ]; # 0..1 When medication should be administered
  # Dosage.asNeeded[x] : 0..1 Take "as needed" (for x). One of these 2
    fhir:Dosage.asNeededBoolean [ boolean ]
    fhir:Dosage.asNeededCodeableConcept [ CodeableConcept ]
  fhir:Dosage.site [ CodeableConcept ]; # 0..1 Body site to administer to
  fhir:Dosage.route [ CodeableConcept ]; # 0..1 How drug should enter body
  fhir:Dosage.method [ CodeableConcept ]; # 0..1 Technique for administering medication
  # Dosage.dose[x] : 0..1 Amount of medication per dose. One of these 2
    fhir:Dosage.doseRange [ Range ]
    fhir:Dosage.doseSimpleQuantity [ Quantity(SimpleQuantity) ]
  fhir:Dosage.maxDosePerPeriod [ Ratio ]; # 0..1 Upper limit on medication per unit of time
  fhir:Dosage.maxDosePerAdministration [ Quantity(SimpleQuantity) ]; # 0..1 Upper limit on medication per administration
  fhir:Dosage.maxDosePerLifetime [ Quantity(SimpleQuantity) ]; # 0..1 Upper limit on medication per lifetime of the
    patient
  # Dosage.rate[x] : 0..1 Amount of medication per unit of time. One of these 3
    fhir:Dosage.rateRatio [ Ratio ]
    fhir:Dosage.rateRange [ Range ]
    fhir:Dosage.rateSimpleQuantity [ Quantity(SimpleQuantity) ]
]
```

Changes since DSTU2

This complex-type did not exist in Release 2

Structure

Name | Flags | Card. | Type | Description & Constraints
Dosage | ? | | Element | How the medication is/was taken or should be taken Elements defined in Ancestors: id, extension
sequence | ? | 0..1 | integer | The order of the dosage instructions
text | ? | 0..1 | string | Free text dosage instructions e.g. SIG
additionalInstruction | ? | 0..* | CodeableConcept | Supplemental instruction - e.g. "with meals" SNOMED CT Additional Dosage Instructions (Example)
patientInstruction | ? | 0..1 | string | Patient or consumer oriented instructions
timing | ? | 0..1 | Timing | When medication should be administered
asNeeded[x] | ? | 0..1 | | Take "as needed" (for x) SNOMED CT Medication As Needed Reason Codes (Example)
asNeededBoolean | | | boolean
asNeededCodeableConcept | | | CodeableConcept
site | ? | 0..1 | CodeableConcept | Body site to administer to SNOMED CT Anatomical Structure for Administration Site Codes (Example)
route | ? | 0..1 | CodeableConcept | How drug should enter body SNOMED CT Route Codes (Example)
method | ? | 0..1 | CodeableConcept | Technique for administering medication SNOMED CT Administration Method Codes (Example)
dose[x] | ? | 0..1 | | Amount of medication per dose
doseRange | | | Range
doseQuantity | | | SimpleQuantity
maxDosePerPeriod | ? | 0..1 | Ratio | Upper limit on medication per unit of time
maxDosePerAdministration | ? | 0..1 | SimpleQuantity | Upper limit on medication per administration
maxDosePerLifetime | ? | 0..1 | SimpleQuantity | Upper limit on medication per lifetime of the patient
rate[x] | ? | 0..1 | | Amount of medication per unit of time
rateRatio | | | Ratio
rateRange | | | Range
rateQuantity | | | SimpleQuantity
Documentation for this format

UML Diagram (Legend)

XML Template

```
<[name] xmlns="http://hl7.org/fhir">
 <!-- from Element: extension -->
 <sequence value="[integer]"/><!-- 0..1 The order of the dosage instructions -->
 <text value="[string]"/><!-- 0..1 Free text dosage instructions e.g. SIG -->
 <additionalInstruction><!-- 0..* CodeableConcept Supplemental instruction - e.g. "with meals"
    --></additionalInstruction>
 <patientInstruction value="[string]"/><!-- 0..1 Patient or consumer oriented instructions -->
 <timing><!-- 0..1 Timing When medication should be administered --></timing>
 <asNeeded[x]><!-- 0..1 boolean|CodeableConcept Take "as needed" (for x) --></asNeeded[x]>
 <site><!-- 0..1 CodeableConcept Body site to administer to --></site>
 <route><!-- 0..1 CodeableConcept How drug should enter body --></route>
 <method><!-- 0..1 CodeableConcept Technique for administering medication --></method>
 <dose[x]><!-- 0..1 Range|Quantity(SimpleQuantity) Amount of medication per dose --></dose[x]>
 <maxDosePerPeriod><!-- 0..1 Ratio Upper limit on medication per unit of time --></maxDosePerPeriod>
 <maxDosePerAdministration><!-- 0..1 Quantity(SimpleQuantity) Upper limit on medication per administration
    --></maxDosePerAdministration>
 <maxDosePerLifetime><!-- 0..1 Quantity(SimpleQuantity) Upper limit on medication per lifetime of the patient
    --></maxDosePerLifetime>
 <rate[x]><!-- 0..1 Ratio|Range|Quantity(SimpleQuantity) Amount of medication per unit of time --></rate[x]>
</[name]>
```

JSON Template

```
{
  "resourceType" : "Dosage",
  // from Element: extension
  "sequence" : <integer>, // The order of the dosage instructions
  "text" : "<string>", // Free text dosage instructions e.g. SIG
  "additionalInstruction" : [{ CodeableConcept }], // Supplemental instruction - e.g. "with meals"
  "patientInstruction" : "<string>", // Patient or consumer oriented instructions
  "timing" : { Timing }, // When medication should be administered
  // asNeeded[x]: Take "as needed" (for x). One of these 2:
  "asNeededBoolean" : <boolean>,
  "asNeededCodeableConcept" : { CodeableConcept },
  "site" : { CodeableConcept }, // Body site to administer to
  "route" : { CodeableConcept }, // How drug should enter body
  "method" : { CodeableConcept }, // Technique for administering medication
  // dose[x]: Amount of medication per dose. One of these 2:
  "doseRange" : { Range },
  "doseQuantity" : { Quantity(SimpleQuantity) },
  "maxDosePerPeriod" : { Ratio }, // Upper limit on medication per unit of time
  "maxDosePerAdministration" : { Quantity(SimpleQuantity) }, // Upper limit on medication per administration
  "maxDosePerLifetime" : { Quantity(SimpleQuantity) }, // Upper limit on medication per lifetime of the patient
  // rate[x]: Amount of medication per unit of time. One of these 3:
  "rateRatio" : { Ratio }
  "rateRange" : { Range }
  "rateQuantity" : { Quantity(SimpleQuantity) }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from Element: Element.extension
  fhir:Dosage.sequence [ integer ]; # 0..1 The order of the dosage instructions
  fhir:Dosage.text [ string ]; # 0..1 Free text dosage instructions e.g. SIG
  fhir:Dosage.additionalInstruction [ CodeableConcept ], ... ; # 0..* Supplemental instruction - e.g. "with meals"
  fhir:Dosage.patientInstruction [ string ]; # 0..1 Patient or consumer oriented instructions
  fhir:Dosage.timing [ Timing ]; # 0..1 When medication should be administered
  # Dosage.asNeeded[x] : 0..1 Take "as needed" (for x). One of these 2
    fhir:Dosage.asNeededBoolean [ boolean ]
    fhir:Dosage.asNeededCodeableConcept [ CodeableConcept ]
  fhir:Dosage.site [ CodeableConcept ]; # 0..1 Body site to administer to
  fhir:Dosage.route [ CodeableConcept ]; # 0..1 How drug should enter body
  fhir:Dosage.method [ CodeableConcept ]; # 0..1 Technique for administering medication
  # Dosage.dose[x] : 0..1 Amount of medication per dose. One of these 2
    fhir:Dosage.doseRange [ Range ]
    fhir:Dosage.doseSimpleQuantity [ Quantity(SimpleQuantity) ]
  fhir:Dosage.maxDosePerPeriod [ Ratio ]; # 0..1 Upper limit on medication per unit of time
  fhir:Dosage.maxDosePerAdministration [ Quantity(SimpleQuantity) ]; # 0..1 Upper limit on medication per administration
  fhir:Dosage.maxDosePerLifetime [ Quantity(SimpleQuantity) ]; # 0..1 Upper limit on medication per lifetime of the
    patient
  # Dosage.rate[x] : 0..1 Amount of medication per unit of time. One of these 3
    fhir:Dosage.rateRatio [ Ratio ]
    fhir:Dosage.rateRange [ Range ]
    fhir:Dosage.rateSimpleQuantity [ Quantity(SimpleQuantity) ]
]
```

Changes since DSTU2

This complex-type did not exist in Release 2

Constraints

Dosage is used in the following places: ActivityDefinition, MedicationDispense, MedicationRequest and MedicationStatement

11.8.0.2 Interpretation of Dosage in different contexts

11.8.1 Examples

URL: https://hl7.org/fhir/STU3/dosage-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.8.1 Dosage Examples

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative

11.8.2 Detailed Descriptions

URL: https://hl7.org/fhir/STU3/dosage-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.8.2 Dosage Detailed Descriptions

Pharmacy Work Group | Maturity Level: 3 | Ballot Status: Trial Use

11.8.2.1 Dosage

Dosage
Definition | Indicates how the medication is/was taken or should be taken by the patient.
Control | 1..1
Summary | true
Dosage.sequence
Definition | Indicates the order in which the dosage instructions should be applied or interpreted.
Control | 0..1
Type | integer
Requirements | If the sequence number of multiple Dosages is the same, then it is implied that the instructions are to be treated as concurrent. If the sequence number is different, then the Dosages are intended to be sequential.
Summary | true
Dosage.text
Definition | Free text dosage instructions e.g. SIG.
Control | 0..1
Type | string
Requirements | Free text dosage instructions can be used for cases where the instructions are too complex to code. The content of this attribute does not include the name or description of the medication. When coded instructions are present, the free text instructions may still be present for display to humans taking or administering the medication. It is expected that the text instructions will always be populated. If the dosage.timing attribute is also populated, then the dosage.text should reflect the same information as the timing.
Summary | true
Dosage.additionalInstruction
Definition | Supplemental instruction - e.g. "with meals".
Control | 0..*
Terminology Binding | SNOMED CT Additional Dosage Instructions (Example)
Type | CodeableConcept
Requirements | Additional instruction such as "Swallow with plenty of water" which may or may not be coded.
Summary | true
Dosage.patientInstruction
Definition | Instructions in terms that are understood by the patient or consumer.
Control | 0..1
Type | string
Summary | true
Dosage.timing
Definition | When medication should be administered.
Control | 0..1
Type | Timing
Requirements | The timing schedule for giving the medication to the patient. The Schedule data type allows many different expressions. For example: "Every 8 hours"; "Three times a day"; "1/2 an hour before breakfast for 10 days from 23-Dec 2011:"; "15 Oct 2013, 17 Oct 2013 and 1 Nov 2013". Sometimes, a rate can imply duration when expressed as total volume / duration (e.g. 500mL/2 hours implies a duration of 2 hours). However, when rate doesn't imply duration (e.g. 250mL/hour), then the timing.repeat.duration is needed to convey the infuse over time period.
Summary | true
Comments | This attribute may not always be populated while the Dosage.text is expected to be populated. If both are populated, then the Dosage.text should reflect the content of the Dosage.timing.
Dosage.asNeeded[x]
Definition | Indicates whether the Medication is only taken when needed within a specific dosing schedule (Boolean option), or it indicates the precondition for taking the Medication (CodeableConcept).
Control | 0..1
Terminology Binding | SNOMED CT Medication As Needed Reason Codes (Example)
Type | boolean|CodeableConcept
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true

Comments | Can express "as needed" without a reason by setting the Boolean = True. In this case the CodeableConcept is not populated. Or you can express "as needed" with a reason by including the CodeableConcept. In this case the Boolean is assumed to be True. If you set the Boolean to False, then the dose is given according to the schedule and is not "prn" or "as needed".
Dosage.site
Definition | Body site to administer to.
Control | 0..1
Terminology Binding | SNOMED CT Anatomical Structure for Administration Site Codes (Example)
Type | CodeableConcept
Requirements | A coded specification of the anatomic site where the medication first enters the body.
Summary | true
Comments | If the use case requires attributes from the BodySite resource (e.g. to identify and track separately) then use the standard extension body-site-instance. May be a summary code, or a reference to a very precise definition of the location, or both.
Dosage.route
Definition | How drug should enter body.
Control | 0..1
Terminology Binding | SNOMED CT Route Codes (Example)
Type | CodeableConcept
Requirements | A code specifying the route or physiological path of administration of a therapeutic agent into or onto a patient's body.
Summary | true
Dosage.method
Definition | Technique for administering medication.
Control | 0..1
Terminology Binding | SNOMED CT Administration Method Codes (Example)
Type | CodeableConcept
Requirements | A coded value indicating the method by which the medication is introduced into or onto the body. Most commonly used for injections. For examples, Slow Push; Deep IV.
Summary | true
Comments | Terminologies used often pre-coordinate this term with the route and or form of administration.
Dosage.dose[x]
Definition | Amount of medication per dose.
Control | 0..1
Type | Range|SimpleQuantity
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | The amount of therapeutic or other substance given at one administration event.
Summary | true
Comments | Note that this specifies the quantity of the specified medication, not the quantity for each active ingredient(s). Each ingredient amount can be communicated in the Medication resource. For example, if one wants to communicate that a tablet was 375 mg, where the dose was one tablet, you can use the Medication resource to document that the tablet was comprised of 375 mg of drug XYZ. Alternatively if the dose was 375 mg, then you may only need to use the Medication resource to indicate this was a tablet. If the example were an IV such as dopamine and you wanted to communicate that 400mg of dopamine was mixed in 500 ml of some IV solution, then this would all be communicated in the Medication resource. If the administration is not intended to be instantaneous (rate is present or timing has a duration), this can be specified to convey the total amount to be administered over the period of time as indicated by the schedule e.g. 500 ml in dose, with timing used to convey that this should be done over 4 hours.
Dosage.maxDosePerPeriod
Definition | Upper limit on medication per unit of time.
Control | 0..1
Type | Ratio
Requirements | The maximum total quantity of a therapeutic substance that may be administered to a subject over the period of time. For example, 1000mg in 24 hours.
Summary | true
Comments | This is intended for use as an adjunct to the dosage when there is an upper cap. For example "2 tablets every 4 hours to a maximum of 8/day".
Dosage.maxDosePerAdministration
Definition | Upper limit on medication per administration.
Control | 0..1
Type | SimpleQuantity
Requirements | The maximum total quantity of a therapeutic substance that may be administered to a subject per administration.
Summary | true
Comments | This is intended for use as an adjunct to the dosage when there is an upper cap. For example, a body surface area related dose with a maximum amount, such as 1.5 mg/m2 (maximum 2 mg) IV over 5 – 10 minutes would have doseQuantity of 1.5 mg/m2 and maxDosePerAdministration of 2 mg.
Dosage.maxDosePerLifetime
Definition | Upper limit on medication per lifetime of the patient.
Control | 0..1
Type | SimpleQuantity
Requirements | The maximum total quantity of a therapeutic substance that may be administered per lifetime of the subject.
Summary | true
Dosage.rate[x]
Definition | Amount of medication per unit of time.
Control | 0..1
Type | Ratio|Range|SimpleQuantity
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Identifies the speed with which the medication was or will be introduced into the patient. Typically the rate for an infusion e.g. 100 ml per 1 hour or 100 ml/hr. May also be expressed as a rate per unit of time e.g. 500 ml per 2 hours. Other examples: 200 mcg/min or 200 mcg/1

minute; 1 liter/8 hours. Sometimes, a rate can imply duration when expressed as total volume / duration (e.g. 500mL/2 hours implies a duration of 2 hours). However, when rate doesn't imply duration (e.g. 250mL/hour), then the timing.repeat.duration is needed to convey the infuse over time period.

Summary | true

Comments | It is possible to supply both a rate and a doseQuantity to provide full details about how the medication is to be administered and supplied. If the rate is intended to change over time, depending on local rules/regulations, each change should be captured as a new version of the MedicationRequest with an updated rate, or captured with a new MedicationRequest with the new rate.

11.8.3 Mappings

URL: https://hl7.org/fhir/STU3/dosage-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

11.8.3 Dosage Mappings

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative

This page provides mappings for Dosage.

11.8.3.1 Mappings for RIM Mapping (http://hl7.org/v3)

Dosage | .outboundRelationship[typeCode=COMP].target[classCode=SBADM, moodCode=INT]
sequence | .text
text | .text
additionalInstruction | .text
patientInstruction | .text
timing | .effectiveTime
asNeeded[x] | .outboundRelationship[typeCode=PRCN].target[classCode=OBS, moodCode=EVN, code="as needed"].value=boolean or codable concept
site | .approachSiteCode
route | .routeCode
method | .doseQuantity
dose[x] | .doseQuantity
maxDosePerPeriod | .maxDoseQuantity
maxDosePerAdministration | not supported
maxDosePerLifetime | not supported
rate[x] | .rateQuantity